

SECTION 13.0

NON-OPERATING ROOM PROCEDURES

CHAPTER 13.1

Non-Operating Room Procedures: Adult

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ANESTHESIA FOR NON-OPERATING ROOM PROCEDURES

GENERAL COMMENTS

- Advances in the fields including radiology, cardiology, and neurology have led to a significant increase in the number of anesthesia procedures performed away from the OR. In line with these changes, the ASA has provided guidelines for the safe delivery of anesthesia at remote locations. Non-operating room anesthesia (NORA) presents significant and often unique challenges for the anesthesiologist. Compared to the general OR population, NORA patients may be less well optimized or even

medically unstable (“too sick for the OR”). The delivery of care may be urgent, while the environment in which that care must be delivered often lacks the amenities and supporting personnel available in an OR. The most common complications of NORA involve airway/respiratory events → brain damage or death, occurring approximately twice as often as in the OR. In most cases, these events were determined to be the result of substandard care and therefore preventable. NORA requires attention to detail, teamwork, and good communication to compensate for the typically resource-poor environment.

- Radiation safety is often of concern in NORA. Fluoroscopy is a significant source of radiation exposure for the anesthesiologist, and distance away from the beam source is the best protection. Six feet of air provides >99% beam attenuation, which is roughly equivalent to 0.5 mm lead (equivalent to the shielding found in heavy “lead” aprons and most shield panels). Although that degree of attenuation is impressive, given the presumed stochastic nature of radiation-induced carcinogenesis, it should not be too comforting.

Anesthetic considerations for non-OR locations (modified from ASA Guidelines) include the following:

- Complete H&P identifying any comorbidity that may require special monitoring or other precautions.
- Primary and backup O₂ sources (e.g., piped O₂ + 1 full E cylinder).
- Adequate and reliable suction.
- Adequate and reliable scavenging system (only for inhalational anesthesia).
- Self-inflating hand resuscitator bag with ability to deliver at least 90% O₂.
- Anesthetic drug supplies and administration equipment.
- Monitoring equipment to allow adherence to the “Standards of Basic Anesthetic Monitoring”.
- Sufficient electrical outlets connected to an emergency power supply.
- Wet locations (e.g., cystoscopy, arthroscopy, labor, and delivery) should be equipped with either an isolated electrical source or circuits with ground-fault interrupters.
- Adequate illumination for patient observation and monitoring equipment (flashlight backup).
- Sufficient space for expeditious access to patient, machine, and support equipment.
- Emergency cart with defibrillator immediately available.
- Immediate access to skilled anesthesia support personnel.
- Provision for appropriate postanesthesia care.

Suggested Readings

1. Chang B, Urman RD: Non-operating room anesthesia: the principles of patient assessment and preparation. *Anesthesiol Clin* 2016; 34(1):223–40.

2. Standards for Basic Anesthetic Monitoring American Society of Anesthesiologists, Schaumburg, IL: 2015.
3. Statement on Non-Operating Room Anesthetizing Locations American Society of Anesthesiologists, Schaumburg, IL: 2013 (reaffirmed 2018).
4. Woodward ZG, Urman RD, Domino KB: Safety of non-operating room anesthesia: closed claims update. *Anesthesiol Clin* 2017; 35(4):569–81.

ELECTROCONVULSIVE THERAPY

PROCEDURAL CONSIDERATIONS

Description

Electroconvulsive therapy (ECT) is the transcutaneous application of electrical stimuli to the brain to produce generalized seizures for the treatment of selected psychiatric disorders, such as severe depression. There are several important aspects of ECT that are of relevance to the anesthesiologist. The first is the uncontrolled motor activity associated with generalized seizures. Prior to introduction of GA, the most common injuries associated with ECT were compression fractures of the vertebral bodies and broken limbs from violent tonic-clonic motor activity. Even with complete paralysis, the masseter muscles are directly stimulated to contract during seizure induction. As a result, the most common injury currently associated with ECT is broken teeth.

A second consequence of ECT induction is that the electrical stimulus can cause contraction of scalp musculature and a brief dilation of meningeal blood vessels, resulting in postictal headaches in up to 40% of patients. Patients <50 yr and those with a Hx of migraine headaches appear most at risk for post-ECT headaches that may occasionally require aggressive pain management. Other consequences of ECT include ↓ SpO₂ especially in obese patients. The use of THRIVE has been shown to prevent desaturation in apneic obese patients undergoing ECT. Many post-ECT patients wake up agitated and require post-ECT sedation (e.g., midazolam 2 mg iv). On occasion, ECT may result in a prolonged (>2 min) seizure requiring propofol (25–100 mg) or midazolam (1–2 mg) to terminate. ECT may also be a significant hemodynamic stressor, even to the extent of causing MI with acute pulmonary edema. Initially, central parasympathetic centers are activated, resulting in ↓↓ HR in ~30% of patients. Brief sinus pauses are not uncommon. The initial parasympathetic effects are followed by sympathetically mediated increases in HR and BP up to 20–30% above baseline. Mean arterial blood pressure has been known to double in some instances. These

cardiovascular responses can persist for minutes to an hour or more after the procedure is completed.

The optimal position for ECT is supine. Occasionally, the head is kept slightly raised to help maintain an adequate airway and decrease anxiety. Patients typically come to ECT quite anxious about the procedure. ECT is generally performed in a PACU or specialized ECT suite. In many centers, outpatients make up the majority of patients seen for ECT. The electrical stimulus is applied through adhesive leads prepared with a contact gel. These leads are usually applied to the forehead in a bitemporal or right unilateral placement. Monitoring typically includes a two-lead electroencephalogram (EEG) and frequently an electromyogram (EMG) to measure motor activity. Before induction, a BP cuff is inflated to act as a tourniquet and prevent neuromuscular blockade in the distal limb. Thus, an arm or leg can be used to measure motor duration of the seizure. A special ECT device is used to generate the appropriate electrical stimulus. Seizures are typically 30–90 sec in duration, and the entire procedure—from the induction of anesthesia to patient awakening—is generally <15 min. The recovery period averages 45–90 min and allows for monitoring of vital signs, as well as the opportunity for the postictal confusion to clear. Patients can wake up mildly confused to frankly delirious and require close nursing supervision. Treatments are typically performed every other day, and the average number of treatments is 6–12 in the acute management of major depression. However, ECT treatments may be tapered rather than stopped abruptly. Thus, the frequency of treatments may go from 3/wk for acute treatment to 1/wk, 2/mo, and finally 1/mo. Maintenance ECT with a frequency ranging from every few weeks to every few months is commonly prescribed for those patients who respond to ECT but fail to benefit from pharmacotherapy.

The most common morbidities associated with ECT include headaches and myalgias. Postictal confusion is the rule, and some anterograde and retrograde memory loss occurs in most patients who have completed an acute course of ECT. Memory loss is typically confined to the period immediately before, during, and immediately after a series of ECT treatments. There tends to be a cumulative memory loss with subsequent ECT treatments within a given series. In most patients, memory deficits largely subside in the first 1–3 mo following treatments. In rare instances, autobiographical memory loss has been reported for months or years after an ECT series has been completed. Long-term memory loss appears to be more common with bilateral lead placement. ECT-related mortality is estimated at ~4/10,000. Cardiac events account for 67% of all ECT-related deaths, with malignant arrhythmias and MIs accounting for most fatalities. Pulmonary events (obstruction, pulmonary edema, or emboli) account for most additional mortality. Cerebrovascular infarctions or hemorrhages have rarely been

reported with ECT.

Variant procedure or approach: Transcranial magnetic stimulation (TMS) uses intense extracranial magnetic fields to induce intracranial electrical current flow. Although the induced currents can be sufficient to evoke motor responses, seizures are rare. Compared to ECT, TMS is painless and hemodynamically benign. Anesthesia is not required.

Usual preop diagnosis

Depression; mania; catatonia; refractory psychosis

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	None
Special instrumentation	Seizure generator and electrodes; EEG/EMG monitors
Unique considerations	Requires muscle relaxation to prevent injury. 50–80 mg of succinylcholine is usually sufficient. Using larger dose of succinylcholine can lead to patient waking up while paralyzed. Bite blocks are required to prevent dental injury. Apply tourniquet (e.g., BP cuff) to one arm before administration of muscle relaxant. A slow heart rate preop may require atropine. A high BP preop may require aggressive hypotensive Rx with beta-blockers, phentolamine, hydralazine, etc. If the seizure duration is not >25 sec, then another seizure may be required. Give more induction agent as needed. Because one arm is not paralyzed, asking the patient to squeeze your fingers will tell you if she/he is awake.
Antibiotics	None
Procedure time	Setup: 5–10 min Treatment: <10 min (seizure duration 25–280 sec)
Postop care	PACU → room or home
Mortality	4/10,000

Morbidity	HA/myalgias/nausea: Common Confusion/memory loss: Common. Be aware that postop hypoxia can present with confusion. Cardiac dysrhythmias: 10–40% (brief asystole common). Keep a close eye on the EKG. MI: Rare Pulmonary edema from excessive iv fluids: not uncommon. Pulmonary aspiration and/or negative pressure pulmonary edema: Rare CVA: Rare
Pain score	2–3 (HA)

■ PATIENT POPULATION CHARACTERISTICS

Age range	≥18 yr. ECT is rarely performed in adolescents and preadolescents; however, patients in their 90s are sometimes candidates for ECT. There is a preponderance of geriatric patients on many ECT services.
Male:Female	1:2
Incidence	The lifetime prevalence of major depression (the primary indication for ECT) is ~17%; 26% of women and 12% of men are affected. <1% of patients with major depression undergo ECT, and ~400,000 ECT procedures are performed in North America annually.
Associated conditions	Substance abuse (30% of all depressed patients meet criteria for alcohol or drug abuse); panic attacks (30% of depressed patients); psychotic symptoms (14% of patients); HTN and sinus tachycardia; dehydration; self-inflicted trauma; pregnancy

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for ECT usually have failed to respond to antidepressants; however, most will continue to take psychotherapeutic medications. Many older patients will be taking other medications for coexisting medical conditions. Drug interactions are an important consideration for the anesthesiologist (see [Appendix F: Tables of Drug](#)

Interactions and Recommendations for Preop Management). The most commonly used medications are listed in Table 13.1-1.

Anesthesia for ECT may seem to be a simple procedure; however, these cases—which seldom take >20 min—can prove very challenging, especially in the geriatric population. ECT can place significant stress on the cardiovascular system; therefore, particular care should be taken to evaluate and optimize the patient’s pretreatment cardiovascular status. ECT usually takes place in remote locations, so the anesthesiologist must ensure that the location is properly equipped and complies with ASA Guidelines for Out-of-OR Procedures and Standards for Basic Anesthetic Monitoring.

Table 13.1-1 Commonly Used Medications in the Treatment of Depression			
Tricyclic	MAOI	SSRI	SNRI and Other Antidepressants
Amitriptyline (Elavil, others)	Isocarboxazid (Marplan)	Fluoxetine (Prozac)	Venlafaxine (Effexor)
	Phenelzine (Nardil)	Paroxetine (Paxil)	Duloxetine (Cymbalta)
Amoxapine (Asendin)	Tranylcypromine (Parnate)	Sertraline (Zoloft)	Nefazodone (Serzone)
Desipramine (Norpramin)	Transdermal selegiline (Emsam)	Fluvoxamine (Luvox) Citalopram (Celexa)	Mirtazapine (Remeron) Bupropion (Wellbutrin)
Doxepin (Sinequan)	Lithium (often used as an adjunctive agent/mood stabilizer in depression)	Escitalopram (Lexapro)	Trazodone (Desyrel)
Imipramine (Tofranil)			Atypical antipsychotics (used adjunctively)
Maprotiline			Olanzapine

(Ludiomil) Nortriptyline (Pamelor)			(Zyprexa) Quetiapine (Seroquel)
Protriptyline (Vivactil)			Aripiprazole (Abilify)
Trimipramine (Surmontil)			Risperidone (Risperdal)
			Ziprasidone (Geodon)
			s-Ketamine (Spravato)

MAOI, monoamine oxidase inhibitor; SSRI, selective serotonin reuptake inhibitor; SNRI, selective norepinephrine/serotonin reuptake inhibitor.

Respiratory	These patients will require airway management and PPV. Hence, preop assessment of the airway must focus on the ease of mask ventilation and the potential need for ET intubation (e.g., airway compromise or severe GERD).
Cardiovascular	A recent MI (<3 mo) is a contraindication to ECT. Relative contraindications include aortic aneurysm, angina, CHF, and thrombophlebitis. The presence of dysrhythmias, a pacemaker, or ICD is not a contraindication for ECT. For the patient with a pacemaker, a means (e.g. a magnet) should be available to convert the pacemaker to an asynchronous mode. Patients on long-term anticoagulation therapy may require pre-ECT dose adjustment (target INR 1.5–2.5), or they should switch to heparin before each ECT session and return to their standard therapy post-ECT. Tests: As indicated from H&P
Gastrointestinal	Patient should be NPO. Patients with Sx of GERD should be pretreated with Na citrate (30 mL po), ± ranitidine (50 mg iv) and metoclopramide (10 mg iv). ET intubation should be considered for all patients at

	risk for aspiration.
Neurologic	ECT is relatively contraindicated in the presence of ↑ ICP, and recent CVA (<3 mo), intracranial mass lesions, or recent intracranial surgery (<3 mo).
Endocrine	Presence of a pheochromocytoma (Sx of which may be confused with a psychiatric disorder) is a contraindication to ECT. <1% of hypertensive patients will have a pheochromocytoma.
Genetic	✓ for family Hx of pseudocholinesterase deficiency. Rocuronium (0.6–1.2 mg/kg) reversed with sugammadex (8–16 mg/kg) is a suitable alternative to succinylcholine. Tests: Dibucaine number (normal ≥ 80) and serum cholinesterase level, if indicated from H&P
Hepatic	Hepatotoxicity has been associated with use of MAOI. Tests: Consider LFTs for patients on chronic MAOI therapy.
Orthopedic	In patients susceptible to bone fracture (e.g., severe osteoporosis, osteoporosis imperfecta), an increased succinylcholine dosage (1.5 mg/kg) is given to ensure profound muscle relaxation. Patients with severe rheumatoid arthritis may have unstable C-spine, and extreme care should be taken during positioning of head and neck.
Ophthalmologic	Retinal detachment is a relative contraindication to ECT. Succinylcholine should be avoided in patients with glaucoma treated with cholinesterase inhibitors (e.g., echothiophate). Mivacurium (0.15 mg/kg) is a suitable alternative.
Pregnancy	Pregnancy is not a contraindication for ECT (even in the third trimester). After the 4th mo, the need for full-stomach precautions requires rapid-sequence induction and ET intubation. Left uterine displacement should be maintained during treatment. Monitor fetal heartbeat.

Psychiatric drugs

Patients receiving tricyclic antidepressants (TCAs) may have an exaggerated pressor response to direct-acting sympathomimetic drugs, with the potential for tachycardia, dysrhythmias, and hyperthermia. The response to indirect-acting sympathomimetic drugs (e.g., ephedrine) may be attenuated in these patients. TCAs also will increase the effects of anticholinergic drugs (e.g., glycopyrrolate, atropine). Patients receiving MAOIs will exhibit exaggerated responses to indirect-acting sympathomimetic drugs (e.g., ephedrine). In addition, in these patients, succinylcholine metabolism is inhibited ($\uparrow$ NMB), and meperidine is contraindicated ($\uparrow\uparrow$ BP, $\uparrow$ Sz, $\uparrow\uparrow$ T). It is probably unnecessary to discontinue TCAs or MAOIs prior to ECT, as long as these interactions can be avoided. Lithium should be discontinued for at least 3 d prior to ECT to avoid delayed recovery and subsequent posttreatment agitation and confusion. Lithium also is associated with $\uparrow$ NMB (succinylcholine and nondepolarizing agents). SSRIs have been associated with prolonged ECT-induced seizure duration and adverse behavioral/neurologic effects following haloperidol administration (use droperidol and metoclopramide with caution). No adverse interactions have been reported between anesthetic agents and SSRIs or SNRIs.

Laboratory

Tests as indicated from H&P

Premedication

Although usually not required, some patients may benefit from an antisialagogue (glycopyrrolate 0.2 mg iv). Patients with Hx of postop N/V will benefit from a prophylactic antiemetic (e.g., ondansetron 4 mg iv). Patients with postseizure muscle pain and headache may benefit from ketorolac (30 mg iv). Some patients will require 500–1000 mg caffeine iv or po to decrease seizure threshold. The iv caffeine effect should be manifest in 5 min. However, po caffeine must be given 30–40 min in advance. Verapamil (not adenosine, which is blocked by

caffeine) should be available to control supraventricular tachycardia (0.07–0.25 mg/kg over 2 min). Labetalol, hydralazine, esmolol, and diltiazem are alternative drugs for control of HR and BP.

◆ INTRAOPERATIVE

Anesthetic technique

A review of previous anesthetic records is very helpful in formulating the anesthetic plan and in anticipating physiologic changes unique to each patient. Usually brief iv anesthesia (mask oxygenation and ventilation) with muscle relaxation (1–2 twitches) is required to prevent patient injury during seizures. Prior to induction, a tourniquet (BP cuff) is applied to the non-iv limb and inflated to a pressure above systolic. This prevents neuromuscular blockade distal to the cuff and permits direct monitoring of seizure activity. Preop BP control (e.g., labetalol 5–20 mg, diltiazem 10–20 mg iv, or esmolol 0.5–1 mg/kg iv in increments) is often necessary.

Induction

Preoxygenation should be attempted in all patients. In some patients who are intolerant of a mask, a less intrusive blow-by technique may be tried. Anesthesia is induced with propofol (1–2 mg/kg), sodium methohexital (0.5–1 mg/kg), or etomidate (0.1–0.2 mg/kg). An induction dose of etomidate will block stress-induced cortisol production for up to 24 h. Propofol (1–1.5 mg/kg) may be used, but may shorten seizure duration. TCAs and MAOIs can increase sleep time. After tourniquet inflation, succinylcholine (0.8–1 mg/kg) is injected to induce paralysis. Hyperventilation is carried out to enhance seizure activity. A bite block is placed, and then the patient is ready for ECT. In barbiturate-tolerant patients, remifentanyl (1–3 mcg/kg) has been used as a means of reducing the barbiturate dose, thereby permitting adequate seizure duration. Patients receiving remifentanyl need minimal postseizure BP control.

Maintenance

Given the brevity of this procedure, maintenance of anesthesia is rarely a concern; however, occasionally, a second or third treatment may be necessary if the

	seizures are of inadequate duration (<25 sec) and quality. In this case, a subsequent dose (10–30 mg) of succinylcholine may be needed. Assisted ventilation is necessary until spontaneous ventilation resumes. Of major concern during the seizure period is the hypertensive response. Some patients may need to be treated prior to induction of anesthesia with either labetalol (5–20 mg iv) or esmolol (10 mg q1min) to control HR/BP. SNP (5–50 mcg/bolus iv) is useful to control BP in refractory cases. Profound ↓ HR may occur in the immediate post-ECT period requiring atropine or a chest thump.	
Emergence	Patients should be awake within 5–10 min postseizure and often are disoriented. Small doses of midazolam (e.g., 0.25–0.5 mg iv) may help to control agitation, and blow-by O₂ may be less stimulating. ASA guidelines for postanesthesia care should be followed.	
Blood and fluid requirements	No blood loss IV: 20 ga × 1 BSS @ TKO	
Monitoring	Standard monitors Tourniquet EEG EMG Transcutaneous CO₂	Seizure activity is usually monitored by a psychiatrist observing the tourniqueted limb and by measuring EMG and EEG activity. (These monitors are usually an integral part of the ECT seizure generator.) Transcutaneous CO₂ monitoring may be useful in patients who are apneic for several minutes.
Positioning	Supine	
Complications	Dysrhythmias	Brief periods of asystole and profound bradycardia are not uncommon (usually related to

Tachydysrhythmias

parasympathetic overactivity). Treatment is rarely necessary. Responds well to esmolol (10–15 mg iv) or lidocaine (1 mg/kg iv), although treatment is usually unnecessary.

↑ BP

↑ BP readily responds to esmolol (10–30 mg) or labetalol (5–20 mg). In refractory cases, 10–50 mcg of SNP may be necessary.

Dental damage
Pulmonary edema
Aspiration

Use of bite block is essential. Dental damage is not prevented by muscle relaxation (direct electrical stimulation of facial and jaw muscles).

POSTOPERATIVE

Complications

HA, myalgias
N/V
Disorientation

Rx: ketorolac 30 mg iv
Rx: ondansetron 4 mg iv
Rx: midazolam 0.25–0.5 mg iv

Memory impairment
MI/ischemia
Dysrhythmias
Pulmonary edema/aspiration
↑ BP

Prolonged ↑ BP is unusual and may suggest the need for further workup.

Suggested Readings

1. Allan CL, Ebmeier KP: The use of ECT and MST in treating depression. *Int Rev Psychiatry* 2011; 23(5):400–12.
2. Brock-Utne JG: *Clinical Anesthesia: Near Misses and Lessons Learnt*, 2nd edition. Springer: 2018, 151–3.
3. Cohran M, DeBattista C, Schmiesing C, et al: Negative pressure pulmonary edema. A potential hazard in patients undergoing ECT. *J ECT* 1999; 15:168–70.
4. Kadiyala PK, Kadiyala LD: ECT: a new look at an old friend. *Curr Opin Anaesthesiol* 2018; 31(4):453–8.
5. Kurup V, Ostroff R: When cardiac patients need ECT—challenges for the anesthesiologist. *Int Anesthesiol Clin* 2012; 50(2):128–40.
6. MacPherson RD, Loo CK: Cognitive impairment following electroconvulsive therapy—does the choice of anesthetic agent make a difference? *J ECT* 2008; 24(1):52–6.
7. Saxena AR, Brock-Utne JG, Brodsky JB: THRIVE and desaturation in obese patients undergoing Electroconvulsive Therapy (ECT). *Anesthesiology Annual Meeting Boston*: October 22–25, 2017.
8. Sharp RP, Welch EB: Takotsubo cardiomyopathy as a complication of electroconvulsive therapy. *Ann Pharmacother* 2011; 45(12):1559–65.
9. Zhang K, Hashimoto K: An update on ketamine and its two enantiomers as rapid-acting antidepressants. *Expert Rev Neurother* 2019; 19(1):83–92.

INTERVENTIONAL NEURORADIOLOGY

PROCEDURAL CONSIDERATIONS

The indications for endovascular therapy for the brain and spine have continued to grow with the technical strides made in both devices and imaging. Endovascular therapy is now considered first-line therapy in the treatment of many intracranial aneurysms. At most centers, it is the treatment of choice for arteriovenous fistulas (AVFs) and is widely used as an adjunctive therapy for arteriovenous malformations (AVMs) and vascular tumors. Recent data have made endovascular therapy the standard of care for treatment of large vessel occlusions in the setting of acute stroke, and it is also used for the treatment of cerebrovascular stenosis. Many of these procedures can be performed with the patient awake; however, GA or deep sedation often is used to minimize patient movement during procedures that require careful catheter and device control for safe operation. The endovascular procedures used in interventional neuroradiology can be divided into three broad categories: (a) embolization, (b) aneurysm therapy, and (c) cerebral revascularization.

Embolization

This therapy may be used for a variety of lesions, including AVMs, dural AVFs, vein of Galen malformations, vascular neoplasms (e.g., meningioma), hemangiomas, glomus tumors, and juvenile nasal angiofibromas, and for treatment of epistaxis.

AVM embolization is often performed as an adjunct to radiosurgery or microsurgery, although in some settings, embolization may be the definitive treatment. Brain AVMs are generally parenchymal lesions with multiple feeding pial arteries and draining veins. A common goal of embolization is to reduce the size and shunt burden presented by the AVM before either radiosurgery or microsurgical resection. Liquid embolic agents are preferred, with n-butyl cyanoacrylate (NBCA) and ethylene vinyl alcohol copolymer (EVOH) being the most widely used materials for the procedure. Embolization often is preceded by neurophysiologic testing, which may include the superselective injection of a barbiturate into the portion of the intracranial circulation being considered for embolization. This may be accompanied by the use of somatosensory and motor-evoked potential testing. Some pre-embolization testing may be performed with the patient awake to allow for more thorough clinical testing, such as language function. However, some endovascular therapists prefer to use GETA to minimize patient motion, and it is needed when motor-evoked testing is performed.

Dural arteriovenous fistulas drain directly into dural veins or sinuses and are most commonly located in the regions of the cavernous, transverse, and sigmoid sinuses. Because of their location, these malformations are supplied by dural or meningeal arteries. They are acquired lesions but often have no clear etiology or antecedent history. Patients may have a variety of symptoms depending on the location, size, and drainage pattern of the dural fistula. Those arising in the transverse or sigmoid sinus region often present with tinnitus. Cavernous sinus fistulae often present with proptosis and ophthalmoplegia. These cavernous sinus fistulae also may be posttraumatic, developing as a direct large-hole fistula between the internal carotid artery and the cavernous sinus. The embolization process for these dural-based lesions differs somewhat from the technique used for pial-based brain AVMs. Arterial embolization may be used, but venous embolization is often used to definitively occlude the fistula. Embolization of meningeal arteries may be preceded by clinical testing for cranial nerve deficits. This usually is accomplished with the superselective injection of lidocaine before embolization, and it is often preferred to have the patient awake for this procedure. If arterial embolization is utilized, a liquid embolic (NBCA or EVOH) is often utilized. Venous embolization is usually done with platinum coil occlusion.

Vein of Galen malformations are congenital lesions that may present in infancy or early childhood. Presenting symptoms include congestive heart failure (CHF)

hydrocephalus, and neurodevelopmental delay. These lesions often require a staged approach to treatment and present a special challenge in the neonate or infant. In general, arterial embolization is performed as the initial endovascular approach, and a liquid embolic agent is used. In some cases, this may be augmented by a venous approach, with embolization using platinum coils.

Tumor embolization usually is performed as an adjunct to the surgical resection of highly vascular tumors (e.g., meningiomas, hemangiomas, hemangioblastomas, glomus tumors, and juvenile nasal angiofibromas). Generally, arterial embolization of meningeal supply vessels is done before surgery, using PVA or trisacryl gelatin microspheres. Physiologic testing with superselective injection of lidocaine often precedes embolization. When there is tumor encasement of a major artery, the patient may also undergo balloon test occlusion followed by permanent occlusion to reduce the risk of intraop bleeding.

Aneurysm Therapy

Endovascular therapy is the treatment of choice for many intracranial aneurysms. The earliest method for endovascular treatment consisted of direct intra-aneurysmal obliteration, filling the aneurysm pouch with detachable platinum coils. A randomized, controlled trial using this technique showed better clinical outcome for patients treated with aneurysm coiling than surgical clipping in the setting of acute subarachnoid hemorrhage.

A spate of new devices and adjunctive techniques have extended the endovascular therapist's ability to treat aneurysms of various sizes and locations that could not be treated with coils alone. Smaller, narrow-necked aneurysms may be treated with a microcatheter through which coils are introduced directly into the aneurysm. Wider-necked aneurysms are more difficult to treat using this technique because coils may prolapse into the parent artery. In this situation, coiling may be done either with balloon remodeling or with the combined use of a stent. Balloon remodeling involves placing a balloon catheter in the parent artery over the ostium of the aneurysm. The balloon is intermittently inflated as coils are inserted to prevent coil prolapse into the parent vessel. Fenestrated stents can also be introduced into the parent artery over the ostium of the aneurysm. The aneurysm is coiled by placing a microcatheter through the stent fenestrations into the aneurysm.

The use of a flow-diverting stent in the parent artery to produce thrombosis of the aneurysm has also become widely used, particularly to treat larger wide-necked aneurysms (see [Fig. 13.1-1](#)). These stents are constructed using fine mesh self-expanding metal. They are placed in the parent artery to reduce blood flow into the

aneurysm sac resulting in gradual thrombosis of the aneurysm while maintaining flow in the parent artery. This treatment is generally reserved for unruptured aneurysms as the patients require treatment with antiplatelet agents. Recently another intra-aneurysmal device (LUNA[®], Medtronic) consisting of a self-expanding and self-retaining spherical braided mesh of nitinol wires has begun to be used, particularly in aneurysms arising at arterial bifurcations.

Parent artery occlusion is still used for some giant or fusiform aneurysms but has more recently been largely replaced by the use of flow-diverting stents. Parent artery occlusion is generally done in a two-step process. Test occlusion is initially performed with a balloon-tipped catheter, and the patient is evaluated using clinical testing and neurophysiologic monitoring. The testing may be done with controlled hypotension to improve the test sensitivity. If the patient tolerates test occlusion, a permanent occlusion is usually done using coils placed into the parent artery.

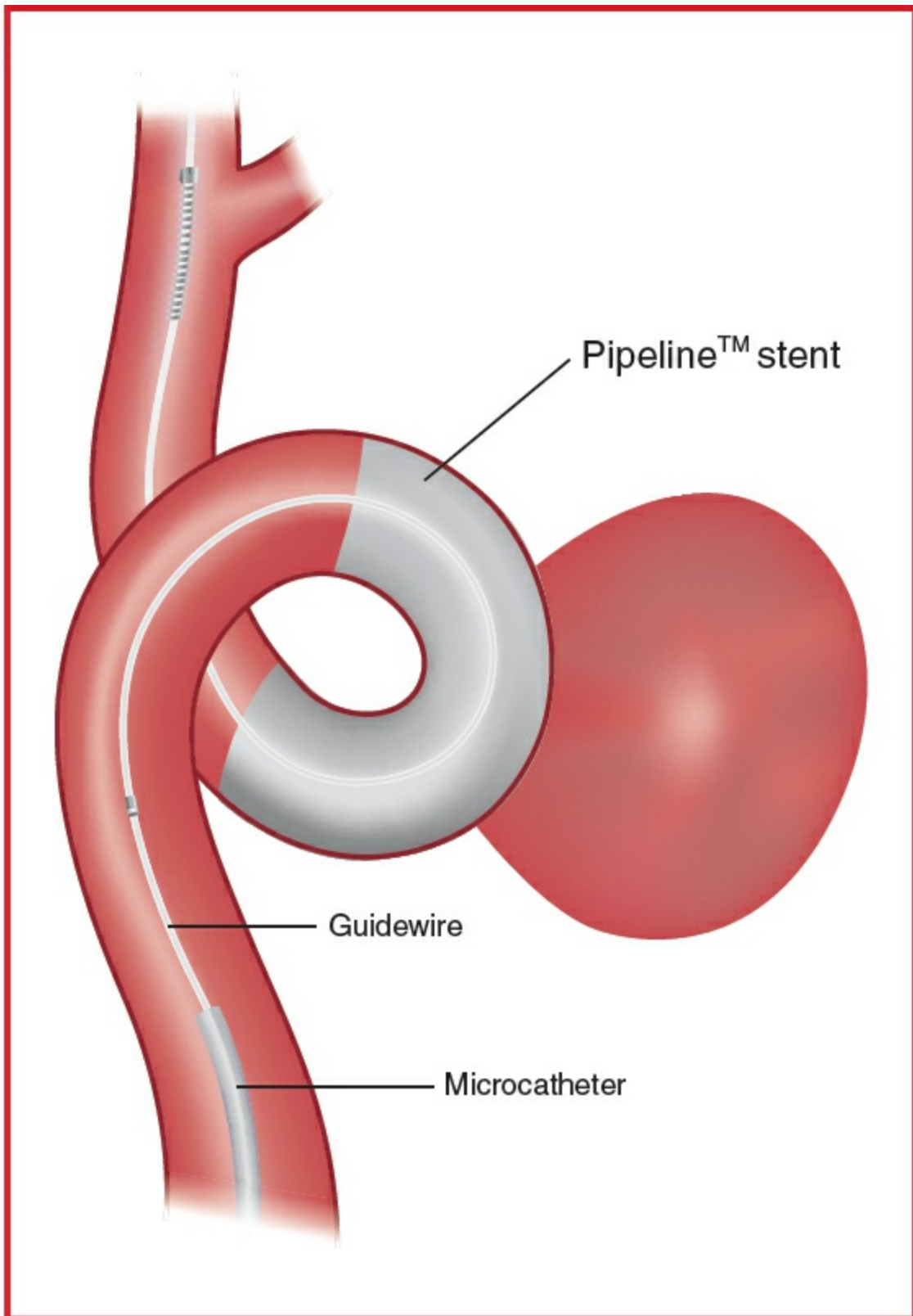


Figure 13.1-1 Example of a fine mesh, flow-diverting stent (Pipeline™) used to treat a wide-neck aneurysm.

Cerebral Revascularization

Acute stroke thrombolysis or thrombectomy is performed to recanalize larger occluded

arteries in the cerebrovascular circulation, including the internal carotid, middle cerebral, vertebral, and basilar arteries. Intravenous tissue plasminogen activator (IV-tPA) had been standard of care when the drug was administered up to 4.5 h following acute stroke onset. Randomized trials using IV-tPA beyond 4.5 h were unable to establish a benefit. In the last decade, endovascular therapeutic devices were introduced and resulted in better recanalization rates and clinical outcomes. These devices include a suction thrombectomy catheter, which can be introduced into the intracranial circulation, and retrievable stents, which are introduced into the thrombus and after entrapping the thrombus are pulled from the circulation. The devices may be used alone or in combination. Recently, five separate randomized trials have shown that endovascular therapy improves outcomes over IV-tPA alone or standard medical therapy in a 6 h (up to 24 h in selected patients) time window with endovascular therapy now being the standard of care for large artery occlusions. Physiologic imaging with either computed tomography (CT) or MR is used at many centers to select patients who will benefit from treatment in these more extended time windows. General anesthesia has become less widely used in these cases as several retrospective studies have suggested better outcomes for patients who are awake during stroke thrombectomy. Unfortunately, these studies are confounded by selection bias, and randomized control trials have not been performed.

Angioplasty and stent placement for symptomatic atherosclerotic stenosis in the cerebrovascular circulation are becoming more widely performed in lieu of medical or direct surgical therapy. Stents with distal protection devices have now been approved for the treatment of cervical carotid artery stenosis. Distal protection devices (e.g., balloon or basket devices) have been shown to reduce the thromboembolic complication rate and are now required for treatment in most cases. A recent large multicenter trial of both symptomatic and asymptomatic carotid stenosis patients found that the composite risk of stroke, myocardial infarction, or death did not differ significantly between stented patients and those undergoing endarterectomy. However, a 2017 systematic review suggests that endarterectomy is the best option for management of asymptomatic carotid stenosis. Arterial stenosis located more distally and intracranial stenotic lesions are treated with angioplasty alone or are stented following angioplasty. There are no efficacy data available for these more distal lesions, and a recent trial of stenting versus medical therapy for intracranial stenosis found that medical therapy had lower rates of stroke and death.

Transcarotid artery revascularization (TCAR): This procedure uses retrograde flow for embolic protection during carotid stent placement. Device placement requires direct access to the affected common carotid artery (CCA) through a small supraclavicular

incision. Dissection is carried down to the CCA, a loose ligature is placed around the CCA and adventitial purse string stitches are placed around the perimeter of the anticipated puncture site. The CCA is punctured and a guide wire is inserted, followed by a dilator and catheter. The CCA is clamped under direct vision, and reverse flow is initiated from the internal carotid artery through a 200 µm filter to a previously placed femoral vein cannula. Transcarotid stent placement then proceeds in the usual fashion with the reversed carotid flow flushing any procedure related embolic debris to the filter for capture.

This procedure can be accomplished with GA or local anesthesia (field block) placed by the surgeon. As usual, with the procedure MAC the anesthesiologist should be prepared to control hemodynamics (art line useful), airway and patient cooperation. ACT goal is 250–300 sec. Some patients will not tolerate the loss of antegrade flow requiring rapid conversion to GA and prompt completion of the procedure.

Vasospasm often accompanies subarachnoid hemorrhage and results in ischemic complications, which are a common cause of morbidity and mortality following aneurysmal rupture. The endovascular therapist is often asked to treat this problem with either drugs or balloon angioplasty. Direct administration of intra-arterial vasodilators, such as verapamil, nimodipine, or nicardipine, has been used particularly for treatment of more distal spasm. More proximal spasm involving the arteries of the circle of Willis is often treated using high-compliance angioplasty balloons.

■ SUMMARY OF PROCEDURES

	Embolization	Aneurysm Therapy	Cerebral Revascularization
Position	Supine	⇐	⇐
Incision	Femoral artery, catheterization (may utilize brachial or radial artery)	⇐	⇐
Unique considerations	BP control; AVM; anticoagulation; EP monitoring	BP control; ± EP monitoring; anticoagulation	⇐
Antibiotics	Usually none; occasionally used with closure device	⇐	⇐

Procedure time	2–5 h	2–4 h	⇐
Closing considerations	Femoral artery compression or closure device	⇐	⇐
EBL	Minimal	⇐	⇐
Postop care	Ward or ICU × 24–48 h	ICU	Ward or ICU
Mortality	1–2% (vascular lesions)	⇐	⇐
Morbidity	Overall: ~5% Thromboembolic stroke Hemorrhage (AVM, AVF)	⇐ ⇐ SAH	⇐ ⇐ Vessel rupture/dissection
Pain score	1–2	1–2	1–2

■ PATIENT POPULATION CHARACTERISTICS

Age range	Neonatal–elderly
Male:Female	1:1
Etiology	Congenital; acquired (traumatic, infectious, degenerative, etc.)
Associated conditions	SAH ± vasospasm; ↑ ICP; HTN/CAD/PVD; blood dyscrasia

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for diagnostic neuroradiologic procedures may require only local anesthesia and sedation. The newer nontoxic and low osmolality contrast agents have improved patient comfort and tolerance for these procedures while minimizing adverse reactions. Patients presenting for therapeutic neuroradiologic procedures may be restless or uncooperative and, therefore, may require GA to tolerate these often lengthy procedures. The advantages of GA, however, must be balanced against the potential

need for interactive neurologic monitoring (e.g., speech, vision, and mental status assessment) that requires the patient to be awake and cooperative. In this case, close consultation among the neuroradiologist, monitoring neurologist, and anesthesiologist is required in formulating the anesthesia plan. In many cases, time is of the essence, and any delay may reduce favorable outcome.

Respiratory	Access to the airway may be limited; therefore, examination should focus on the need for elective ET intubation. Patients with chronic cough may require GA to ensure immobility. Tests: As indicated from H&P
Cardiovascular	Patients with recent intracranial hemorrhage may demonstrate ECG abnormalities (PVCs in 30–80%; ST-T wave changes in >50%), which need to be differentiated from new ischemic heart disease (echo, cardiac enzymes). Tests: ECG; other tests as indicated from H&P
Neurologic	Symptoms vary with the location, size, and type of lesion. Aneurysms seldom produce neurologic symptoms unless they leak or rupture, whereas tumors are commonly associated with symptoms of ↑ ICP (HA, N/V, altered mental status, papilledema). Patients with recent cerebral hemorrhages are likely to be medicated with calcium channel blockers (e.g., nimodipine, nicardipine) to ↓ arterial vasospasm. Patients with ↑ ICP or cranial trauma usually will need GETA and mechanical ventilation.
Premedication	Preop sedation may mask the Sx of ↑ ICP or intracranial hemorrhage. Patients at high risk for contrast-media reactions (e.g., patients with previous contrast reaction, allergy to iodine) should receive prophylactic treatment usually consisting of prednisone, 50 mg po q6h × 3, starting 18 h before the study, and diphenhydramine, 50 mg po, 1 h before the procedure.

◆ INTRAOPERATIVE

Anesthetic technique

MAC may be adequate for patients undergoing diagnostic procedures and necessary for patients requiring neurologic assessment during more invasive procedures; otherwise, use GETA or be prepared to rapidly convert to GETA during the procedure. Although several reports suggest that patients receiving GETA for neurointerventional procedures have worse outcomes compared to those receiving sedation, these observations are not based on randomized controlled studies and are inherently biased since the patients most likely to require only sedation are also the most likely to be stable and uncomplicated at the time of the procedure.

Induction	Standard induction. Patients with $\uparrow$ ICP should be hyperventilated to an ETCO_2 of ~ 30 mm Hg. In patients with vascular lesions that may leak or rupture, BP responses to laryngoscopy and intubation should be blunted (e.g., remifentanyl: 3–5 mcg/kg iv 1–2 min in advance).	
Maintenance	Standard maintenance. Muscle relaxation is usually mandatory to control ventilation and minimize the chance of movement. Maintain normocapnia, although hyperventilation may be necessary to $\downarrow$ ICP and may also enhance the quality of the angiogram. Heparin (ACT goal: 250–350 sec; $t_{1/2} = \sim 30$ min) may be requested. Protamine should be immediately available.	
Emergence	Prompt awakening is important to permit neurologic evaluation. Ondansetron (4 mg iv) is useful to $\downarrow$ PONV. Extubate when airway reflexes have returned after reversal of muscle relaxant. Continuous control of BP may be necessary during emergence phase. Patients typically are transported to ICU immediately following the procedure.	
Blood and fluid requirements	IV: 18 ga $\times$ 1–2 BSS @ 3–5 mL/kg/h	Avoid glucose-containing solutions unless the patient is hypoglycemic (< 70 mg/dL).
Monitoring	Standard monitors	BP can be monitored from

	Arterial line Urinary catheter ± neuromonitoring Glucose	femoral line placed by radiologist. However, another arterial line often will be necessary for postop monitoring in the ICU. Place urinary catheter if procedure is lengthy (>3 h). Keep isoflurane or sevoflurane <0.5 MAC and N₂O <50% to minimize interference with EP monitoring. Supplement with remifentanyl infusion as necessary. TIVA may be requested in the erroneous belief that motor or sensory EPs cannot be obtained while using inhalational agents. Keep serum glucose 70–140 mg/dL. Use insulin infusion to control hyperglycemia.
Control of BP	SNP (0.2–2 mcg/kg/min) Maintain normovolemia. Esmolol (50–200 mcg/kg/min)	BP control may be necessary during intracranial catheter manipulation, embolization, and postembolization. Optimal BP management remains unclear. Close communication with the radiologist is important. In general, in patients with acute ischemic stroke, keep SBP 140–180 mm Hg. In acute hemorrhagic stroke, lowering SBP to 140 mm Hg is recommended. SNP/esmolol may be necessary and can be infused through a second peripheral

		iv. Clevidipine (1–21 mg/h; $T_{1/2} = 5–15$ min) can be used in place of SNP ($t_{1/2} = <2$ min), although it is significantly more costly (2019).
Positioning	✓ and pad pressure points ✓ eyes	X-ray table may not be well padded → nerve damage.
Complications, contrast related	Common reactions: - N/V - Itching - Urticaria - Sensation of warmth - Pain - Anxiety - Rash Neurotoxic Sx: - Hemiplegia - Blindness - Aphasia ↓ consciousness Major allergic reactions: - Bronchospasm ↓ BP - Cardiac arrest - Pulmonary edema - Laryngeal edema - Dysrhythmias	These reactions occur in >5% of patients and may require no treatment apart from reassurance or a mild anxiolytic. Mild allergic reactions may be treated with diphenhydramine 25–50 mg iv. Monitor patients for progression of Sx → need for more aggressive therapy. These reactions may be related to the hyperosmolarity of the agent. If persistent, procedure should be terminated. Rx may require steroids, and vasopressors to improve perfusion. In the anesthetized patient, these Sx will be masked. Epinephrine (0.25–0.5 mg iv) should be given immediately for severe reactions. Rx of anaphylaxis includes the following: eliminate antigens (e.g., contrast agent, latex); secure airway; administer 100% O₂, iv fluids,

		epinephrine, and diphenhydramine + ranitidine. Supplemental Rx may include steroids (e.g., hydrocortisone 5 mg/kg), atropine, NaHCO ₃ , vasopressin, and epinephrine infusion.
Complications, other	Hemorrhage	Reverse heparin (e.g., 1 mg protamine/100 mg heparin after 30–60 min). Intracranial bleeding may require immediate transport to OR for surgical repair.
	Vasospasm	Rx: vasodilators (e.g., NTG or papaverine) delivered by catheter, or balloon angioplasty.
	Occlusion of vessel	Deliberate hypertension may be beneficial. Rx: angioplasty. (Recanalization and stenting may be used for thrombotic occlusions.)

POSTOPERATIVE

Complications	Neurologic deficits	CT scan for evaluation, as prompt neurosurgical intervention may be required.
	Vasospasm	May require Ca ⁺⁺ channel blocker (e.g., nimodipine). Consult with neurosurgeon.

Suggested Readings

1. Avitsian R, Machado SB: Anesthesia for endovascular approaches to acute ischemic stroke. *Anesthesiology Clin* 2016; 34:497–509.
2. Becske T, Kallmes DF, Saatci I, et al: Pipeline for uncoilable or failed aneurysms:

results from a multicenter clinical trial. *Radiology* 2013; 267:858–68.

3. Berkhemer OA, Fransen PS, Beumer D, et al: A randomized trial of intra-arterial treatment for acute ischemic stroke. *N Engl J Med* 2015; 372(1):11–20.
4. Brinjikji W, Paternak J, Murad MH, et al: Anesthesia-related outcomes for endovascular stroke revascularization: a systematic review and meta-analysis. *Stroke* 2017; 48:2784–91.
5. Brott TG, Howard G, Roubin GS, et al: Long-term results of stenting versus endarterectomy for carotid-artery stenosis. *N Engl J Med* 2016; 374:1021–31.
5. Derdeyn CP, Chimowitz MI, Lynn MJ, et al: Aggressive medical treatment with or without stenting in high-risk patients with intracranial artery stenosis (SAMMPRIS): the final results of a randomised trial. *Lancet* 2014; 383:333–41.
7. Lee CZ, Young WL: Anesthesia for endovascular neurosurgery and interventional neuroradiology. *Anesthesiol Clin* 2012; 30(2):127–47.
3. Lobanova I, Qureshi AI: Blood pressure goals in acute stroke-how low do you go? *Curr Hypertens Rep* 2018; 20(4):28.
9. Malas MB, Leal Lorenzo JJ, Nejim B, et al: Analysis of the ROADSTER pivotal and extended-access cohorts shows excellent 1-year durability of transcatheter stenting with dynamic flow reversal. *J Vasc Surg* 2019; 69(6):1786–1796.
0. Moresoli P, Habib B, Reynier P, et al: Carotid stenting versus endarterectomy for asymptomatic carotid artery stenosis: a systematic review and meta-analysis. *Stroke* 2017; 48(8):2150–7.
1. Saver JL, Goyal M, Bonafe A, et al: Stent-retriever thrombectomy after intravenous t-PA vs. t-PA alone in stroke. *N Engl J Med* 2015; 372(24):2285–95.
2. Scharf J, Dempfle CE: Anticoagulation in neurointerventions: basic pharmacology and pathophysiology, current status, practical advice. *Clin Neuroradiol* 2012; 22(1):3–13.
3. Wang A, Stellfox M, Moy F, et al: General anesthesia during endovascular stroke therapy does not negatively impact outcome. *World Neurosurg* 2017; 99:638–43.

DIRECT CURRENT CARDIOVERSION

Bing Liem

PROCEDURAL CONSIDERATIONS

Description

Direct-current (DC) cardioversion is a treatment for various cardiac arrhythmias (e.g., atrial fibrillation [AF], atrial flutter [AFL], other supraventricular tachyarrhythmias [SVT], or, rarely, slow ventricular tachycardia [VT]) that uses a brief, dosed discharge

of electricity across the heart. Depending on the seriousness of the arrhythmia (with respect to its effect on hemodynamic stability), cardioversion to restore normal rhythm may be performed emergently or electively (Fig. 13.1-2).

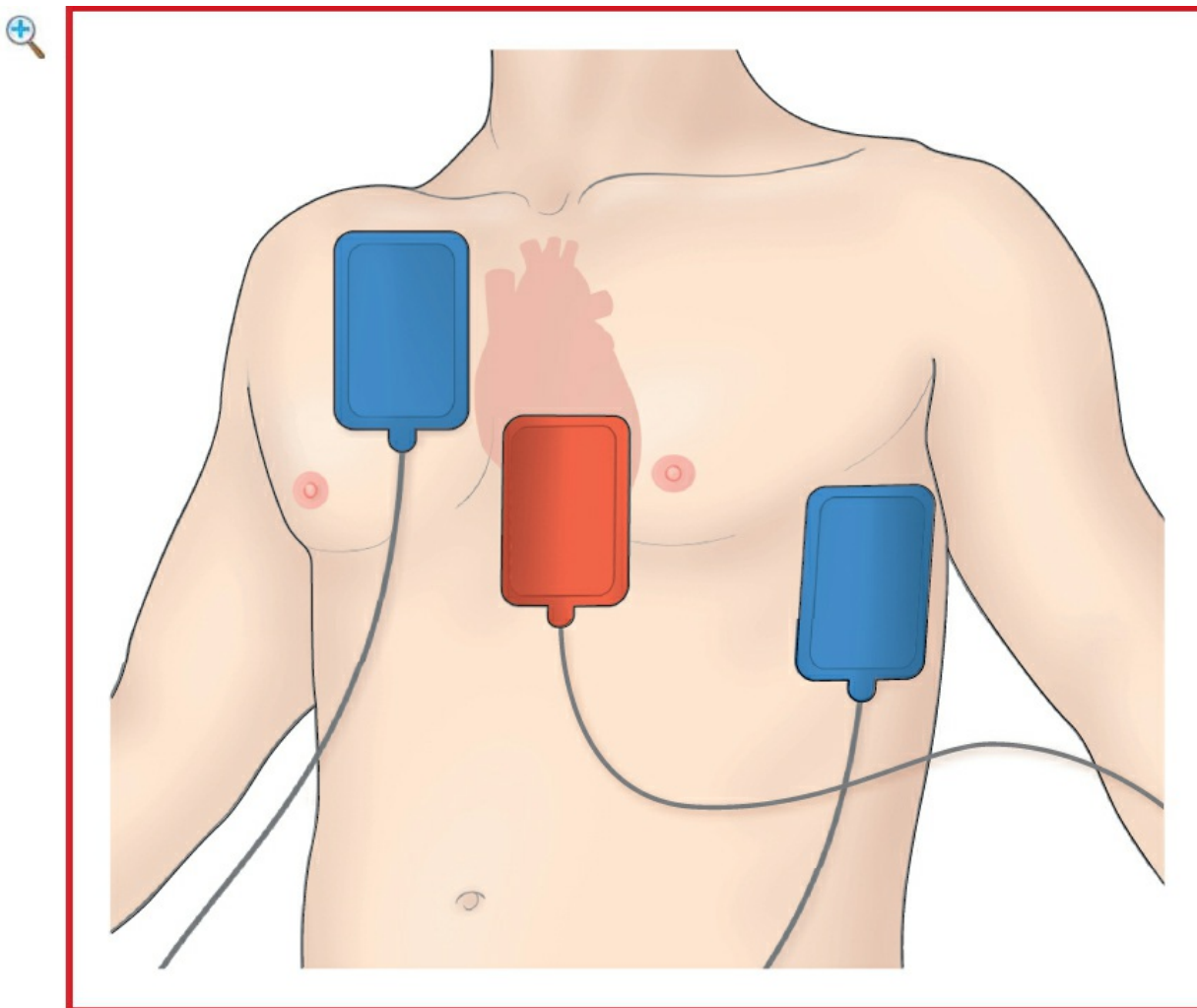


Figure 13.1-2 Positions of defibrillation pads are shown; the AP positions in red and anterolateral in blue. (Modified from Lynn P. Taylor's *Clinical Nursing Skills*, 5th edition. Wolters Kluwer, Philadelphia: 2019, 970.)

Unlike emergent shock, which is typically performed for ventricular fibrillation (VF), elective cardioversion should always be performed with the patient under deep sedation or brief GA to avoid discomfort. The shock energy is delivered across the heart either from an external source (a defibrillator unit) or internally (from a previously implanted ICD/CRT-D). External cardioversion requires placement of a pair of pads that should be positioned to ensure current flow through the heart. The most common positions are anteroposterior (AP) and basilar-apical/anterolateral (right shoulder and left lateral chest). In one randomized study, the AP position was shown to be more effective than the anterolateral position. Biphasic waveform shock energy is typically 25–200 J (25–

100 J for AFL and SVT and 150–200 J for AF). Most external defibrillators deliver a maximum of 200 J. Special units can deliver up to 360 J, for circumstances requiring higher energy (e.g., very large patients or patients with severe hypertrophic cardiomyopathy). Monophasic waveform is no longer used. When internal cardioversion is performed, the implanted unit is typically set to deliver its full energy (30–40 J biphasic waveform). In either case, the shock is delivered synchronous with the QRS complex as to avoid the vulnerable electrical period (during the T-wave, which may risk induction of VF). In patients with implanted devices (pacemaker [PM] or ICD/CRT-D), the pads should be placed at least 6 inches away from the implanted generator to avoid potential damage to the implanted system. The implanted device should be interrogated before and after the procedure. If unsuccessful, cardioversion may need to be repeated using higher energy, using an alternate pad positions or after infusion of ibutilide (class III agent) to lower defibrillation threshold. If the arrhythmia immediately recurs, cardioversion is repeated after infusion of an antiarrhythmic drug to promote maintenance of normal rhythm.

Usual preop diagnoses: AF, AFL, SVT, and slow VT along with underlying cardiovascular diseases such as hypertension, cardiomyopathy (ischemic or nonischemic), valvular disease(s) (most commonly mitral valve regurgitation), and congenital heart disease(s).

■ SUMMARY OF PROCEDURE

Position	Supine with defibrillator pads positioned AP or anterolateral
Unique considerations	In patients with AF, adequate anticoagulation: INR = 2–3 for those on warfarin and uninterrupted use for those on target-specific oral anticoagulants (NOAC) for at least 3 wk prior; otherwise, TEE should be considered to rule out left atrial thrombus.
Antibiotics	None
Procedure time	≤30 min
Postop care	Monitoring of cardiac rhythm in treatment room
Mortality	0.1%
Morbidity	Skin burns: <15% (1st degree); <2% (2nd degree); lesser incidence with biphasic waveform Embolic event: 2% (↑ risk with mitral valve disease)

Acute pulmonary edema: 1%
 More serious arrhythmia: 1%
 Myocardial damage: Incidence unknown, but estimated to be very low—proportional to delivered energy

Pain score 2–3

■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages
Male:Female	3:1 male predominant but no recent statistics available
Incidence	100,000/yr in the United States
Etiology	Reentry arrhythmia substrates (e.g., AF, AFL, SVT, VT) from hypertensive heart disease, cardiomyopathy (ischemic or nonischemic); idiopathic.
Associated conditions	Systolic or diastolic heart failure, CAD, cardiomyopathy, HTN, valvular heart disease, COPD, obesity, CVA, diabetes, obstructive sleep apnea, obesity

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for cardioversion fall into one of two categories: elective or emergent. The presence or absence of hemodynamic instability will define the category. In the emergency patient, full-stomach precautions may be necessary. Elective cardioversions usually are carried out on patients who have failed drug therapy. In patients with recent onset AF, optimal timing for cardioversion is reported to be 24–48 h after onset.

Respiratory	Preop evaluation of the airway should focus on the need for elective ET intubation (patients with GERD, difficult mask fit, or airway compromise).
Cardiovascular	Relative contraindications to elective cardioversion include ↓ K ⁺ , inadequate anticoagulation, presence of β-blockade, AV block, and rare digitalis toxicity (toxic = >3 ng/mL). The presence of significant CHF, CAD,

	or valvular disease may predispose this patient population to ↓↓ BP in response to anesthetic agents. Consider use of etomidate (0.1–0.2 mg/kg). Patients at ↑ risk for an embolic event include those with Hx of embolization within 2 yr, mitral stenosis, intra-arterial thrombus, CHF, or hyperthyroidism. In these patients, ensure adequate anticoagulation (PT 1.5–2 × baseline, INR 2.0–3.0). It has been suggested that NTG patches near the electrodes be removed prior to cardioversion to avoid risk of explosion. Tests: ECG; TEE (✓ for thrombus and size of atrium); digitalis level (toxic = >3 ng/mL → refractory VF following cardioversion); electrolytes; INR
Endocrine	Hyperthyroidism → AF
Gastrointestinal	Full-stomach precautions may be necessary in the emergency patient.
Neurologic	✓ Hx for TIAs or CVAs → ↑ risk of embolic event. Pre- and postprocedure neurologic exams should be done.
Hematologic	✓ need for anticoagulation (see above).
Laboratory	Other tests as indicated from H&P
Premedication	Usually not needed. For the emergency patient, take full-stomach precautions.

INTRAOPERATIVE

Anesthetic technique

Brief GA with mask oxygenation and ventilation

Induction	Preoxygenate patient. For many patients, etomidate (0.1–0.2 mg/kg iv) is the agent of choice (NB: etomidate-induced clonus → ECG artifact). Propofol (1.0–1.5 mg/kg iv slowly) until loss of lid reflex can be used in hemodynamically stable patients but may be associated with slower recovery. Supplemental analgesia
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	may be provided by fentanyl (e.g., 0.5–1 mcg/kg preinduction) or remifentanyl (1–2 mcg/kg), thereby reducing anesthetic requirements.	
Maintenance	Occasionally necessary to repeat cardioversion. Additional small doses of propofol, etomidate, or remifentanyl may be required.	
Emergence	Patients should awaken rapidly with full recovery of airway reflexes. Outpatients are usually discharged to home within 1–2 h.	
Blood and fluid requirements	No blood loss IV: 20 ga × 1 NS/BSS @ TKO	
Monitoring	Standard monitors	Avoid placement of ECG electrodes in precordial area.
Positioning	Hospital bed, supine	Procedure takes place at patient's bedside.
Complications	Loss of airway VF ↑↑ BP/myocardial ischemia Severe bradycardia Thermal injury ↓ CO	Use airway manipulation ± artificial airways; always be prepared to intubate. Use ACLS protocols. Cardioversion → catecholamine surge → acute MI in susceptible patient population. Rx: atropine (e.g., 0.4 mg iv) Ensure good electrode/skin contact. 2° anesthetic drugs or myocardial stunning from cardioversion. Rx: inotropic support (e.g., ephedrine)

POSTOPERATIVE

Complications	Recall	Especially in hemodynamically fragile patients. Discuss possibility with patient in advance.
	Systemic embolization	Neurologic exam should be repeated postcardioversion.
	↓ BP/↓ CO/CHF new dysrhythmia	Atrial contraction may not be effective following cardioversion → ↓ CO/↓ BP.
	Renal dysfunction	Incidence: 17%. ✓ renal function labs.
Pain management	Minimal	Myalgias not uncommon; consider iv ketorolac.
Tests	ECG	Verify NSR.

Suggested Readings

1. Calkins H, Hindricks G, Cappato R, et al: 2017 HRS/EHRA/ECAS/APHRS/SOLAECE expert consensus statement on catheter and surgical ablation of atrial fibrillation. Heart Rhythm 2017; 14:e276–433.
2. Desai PM, Kane D, Sarkar MS: Cardioversion: what to choose? Etomidate or propofol. Ann Card Anaesth 2015; 18(3):306–11.
3. Hellman T, Kiviniemi T, Nuotio I, et al: Optimal timing for cardioversion in patients with atrial fibrillation. Clin Cardiol 2018; 41(7):966–71.
4. Hellman Y, Cohen MJ, Leibowitz D, et al: The incidence and prognosis of renal dysfunction following cardioversion of atrial fibrillation. Cardiology 2013; 124(3):184–9.
5. Kirkhhoh P, Eckardt L, Loh P, et al: Anterior-posterior versus anterior-lateral electrode positions for external cardioversion of atrial fibrillation: a randomized trial. Lancet 2002; 360:1275–9.
5. Nichol G, Sayre MR, Guerra F, et al: Defibrillation for ventricular fibrillation: a shocking update. J Am Coll Cardiol 2017; 70(12):1496–509.

IMPLANTATION OF CARDIOVERTER-DEFIBRILLATOR (ICD) OR CARDIAC RESYNCHRONIZATION THERAPY DEFIBRILLATOR

(CRT-D)

PROCEDURAL CONSIDERATIONS

Description

The implantable cardioverter-defibrillator (ICD) is an effective device for the prevention of premature death from VT or VF. The results of randomized clinical trials involving survivors of cardiac arrest and those considered at risk for sudden death showed the superiority of ICD therapy over conventional medical therapy in lowering the incidence of sudden death and overall mortality. Hence, ICD therapy has become the gold standard for patients considered at risk for sudden death (primary prevention) or those with a history of spontaneous or inducible sustained VT/VF (secondary prevention). Since its inception in 1983, there have been significant advances in ICD technology. The more efficient biphasic shock waveform allows for a much lower defibrillation threshold and, hence, a lower required energy delivery, which, in turn, allowed for dramatic reduction in the generator's size and the need for epicardial electrodes. Furthermore, the firmware now provides significantly broader spectrum in functionality. Newer devices can incorporate the full capabilities of a permanent pacemaker along with resynchronization therapy (see below), the ability to differentiate VT from SVT, and deliver antitachycardia pacing for VT or shock for VF. Therapy for atrial tachyarrhythmias (atrial tachycardia and flutter) with atrial overdrive pacing is also available in select devices. The need for GA in ICD implantation has been diminishing and deep sedation is required mainly during defibrillation threshold testing (DFT). DFT ([Fig. 13.1-3](#)) may be unnecessary since clinical trials have shown that ICDs rarely fail in defibrillating VF and, furthermore, that unnecessary ICD shocks are associated with negative cardiovascular outcomes, including increased overall mortality. Thus, deep sedation or GA is necessary only for patients with other cardiovascular and pulmonary comorbidities.

A new ICD system using entirely subcutaneous approach for the lead and generator placement is now available ([Fig. 13.1-4](#)). The lead is implanted subcutaneously along the sternum and the generator placed in the lateral chest. The need for DFT with this system is higher than for the standard ICD; hence, deep sedation or GA is required.

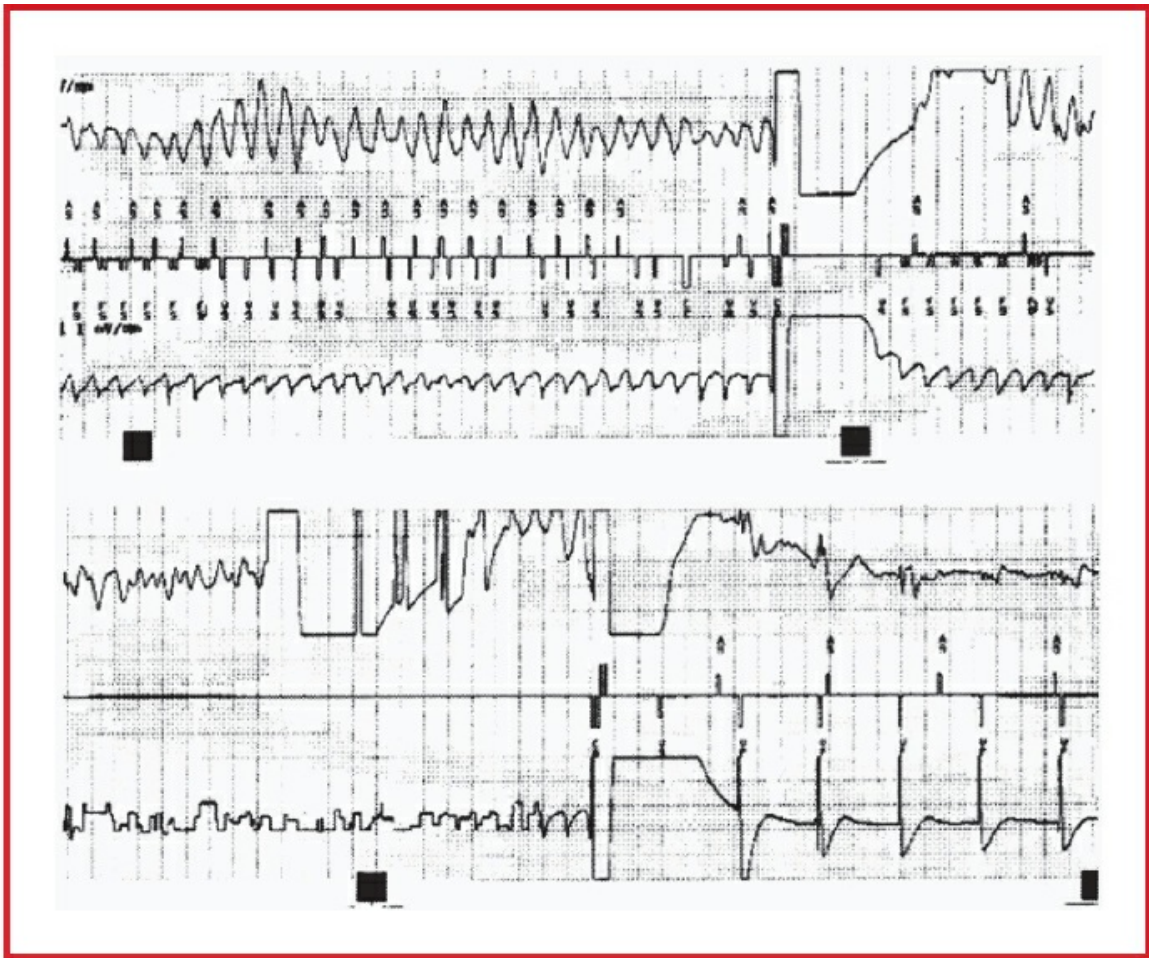


Figure 13.1-3 Induced VF (with underlying AF) is shown on top panel, whereby the first (24-J) shock failed to terminate VF but terminated AF (as indicated by the ICD annotation). The second shock (34-J) terminated VF, resulting in V-paced rhythm with underlying sinus and AV block. Total downtime during VF was 16 sec. Downtime is dependent on the detection time and charge time, which, in turn, depend on the energy programmed.

Cardiac resynchronization therapy (CRT) was developed for symptomatic systolic heart failure (HF) patients with electrical and/or mechanical dyssynchrony, typically an intraventricular conduction delay (e.g., LBBB) causing a mechanical dyssynchrony → systolic inefficiency. Resynchronization of right and left ventricular (LV) depolarization requires implantation of a LV lead into one of the tributaries of the coronary sinus in addition to the other standard pacing leads. The LV lead placement can increase the procedure time significantly, and combined with the more compromised underlying systolic function, monitored deep sedation or GA may be the more appropriate approach for this procedure.

Usual preop diagnosis

Documented, induced, or high-risk VF; moderate-severe heart failure

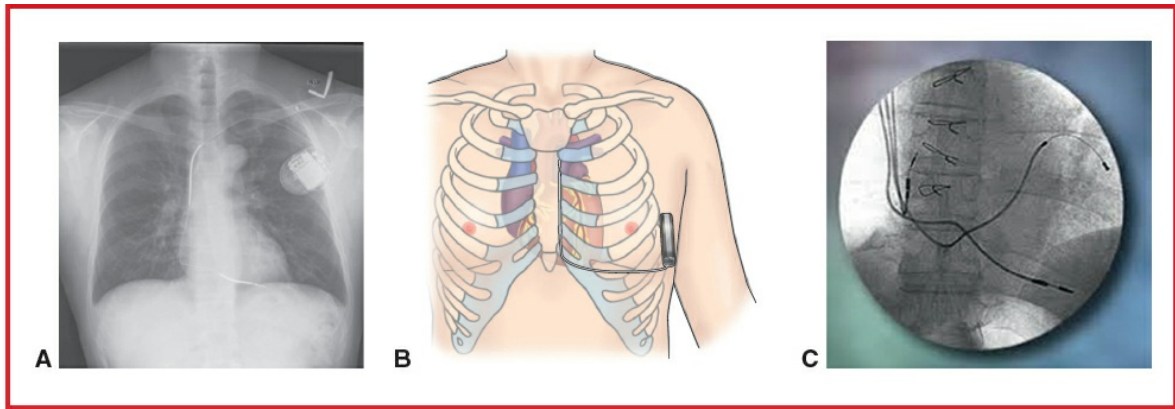


Figure 13.1-4 Shown here is (A) the radiograph of a single-lead ICD with dual-coil system, (B) subcutaneous ICD system (Modified from Hinkle JL, Cheever KH Brunner & Suddarth's Textbook of Surgical Nursing 14th edition. Wolters Kluwer, Philadelphia: 2018, 697.), and (C) the leads in a CRT system.

■ SUMMARY OF PROCEDURE

Position	Supine with defibrillator pads positioned AP or basilar-apical
Incision	Left pectoral
Special instrumentation	ICD pulse generator, lead(s), and testing system (manufacturer specific)
Unique considerations	VT/VF induction with associated ↓ CO, ↓ BP
Antibiotics	Cefazolin 2 g iv
Surgical time	1–2 h (more for ICD with cardiac resynchronization [CRT-D])
Closing considerations	Routine subcutaneous/submuscular pocket closure
EBL	5–20 mL
Postop care	Monitoring of arrhythmia, pocket bleeding/hematoma, pneumothorax
Mortality	0.1% (transvenous); ≤5% (thoracotomy)
Morbidity	New-onset arrhythmias: ≤5% Pneumothorax: ≤5% (transvenous) Pericardial effusion/tamponade: ≤1% (transvenous)

Pain score 3 (6–9, thoracotomy)

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any
Male:Female	Male preponderance
Incidence	>50,000/yr in the United States
Etiology	Reentry substrates from remote myocardial fibrosis, channelopathies (e.g., Long QT and Brugada syndromes)
Associated conditions	LV dysfunction; CAD; cardiomyopathy; HTN; valvular disease

Suggested Readings

1. Al-Khatib SM, Stevenson WG, Ackerman MJ, et al: 2017 AHA/ACC/HRS Guidelines for management of patients with ventricular arrhythmia and the prevention of sudden cardiac death. *Heart Rhythm* 2018;138(13):e210–71.
2. Boersma L, Barr C, Knops R, et al: Implant and midterm outcomes of the subcutaneous implantable cardioverter-defibrillator registry. The EFFORTLESS study. *J Am Coll Cardiol* 2017; 70:830–41.
3. Healey JS, Hohnloser SH, Glikson M, et al: Cardioverter defibrillator implantation without induction of ventricular fibrillation: a single-blind, non-inferiority, randomized controlled trial (SIMPLE). *Lancet* 2015; 385:785–91.

EXTRACTION OF PACEMAKER AND IMPLANTABLE CARDIOVERTER-DEFIBRILLATOR (ICD) LEADS

■ PROCEDURAL CONSIDERATIONS

Almost all PM and ICD systems are implanted utilizing endovascular lead(s). These leads have excellent longevity, typically beyond 25 yr, and hence in the case of a PM implantation for sick sinus syndrome (SSS), most patients will not need a lead “replacement” procedure. However, in younger patients (e.g., PM implantation in the pediatric population and ICD implantation in the young athlete), “replacement” may become necessary. Unfortunately, some leads may undergo premature degradation due to a manufacturing deficiency. In the majority of lead “replacement” (now more

commonly called “revision”) procedures, the simple addition of a new lead will suffice, but in some cases where multiple leads are already present, a “debulking” procedure is necessary to avoid vascular congestion or unwanted lead-to-lead interaction. Hence, removal of some, or all, of those leads would be necessary. In some cases (e.g., refractory sepsis or bacteremia), lead removal is considered mandatory. There are other rare scenarios where lead removal is at least strongly recommended. Examples include progressive tricuspid valve regurgitation that is suspected to be caused by the presence of multiple RV leads and where the presence of multiple leads has caused SVC syndrome.

Leads that have been implanted for longer than 1 yr are likely to have adhered to the adjacent vascular wall or endocardial surface. Adhesion sites are typically located in the subclavian vein, the junction of the innominate vein and SVC, along the SVC, the tricuspid valve, and the RV tissue (Fig. 13.1-5). Therefore, removal of chronic leads typically requires fibrotic tissue cutting tools in the form of telescoping sheaths. The forward advancement of the sheaths is done either by manual force or with the aid of thermal tissue ablation (using LASER or radiofrequency [RF] heating). The areas at the junction of the innominate vein and the superior vena cava and along the superior vena cava are of concern in terms of risks for vascular tear because the sheath may not conform to the lead angulation. In utilizing any of these methods, vascular injury is inevitable and may result in transmural tear with rapid exsanguination. Thus, lead extraction procedures should be performed with complete preparation for immediate **emergency sternotomy** with a surgical team on active standby. For these reasons, GETA is a necessity. In addition, the patient should be prepped and draped for possible sternotomy and surgical instruments, and a CPB machine should be in the room with the perfusionist team nearby. Blood products should also be immediately available. Vascular access should be large bore and a central line is advisable. Continuous blood pressure monitoring via an arterial line is mandatory and continuous TEE or standby TTE should be used. The surgical approach of the procedure is similar to a standard implantation or pulse generator change. The patient will have continuous ECG monitoring and external defibrillator/PM pads applied. The pulse generator is removed from the pocket as usual and the leads are released from encapsulation. For the patient who is PM dependent, a temporary PM is mandatory. The temporary PM lead should be placed through a venous entry other than the one being used for the extraction. If temporary pacing support is expected to be necessary for several days or weeks (e.g., in the case of an infected system when reimplantation is delayed), it has become customary to utilize a permanent PM lead with an active fixation, as this will provide more reliable backup pacing. The lead is connected to a permanent PM generator securely taped against the skin, thus improving patient ambulation. The right IJ venous entry is the

best route for placement of this “temporary-permanent” pacing system. In the event of incomplete lead removal, retrieval of the remnant is best done using a femoral approach. Several retrieving tools, such as snares and baskets (Fig. 13.1-6), are available in the market.

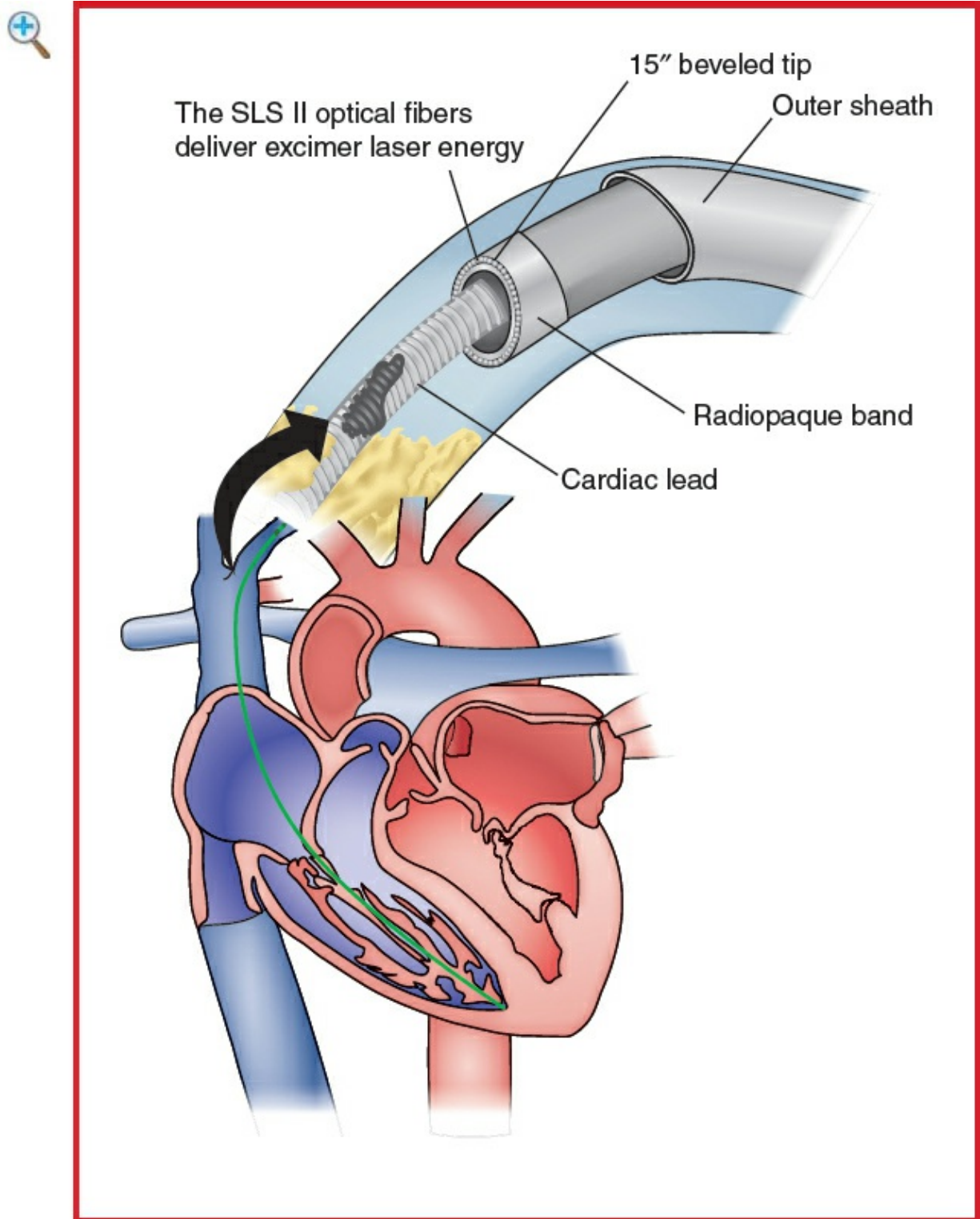


Figure 13.1-5 The laser sheath is placed over the lead to be extracted and advanced into the heart. The laser is then used to ablate scar tissue surrounding the lead to facilitate extraction.

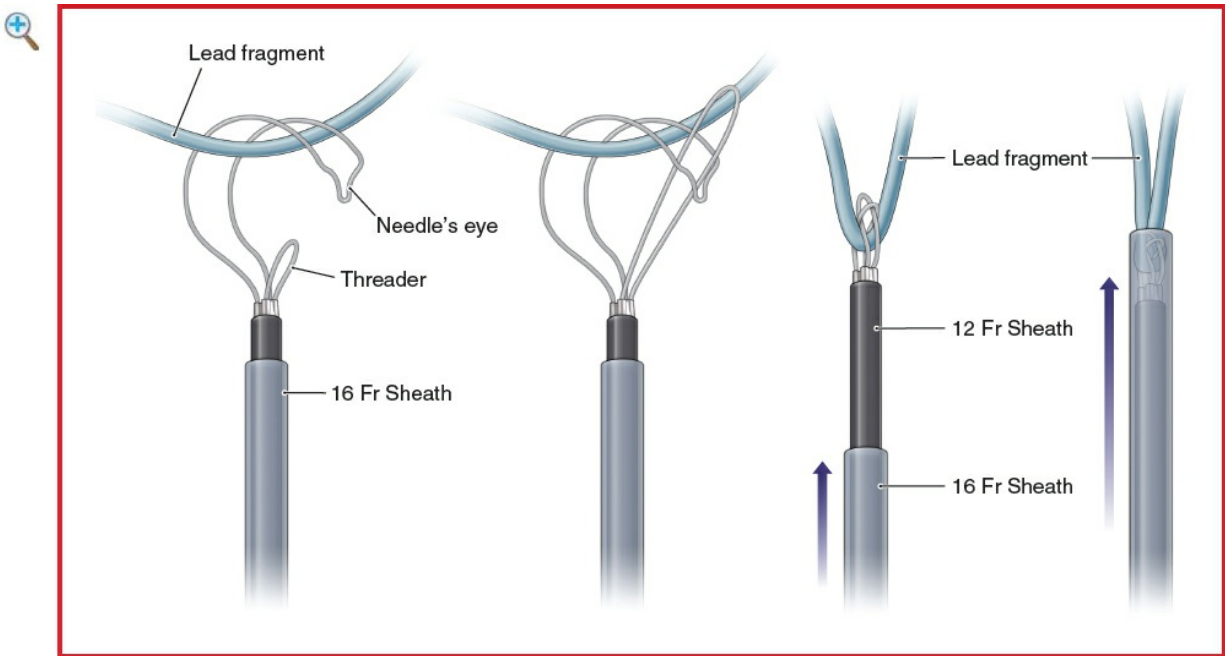


Figure 13.1-6 An example of a lead retrieval tool in use to capture and remove a lead fragment.

Suggested Readings

1. Bongiorno MG, Segreti L, Di Cori A, et al: Overcoming the current issue surrounding device leads: reducing the complications during extraction. *Expert Rev Med Devices* 2017; 14(6):469–80.
2. Kratz JM, Toole JM: Pacemaker and internal cardioverter defibrillator lead extraction: a safe and effective surgical approach. *Ann Thorac Surg* 2010; 90(5):1411–7.
3. Kusumoto FM, Schoenfeld MH, Wilkoff BL et al: 2017 HRS expert consensus statement on cardiovascular implantable electronic device lead management and extraction. *Heart Rhythm* 2017; 14:e503–51.
4. Sonny A, Wakefield BJ, Sale S, et al: Transvenous lead extraction: a clinical commentary for anesthesiologists. *J Cardiothorac Vasc Anesth* 2018; 32(3):1101–11.
5. Wilkoff BL, Byrd CL, Love CJ, et al: Pacemaker lead extraction with the laser sheath: results of the Pacing Lead Extraction with Excimer Sheath (PLEXES) Trial. *J Am Coll Cardiol* 1999; 33:1671–6.
5. Wilkoff BL, Love CJ, Byrd CL, et al: Transvenous lead extraction: Heart Rhythm Society expert consensus on facilities, training, indications, and patient management. *Heart Rhythm* 2009; 6:1085–104.

CATHETER ABLATION OF ATRIAL FIBRILLATION

PROCEDURAL CONSIDERATIONS

Description

Catheter ablation involves placement of catheters onto the desired cardiac chambers via venous and/or arterial access (most commonly the femoral blood vessels) and more recently via pericardial space. Through the electrodes at the tip of the catheters, intracardiac (or epicardial in the case of pericardial approach) signals can be recorded, arrhythmias can be monitored and/or induced using specific pacing protocols, and small areas of cardiac tissue can be ablated using various modalities. The most common modality is radiofrequency (RF) energy. Other emerging ablation techniques include cooling with cryoenergy and heating using ultrasonic, LASER, and microwave modalities. For atrial fibrillation (AF), the ablation is aimed at the triggers and substrates. The triggers can be found at various sites, most commonly at the entries of the pulmonary veins, superior vena cava, and coronary sinus into the left and right atria. The substrates for AF are believed to be in various areas on the left atrial wall, usually the posterior wall ([Fig. 13.1-7](#)).

Unlike catheter ablation procedures for other arrhythmias with a more focused area of interest, AF ablation requires energy delivery at a significantly greater number of sites. In most cases, the mapping and ablation are guided by a three-dimensional tracking system. The procedure also requires access to the left atrium, typically via a transseptal catheterization, which is usually guided by intracardiac echocardiogram. In the typical scenarios, two transseptal sheaths are placed in the left atrium to facilitate recording, pacing, and application of ablation. Typically, pulmonary vein isolation (PVI) is the basic approach. Left appendage isolation is also being performed in some centers. If elimination of left AFL substrate is a target, a significant amount of additional ablations are necessary. Typically, a “roof” line is created as well as a line connecting the left lower pulmonary vein to the mitral valve annulus. More recently, single-shot ablation devices to isolate pulmonary veins (e.g., cryoballoon) have become available ([Fig. 13.1-8](#)). These devices allow a single application of energy to produce a circumferential isolating lesion.

The standard procedure typically requires several (2–4) hours to complete. Cryoballoon procedures can be significantly shorter. The patient is continuously anticoagulated with iv heparin. Many centers are now performing the procedure with the patient already fully anticoagulated prior to the start of the procedure. One or more cardioversions may also be necessary, especially in patients with persistent AF. Furthermore, many centers are now using high-dose isoproterenol and adenosine after the ablation to assess likelihood of recurrence.

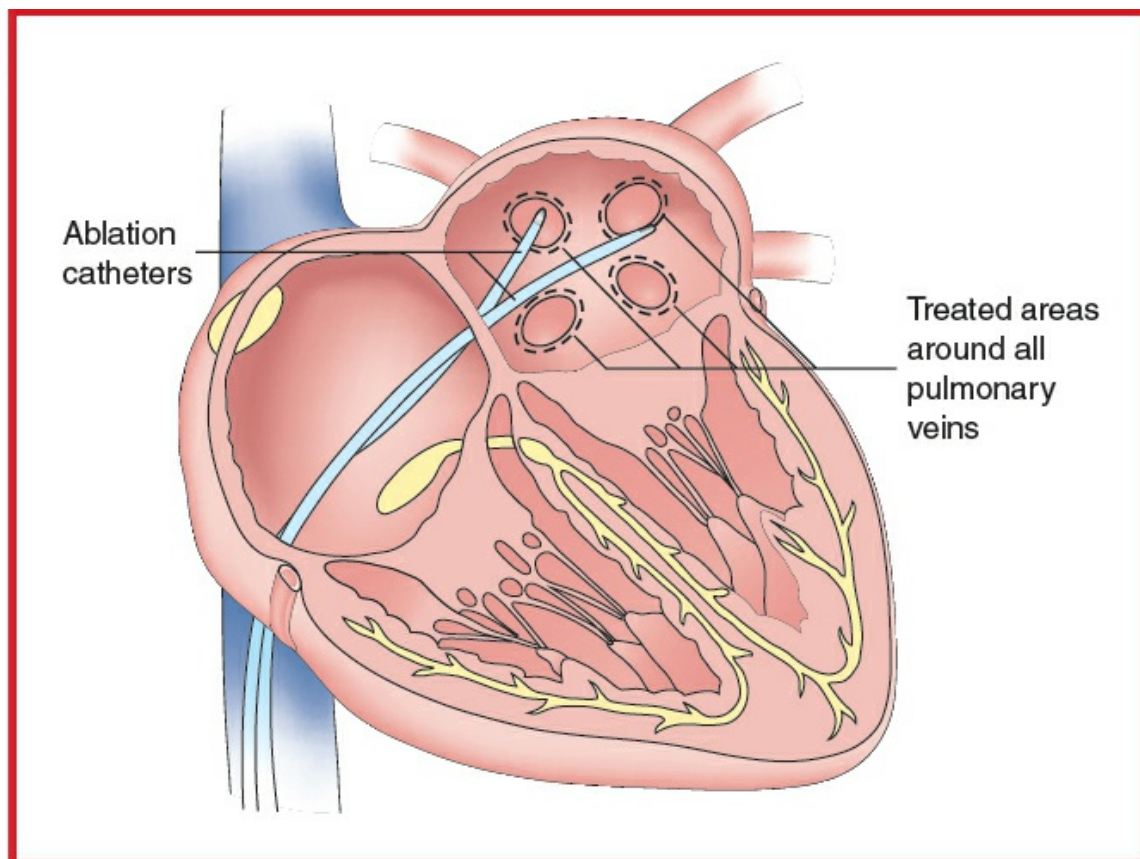


Figure 13.1-7 A transvenous transseptor catheter electrode is used to ablate target areas around the pulmonary vein entry sites to treat atrial fibrillation. Other common target sites include atrial walls as well as SVC and coronary sinus inlets.

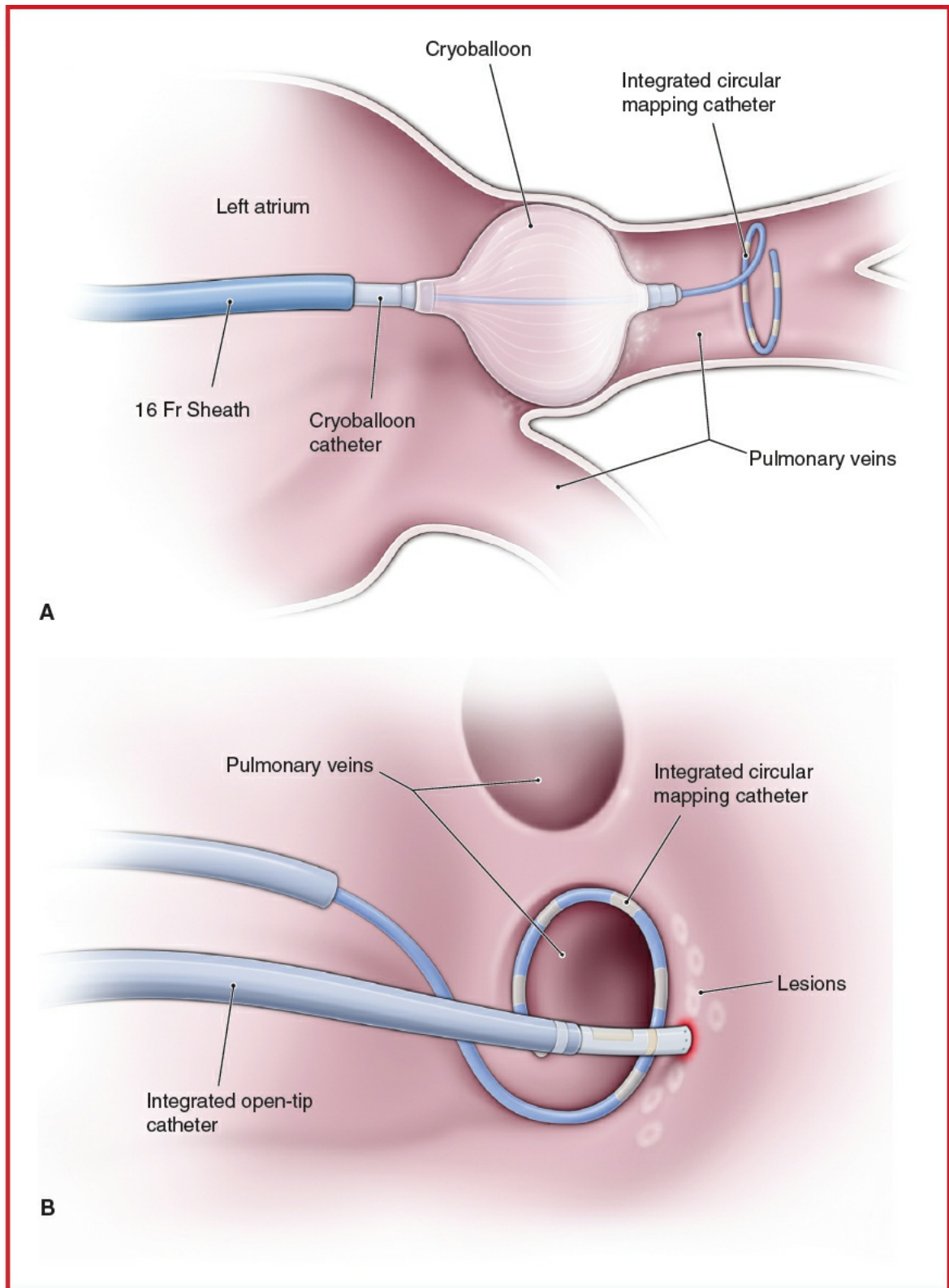


Figure 13.1-8 A: Cryoballoon ablation. **B:** Catheter tip RF ablation. Shown on panel(A) is localization mapping of an RVOT VT by mapping a single PVC with the same morphology of the VT and on panel (B) the cluster of ablation points.

GETA has been the standard modality for providing comfort and safety for patients. In

addition to the length of the procedure, there is a higher risk for cardiac and vascular perforation compared to a simple ablation procedure. The significant ablation volume is associated with an increased chance of left atrial perforation. A misdirected transseptal catheterization in itself can cause perforation of the aorta, posterior wall of the left atrium, and other structures. An emergency thoracotomy may be necessary. In addition to safety, GETA provides a stationary patient, which is critical in the acquisition and maintenance of three-dimensional map accuracy. Transesophageal temperature monitoring has become a standard of care and would be poorly tolerated in unanesthetized patients.

When feasible, AF ablation procedures should be performed in a hybrid procedure room such that, if needed, emergency thoracotomy can be performed with minimal delay. Under expert operators, the need for emergency thoracotomy is low (under 1–2%), and hence, surgical “standby” is not usually necessary, but this precautionary step is advisable for centers with low patient volumes.

Suggested Readings

1. Briceno DF, Madan N, Villablanca PA, et al: Periprocedural anticoagulation for catheter ablation of atrial fibrillation: practical implications for perioperative management. *J Cardiothorac Vasc Anesth* 2017; 31(4):1519–26.
2. Chen S, Schmidt B, Bordignon S, et al: Atrial fibrillation ablation using cryoballoon technology: recent advances and practical techniques. *J Cardiovasc Electrophysiol* 2018; 29(6):932–43.
3. Kuck KH, Brugada J, Fürnkranz A, et al: Cryoballoon or radiofrequency ablation for paroxysmal atrial fibrillation. *N Engl J Med* 2016; 374:2235–45.

CATHETER ABLATION OF VENTRICULAR TACHYCARDIA

PROCEDURAL CONSIDERATIONS

Description

The pathophysiology of VT is diverse. Catheter ablation of VT encompasses a wide spectrum of procedural complexity, and planning must incorporate the patient’s underlying cardiovascular status. The most “benign” forms of VT have a focal origin, most commonly in the right ventricular outflow tract (RVOT). The mechanism is usually automatic or triggered activity and hence tends to be self-terminating. Other sites with a similar mechanism are in the LV (e.g., LVOT) and the left fascicles. These patients

typically have preserved LV systolic function. The most “malignant” forms of VT are associated with underlying structural myocardial damage (e.g., prior MI) and nonischemic (“idiopathic”) cardiomyopathy. A less common form of cardiomyopathy is arrhythmogenic right ventricular dysplasia/cardiomyopathy (ARVD/C). VT mechanism is reentry, and therefore, it tends to be sustained and results in hemodynamic instability. Other (less common) VTs are those from sarcoidosis and Chagas disease. In patients with congenital disorders, VT can be found in many scenarios, such as noncompaction cardiomyopathy and postop tetralogy of Fallot (ToF).

Unlike the approach to ablation for AF (which is largely anatomical), ablation for VT is similar to that of catheter ablation for SVT. Induction of “clinical” VT is necessary to confirm the site that is critical to the initiation and/or maintenance of the tachycardia. The incorporation of 3D mapping has been a very useful advancement in this quest, but nonetheless, VT typically needs to be induced multiple times during the procedure. This is usually of minimal hemodynamic consequence in the case of the more benign VTs, but in patients with compromised systolic function, induction of sustained VT causes significant hemodynamic sequelae; hence, inotropic support is frequently necessary, such as iv phenylephrine or occasionally a transvenous LVAD device. Repeated cardioversions are frequently involved when overdrive pacing fails, which may cause more systolic dysfunction. While these patients typically already have an implanted ICD, external cardioversion is preferred for better timing. VT induction is again repeated after the ablation has been completed.

Once mapping of the critical focus has been completed, ablation is carried out. The majority of ablation is done via an endocardial approach with/without transseptal access. Some VTs (Chagas and 40% of nonischemic cardiomyopathy) have a critical site on the epicardial surface and therefore require pericardial entry (typically using a subxiphoid approach). For VTs with localized origin (RVOT, LVOT, FT), a small clusters of ablations will usually suffice ([Fig. 13.1-9](#)), keeping in mind that some outflow tract origins are located at one of the aortic valve cusps. For these procedures, MAC anesthesia would suffice and also is frequently preferred to allow for “lightening up the sedation,” which is frequently necessary to promote induction of these types of VT. For re-entry VT caused by ischemic and nonischemic cardiomyopathy, the critical area is usually an “isthmus” of surviving myocardium in between scars ([Fig. 13.1-10](#)). In many cases, a very broad area of “diseased” myocardium (marked by a large area of low voltage) is the cause of VT, especially polymorphic VT; thus, a broad area of ventricular myocardium will be ablated (known as homogenization technique, [Fig. 13.1-11](#)). Procedural duration varies significantly, and it is not unusual to encounter long procedure times. Clearly, GETA would be required. Postop care for these patients can

also be challenging as prolonged hemodynamic compromise as well as residual VT episodes can require delayed extubation.

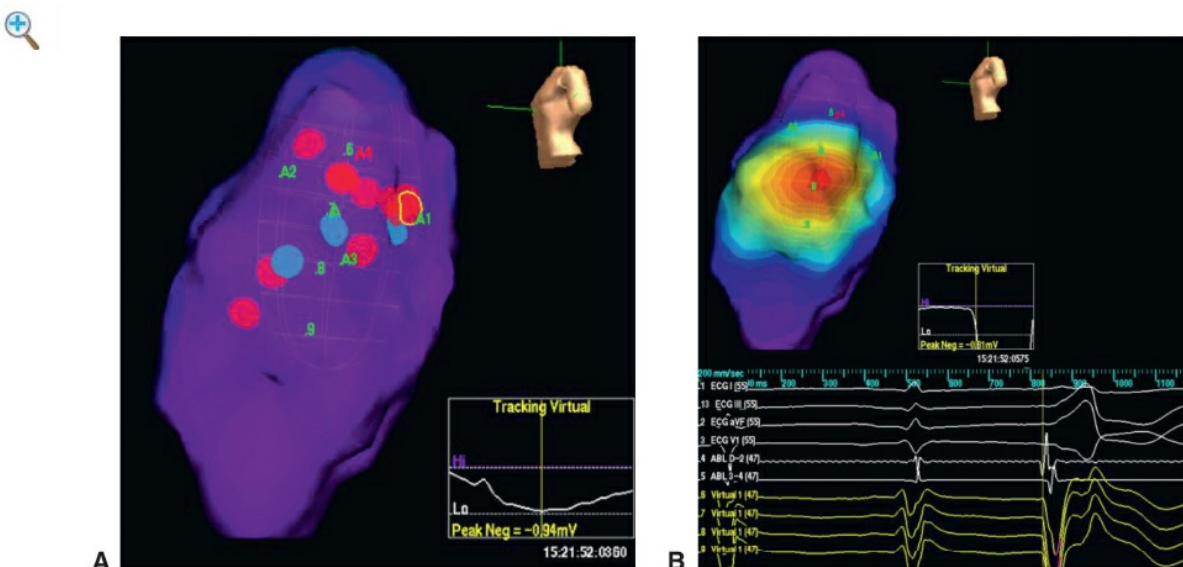


Figure 13.1-9 Shown on panel (A) is localization mapping of an RVOT VT by mapping a single PVC with the same morphology of the VT and on panel (B) the cluster of ablation points.

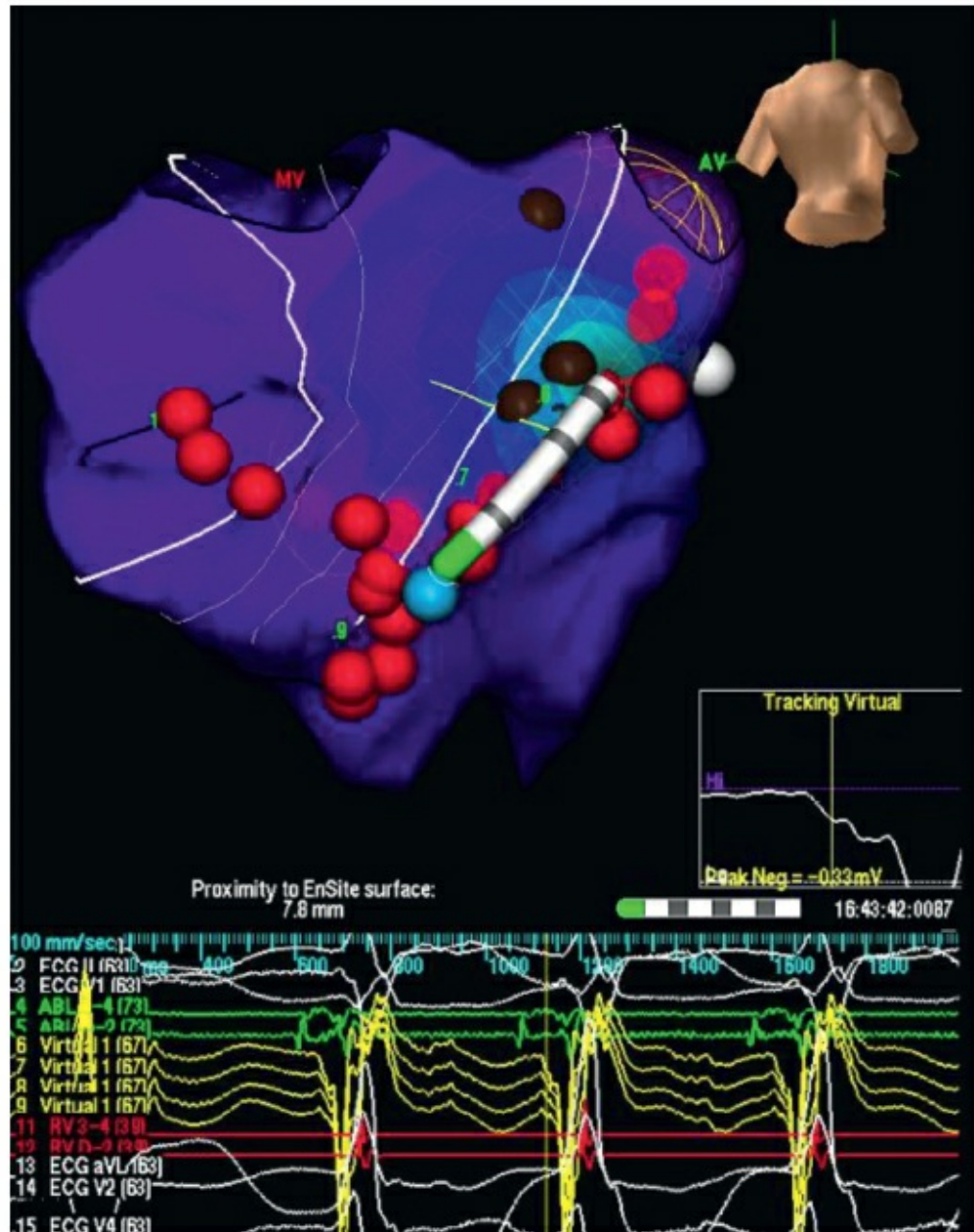


Figure 13.1-10 In the top panel, an “isthmus” ablation was performed based on sites of early activation. In the electrogram (EGM) series (lower panel), the ablation catheter recorded (ingreen) an early sharp depolarization preceding the onset of virtual unipolar signals collected by the multipolar noncontact recording catheter (in yellow).

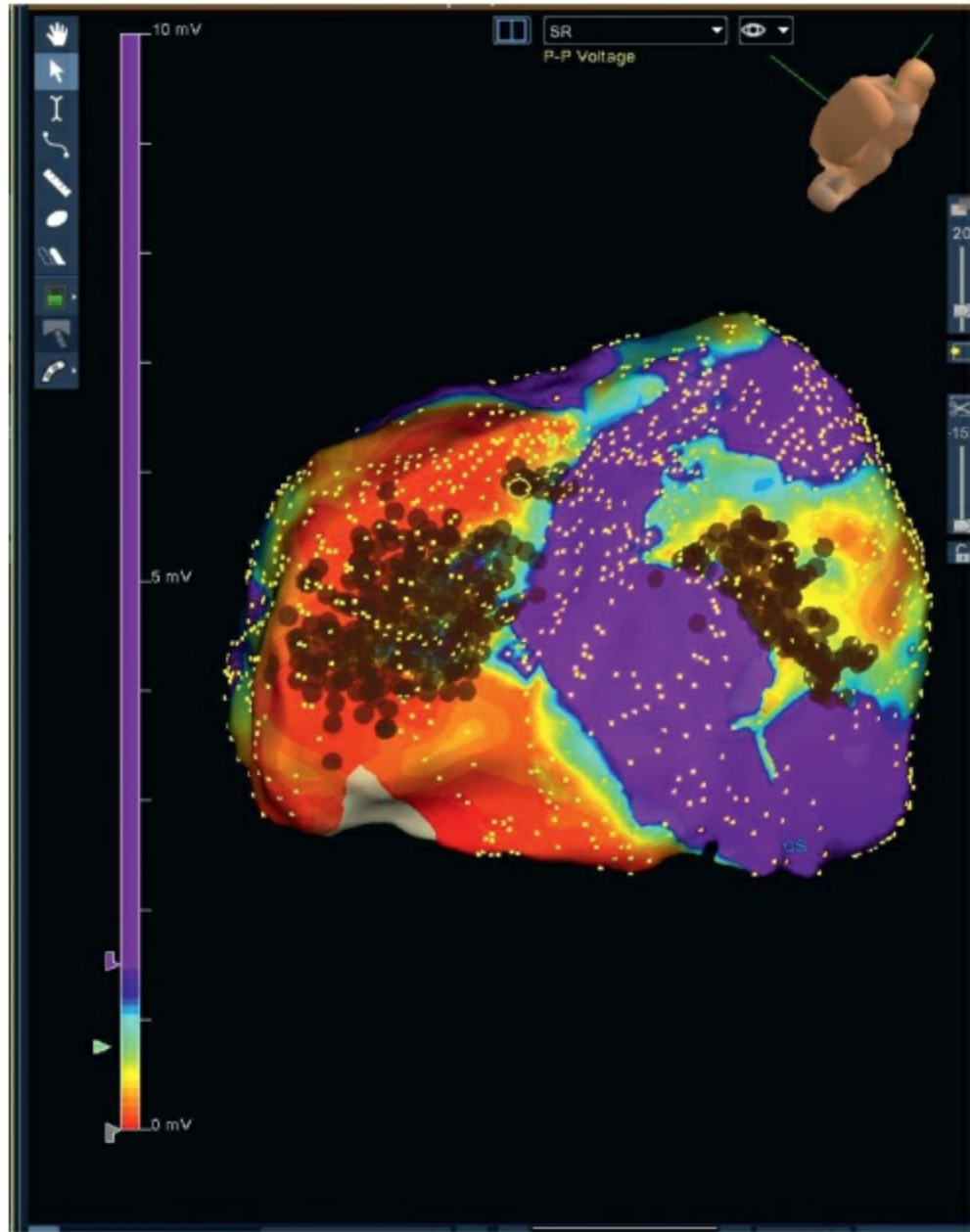


Figure 13.1-11 Shown here is a broad epicardial ablation (brown dots) of areas with low voltage (denoted by colors that are below the set threshold as marked by the arrow on the voltage scale); an example of “homogenization” ablation. The yellow points marked voltage recording points.

Suggested Readings

1. Al-Khatib SM, Stevenson WG, Ackerman MJ, et al: 2017 AHA/ACC/HRS Guideline for management of patients with ventricular arrhythmia and the prevention of sudden cardiac death. *Heart Rhythm* 2018; 15(10):e73–189.
2. Della Bella P, Brugada J, Zeppenfeld K, et al: Epicardial ablation for ventricular tachycardia. A European multicenter study. *Circ Arrhythm Electrophysiol* 2011;

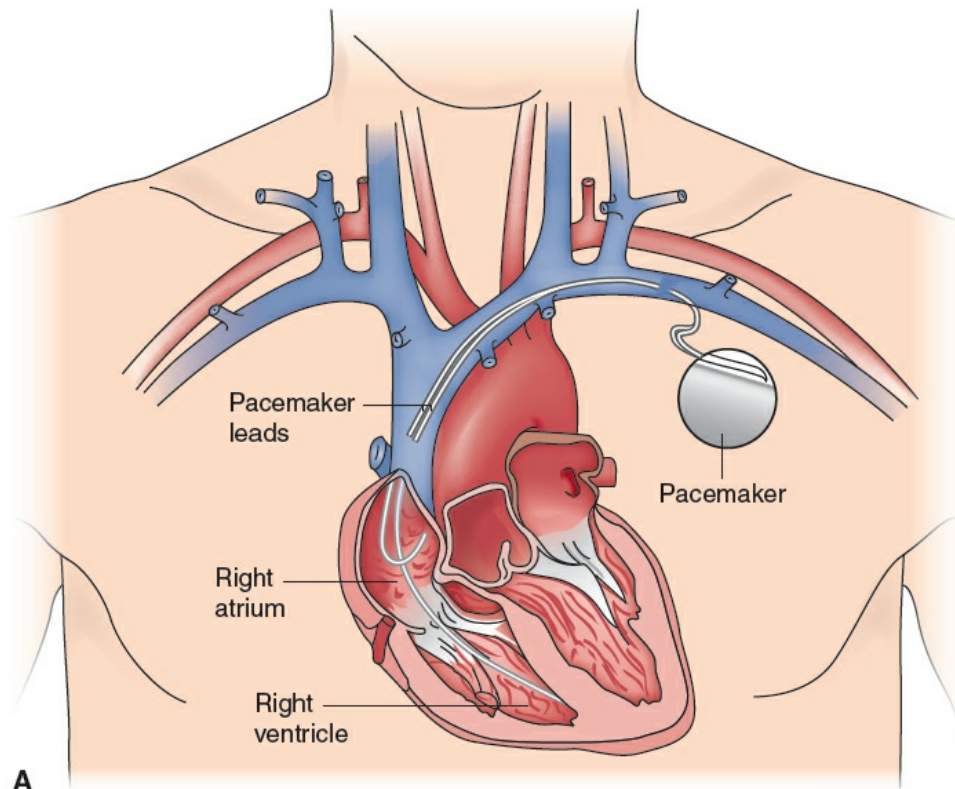
4:653–9.

3. Josephson ME, Anter E: Substrate mapping for ventricular tachycardia. Assumptions and misconceptions. J Am Coll Cardiol Clin Electrophysiol 2015; 1:341–52.

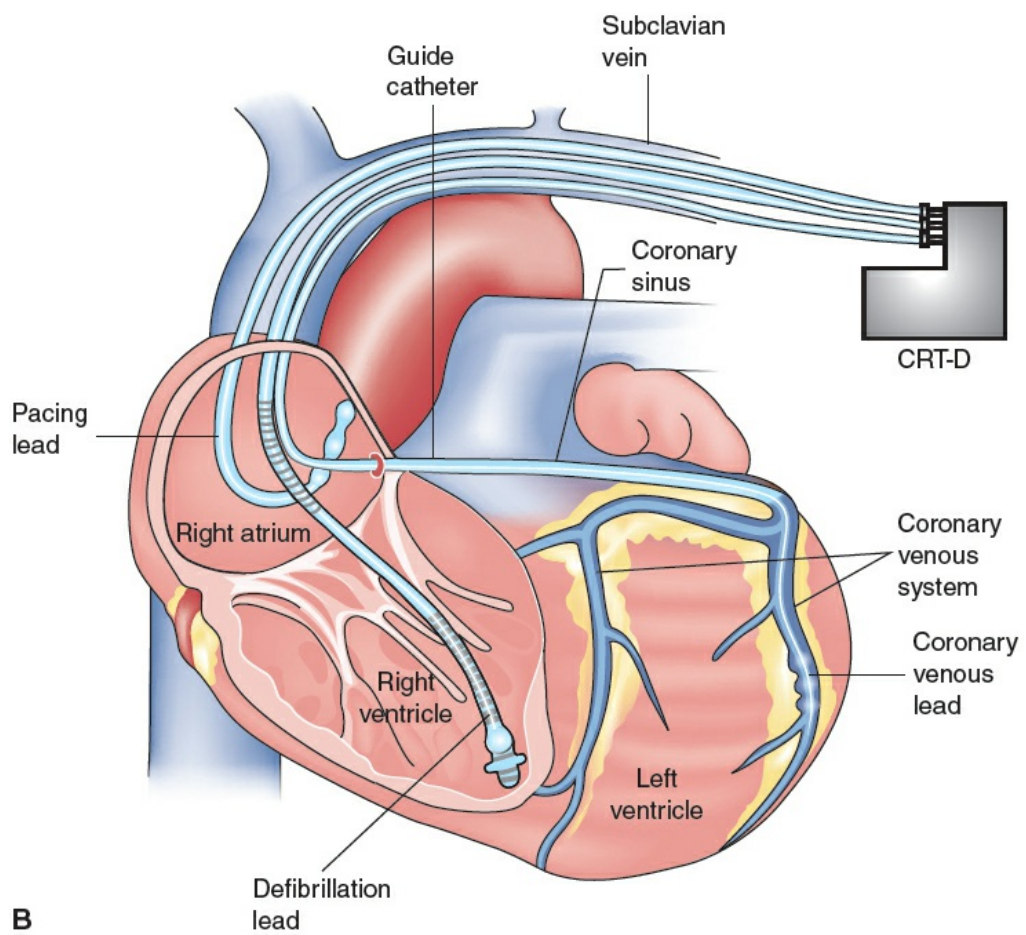
ANESTHETIC CONSIDERATIONS FOR ICD, CRT-D, AND PACEMAKER PLACEMENT; LEAD EXTRACTION; CATHETER ABLATION

PREOPERATIVE

Patients presenting for **ICD/CRT-D placement** may be divided into three populations, based on symptomatology, associated pathology, and probable outcome: (a) **Supraventricular dysrhythmias** (e.g., Wolff-Parkinson-White [WPW] syndrome): usually young and otherwise healthy patients. May be associated with Ebstein anomaly (tricuspid valve defect → RV failure), mitral valve disease, or CAD. Low periop mortality (1%). (b) **Ventricular dysrhythmias**: usually older patients with significant ventricular dysfunction (EF = 10–35%) and other pathologies, such as CAD or cardiac failure. These patients have either survived an episode of VT/VF or are otherwise a risk for sudden death. (c) **Congestive heart failure (CHF)** without significant dysrhythmias, usually 2° dilated cardiomyopathy. These patients have very low EFs (10–25%), LBBB, and Hx of failed conventional CHF treatment. The CRT-D device is implanted in these patients to provide paced “resynchronization” of the contractions of the LV and RV. This requires a third pacer lead placed into the coronary sinus to pace the LV independent of the RV (see [Fig. 13.1-12B](#)). In this case, the device is not placed primarily for its defibrillator function, but for its ability to pace both ventricles synchronously (biventricular pacing). Patients presenting for **permanent pacemaker insertion** may have a variety of dysrhythmias, including SSS, heart blocks (2nd and 3rd degree), and tachycardias refractory to medication. CRT patients may be quite frail with significant comorbidities.



A



B

Figure 13.1-12 Placement of cardiac leads.

Patients presenting for **lead extraction** typically have infected leads (often methicillin-resistant *Staphylococcus*) or, less commonly, malfunctioning leads.

Patients presenting for **catheter ablation** usually have paroxysmal or chronic refractory AF or episodic polymorphic VT or VF related to structural heart disease. They may be at significant risk for death (VT/VF) or thromboembolic events including stroke (AF). AF patients may be receiving a variety of medications, including beta-blockers (e.g., carvedilol), calcium channel blockers (e.g., diltiazem) (less often digitalis or amiodarone), aspirin, clopidogrel, warfarin, direct thrombin inhibitors (e.g., dabigatran), or direct Factor Xa inhibitors (e.g., rivaroxaban, apixaban). VT/VF patients will often have a repeatedly activated ICD in place and be receiving amiodarone or sotalol.

Respiratory	May have associated pulmonary disease 2° smoking Tests: CXR; consider PFT, as indicated from H&P
Cardiovascular	ICD: ICD patients are left on full medication because it usually is impossible to wean them from antidysrhythmic medications before surgery. In most studies, patients have a mean LV EF of 35% and a New York Heart Association functional class of II or III. Supraventricular dysrhythmias: Look for precipitating factors in the dysrhythmia and any methods that have been used to terminate the dysrhythmia. Usually, drug therapy is terminated before surgery to make a dysrhythmia inducible. Note type of drugs used to terminate a dysrhythmia. Ventricular dysrhythmias: Ask about any methods that have been used to terminate dysrhythmia. Look for associated conditions, including CAD, CHF, cardiomyopathy, LV aneurysm, HTN, mitral insufficiency, and diabetes. Generally, these patients have poor LV function with ↑ sensitivity to myocardial depressants. It is important to note that they may be on combinations of antidysrhythmics. Many of these drugs have

significant negative inotropic effects. Of special note is amiodarone, which has been associated with intractable bradydysrhythmias, refractory vasodilation, and difficulty in weaning from CPB.

Pacemaker: The anesthesiologist should be aware that there is an NASPE code (North American Society of Pacing and Electrophysiology) that describes pacemaker function with a three- to five-letter code (see table below).

Position I Pacing Chamber(s)	Position II Sensing Chamber(s)	Position III Response(s) to Sensing	Position IV Programmability	Position V Multisite Pacing
O = None	O = None	O = None	O = None	O = None
A = Atrium	A = Atrium	I = Inhibited	R = Rate Modulation	A = Atrium
V = Ventricle	V = Ventricle	T = Triggered		V = Ventricle
D = Dual (A + V)	D = Dual (A + V)	D = Dual (T + I)		D = Dual (A + V)

For example, a VVI notation indicates that the ventricle is paced and sensed and there is inhibition by native beats.

Catheter ablation: Most commonly, presenting patients have medically refractory or paroxysmal atrial fibrillation or VT/VF and are at increased risk for thromboembolic events or sudden death. Preop antidysrhythmic and anticoagulant medications should be managed in consultation with the prescribing physician and the interventional cardiologist.

Infected leads: Patients with infected leads may have endocarditis, tricuspid valve dysfunction, and sepsis.

Tests: ECG: ✓ dysrhythmia, ischemia, electrophysiologic report. Echo: ✓ ventricular function, wall motion abnormalities, valvular problems. Cardiac angiography: ✓ ventricular function, CAD, valvular disease, LV aneurysm. ICD: identify and interrogate device.

Renal	Patients with poor ventricular function may have associated renal compromise. Electrolyte abnormalities (K ⁺ , Mg ⁺⁺) may be associated with ↑ cardiac irritability and should be corrected preop.
Hematologic	Tests: BUN; Cr; electrolytes Hb/Hct; coag tests (PT, PTT, Plt)
Laboratory	Other tests as indicated from H&P
Premedication	For adults: midazolam 0.5–2.0 mg iv, with careful observation and supplemental O ₂ .

INTRAOPERATIVE

Anesthetic technique

Local anesthesia with sedation, GA with LMA, or GETA, as indicated. Catheter ablation and placement of most ICDs, CRT-Ds, and PMs are done in the cardiac catheterization suite. ICDs are very small and are implanted in the same position as a PM. If patients are orthopneic due to their CHF, or if the procedure is projected to be long, GA is often preferable. Also, elderly patients may become disoriented with sedation and, thus, may require urgent GA. Lead extraction is often done under GETA.

Induction	Exact type of induction depends on the patient's medical condition. Sedation can be provided with small doses of midazolam (1–2 mg) ± fentanyl (25–50 mcg) titrated to effect. An alternative technique is to use a propofol infusion (e.g., 25–75 mcg/kg/min). Because local anesthesia is provided by the surgeon, the procedure usually is not painful and does not require postop pain control. It is important to avoid oversedating these patients because they will tend to become disoriented and uncooperative. Even if local anesthesia with sedation is provided for the placement of leads and the device, a brief period of GA is usually required for device testing. This can be provided easily with mask ventilation and induction with propofol (e.g., 1 mg/kg iv) or etomidate (e.g., 0.1 mg/kg iv), similar to anesthesia for cardioversion procedures. For those patients requiring GA for the entire procedure, induction with propofol (1–2 mg/kg) or etomidate (0.1–0.3 mg/kg) is often used.
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	Muscle relaxants are not required unless intubation is planned. Narcotics usually are not required because local anesthesia is used.	
Maintenance	As previously discussed, verification of correct lead placement usually involves the induction of VF or VT and the testing of the device's capability to restore NSR. External defibrillation should be available at all times, as should antidysrhythmics (e.g., lidocaine and amiodarone). While the device is tested, the patient should be breathing 100% O ₂ . Multiple testing cycles can result in depressed LV function, and inotropes may be needed.	
Emergence	These patients are extubated (if GETA used). Recovery is in a PACU. If, however, multiple test shocks are needed or the heart displays evidence of injury (need for inotropes, ST-segment abnormalities), then extubation may need to be deferred to ICU.	
Blood and fluid requirements	IV: 16–18 ga × 1 BSS @ 2–3 mL/kg/h	Care should be taken to minimize iv fluids in CHF patients.
Monitoring	Standard monitors ± Arterial line ± CVP/TEE External defibrillation Temperature	A CVP catheter or TEE may be used depending on LV function. ICD patients generally require only an arterial line. Because antidysrhythmics may affect the testing procedure, they should be avoided when possible. Defibrillation or cardioversion is treatment of choice. Normothermia should be maintained.
Positioning	Supine	

	✓ and pad pressure points	
	✓ eyes	
Complications	Pneumohemothorax Pericardial effusion/tamponade HTN/CHF Coronary sinus rupture/central vein laceration	Rarely, cardiac rupture or central vein laceration may occur during lead extraction. Significant blood loss may occur. Coronary sinus rupture has been reported during biventricular lead placement and may require open repair.

▼ POSTOPERATIVE

Complications	Recurrent dysrhythmias Hemorrhage Myocardial ischemia	Pacemaker lead thrombosis → PE has been reported.
Pain management	Usually managed with oral analgesics	
Tests	CXR ECG Electrophysiologic testing Electrolytes	✓ line/lead placement, r/o pneumohemothorax. ✓ for ischemia, dysrhythmias.

Suggested Readings

1. Dukkipati SR, et al: Catheter ablation of ventricular tachycardia: indications strategies, and outcomes—Part I and II. J Am Coll Cardiol 2017; 70(23):2909–41.
2. Lee G, Sanders P, Kalman JM: Catheter ablation of atrial arrhythmias: state of the art. Lancet 2012; 380(9852):1509–19.
3. Miller JD, Yousuf O, Berger RD: The implantable cardioverter-defibrillator: an update. Trends Cardiovasc Med 2015; 25(7):606–11.

4. Sonny A, Wakefield BJ, Sale S, et al: Transvenous lead extraction: a clinical commentary for anesthesiologists. J Cardiothorac Vasc Anesth 2018; 32(3):1101–11.
5. Stahlberg M, Lung LH, Zabarovskaja S, et al: Cardiac resynchronization therapy: a breakthrough in heart failure management. J Intern Med 2012; 272(4):330–43.
6. Upadhyay GA, Mela T, Singh JP: The burden of inappropriate shocks in young people and how to avoid them. Heart 2012; 98(19):1457–66.

TRANSJUGULAR INTRAHEPATIC PORTOSYSTEMIC SHUNT (TIPS)

PROCEDURAL CONSIDERATIONS

Description

Hepatic cirrhosis is a progressive disease that eventually results in portal HTN and the development of varices at a variety of sites. Bleeding from esophageal varices is a serious complication of portal HTN, occurring in 25% of patients within 1 yr of diagnosis (~50% mortality). Prior to 1989, surgically placed shunts were used to direct high-pressure portal blood into the systemic venous circulation in patients with recurrent bleeding after endoscopic sclerotherapy or banding. The **transjugular intrahepatic portosystemic shunt (TIPS)** procedure has almost completely replaced surgical shunts.

TIPS is a percutaneous shunt between the portal and systemic circulations. The shunt is created between the hepatic vein and portal vein within the liver parenchyma, maintained by placement of a stent graft. This creates a low-resistance conduit to decompress the portal circulation, thereby decreasing variceal blood flow and ascites formation. Although it can be performed with conscious sedation, balloon dilation of the tract is extremely painful, typically requiring GA.

TIPS was developed initially by Rosch in dog studies in 1969. Palmaz and Richter performed the first successful TIPS in humans in 1989, using metallic stents to maintain the patency of the tract. Since then, TIPS has become the procedure of choice for patients who fail sclerotherapy and banding. In addition, it addresses another common problem associated with cirrhosis: refractory ascites.

Through the right IJ, a 10-Fr sheath is placed in the upper IVC. A catheter/guide wire combination is used to select the right hepatic vein. A wedged hepatic venogram using CO₂ is performed. Injection of iodinated contrast can result in rupture of the liver

capsule and exsanguination. The wedged venogram refluxes contrast through the sinusoids and into the portal vein, thereby providing a map. The catheter is exchanged over a stiff wire for a metallic introducer/needle. A variety of needle kits are available. All include a long introducer sheath and coaxially inserted curved-tip needle or metal cannula, with directional indicator, to help steer the needle toward the right portal vein.

Portal venous pressures are measured, and the pressure gradient between the portal vein and right atrium is determined. Following a portal venogram, the catheter is exchanged for an angioplasty balloon and the tract dilated (Fig. 13.1-13C). The use of covered stents for TIPS creation led to significantly improved patency rates, and for this reason, the Viatorr® (Gore, Flagstaff, AZ) self-expanding covered stent has largely replaced the bare metal stents. The Viatorr is positioned across the tract, and the uncovered portion is first deployed within the portal venous system (Fig. 13.1-13D). The covered portion is then pulled into the tract and deployed in a separate step. The stent may be dilated to a diameter of 8–10 mm using an angioplasty balloon, depending on the stent diameter and desired portosystemic gradient. If necessary, a second stent is deployed to cover any remaining unstented portions of the hepatic tract. Following TIPS placement, porta venogram and pressure gradients are remeasured. Ideally, the pressure gradient following shunting should be between 6 and 12 mm Hg. If the gradient is too high, there is a risk of rebleeding; if too low, there is overshunting of blood, thus bypassing the entire portal venous system and increasing the risk of encephalopathy and liver failure. After successful creation of the shunt, all devices, including the right jugular sheath, are removed, and hemostasis is achieved. Patients are closely monitored in an ICU or step-down unit for 24–48 h.

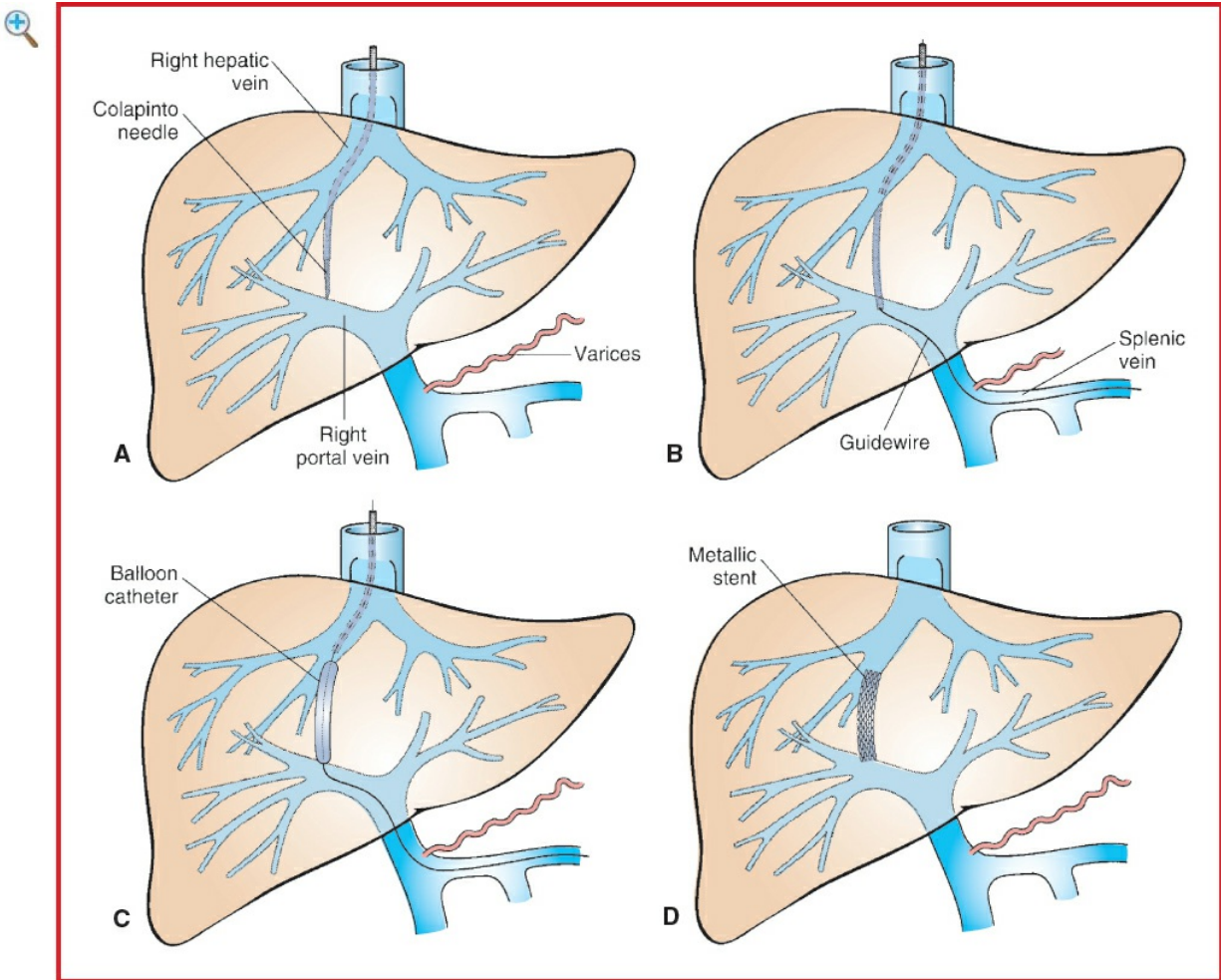


Figure 13.1-13 TIPS placement. **A:** Sheathed Colapinto needle is advanced out of hepatic vein into portal vein branch. Varices are present. **B:** Guide wire is advanced through the needle sheath into splenic vein. **C:** Parenchymal liver tract is dilated using a balloon angioplasty catheter. **D:** The metallic stent is deployed within the shunt tract. (Redrawn from Haskal ZJ, Ring F. Current Techniques in Interventional Radiology. Current Science, Philadelphia: 1994. Reprinted with permission from Ziv J Haskal, MD.)

Direct IVC-to-portal shunt (DIPS) is a newer alternative to TIPS. Simultaneous venous access is required from both the internal jugular vein and common femoral vein. For DIPS, an intravascular ultrasound (US) probe is used to guide needle puncture from the IVC, through the caudate lobe, and into the main portal vein. Because the tract traverses extrahepatic regions, placement of a covered stent is mandatory. A theoretically fewer number of passes is needed to access the portal vein because the puncture is US guided; however, there is likely a greater risk of intraperitoneal hemorrhage with each puncture attempt.

If TIPS/DIPS is performed for the treatment of bleeding varices, it may be necessary to embolize and sclerose the varices if they persistently fill following shunt creation. This may be accomplished using sclerosing agents (e.g., ethanolamine) with an occlusion

balloon and embolic agents (e.g., metallic coils, Gelfoam™), depending on operator preference.

Usual preop diagnosis

Bleeding esophageal varices (as a result of portal HTN); ascites; Budd-Chiari and hepatorenal syndromes

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Right IJ access
Special instrumentation	Rosch-Uchida, Colapinto, or Hawkins needle set; angioplasty balloons; endovascular stent grafts, marker pigtail, CO ₂ injection apparatus, intravascular ultrasound (IVUS) for DIPS, embolic agents (e.g., coils, Gelfoam, ethanolamine) for variceal embolization/sclerosis if necessary
Unique considerations	May need FFP
Antibiotics	Cefazolin 2 g iv
Procedure time	2–6 h
EBL	0–3000 mL
Postop care	ICU or step-down unit; careful fluid management
Mortality	Emergency: 50–100% Postprocedure: See Tables 13.1-2–13.1-5
Morbidity	Encephalopathy: 18–30% Late liver failure: 10–25% Shunt occlusion: 10%/yr Liver capsule puncture → intraperitoneal hemorrhage (continue procedure to decompress portal venous system → ↓ bleeding) Hepatic artery puncture: may require embolization Allergic reactions MI (related to ↑ CVP) Renal failure (usually transient)

Pain score 7–8 (first few hours only)

Table 13.1-2 MELD Score Calculation

The Model for End-Stage Liver Disease (MELD) score is calculated using three objective indicators—serum bilirubin, serum creatinine level (Cr), and international normalized ratio (INR)—according to the United Network for Organ Sharing modification of the original formula:

$$\text{MELD} = 9.6 \times \log e (\text{cr}) + 3.8 \times \log e (\text{Bilirubin}) + 11.2 \times \log e (\text{INR}) + 6.4$$

The MELD score can easily be calculated by accessing the following website:
<http://www.mayoclinic.org/meld/mayomodel7.html>

Table 13.1-3 Overall Mortality Based on MELD Score

MELD Score	No. of Patients	Mortality Rate (%)		
		30 d	3 mo	6 mo
≤10%	28	0 (0, 0)	0 (0, 0)	0 (0, 0)
11–17	83	7.3 (1.7, 12.9)	16.0 (8.0, 24.0)	24.9 (15.1, 34.7)
18–24	40	17.9 (5.9, 29.9)	34.8 (19.4, 50.2)	38.6 (22.4, 54.9)
≥25	15	42.6 (16.7, 68.4)	65.5 (39.9, 91.2)	74.2 (50.0, 98.3)

Patients undergoing elective TIPS creation with a MELD score of 18 or more had a significantly lower 3-mo survival rate than those with a MELD score of 17 or less. Numbers in parentheses are 95% confidence intervals.

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Table 13.1-4 Mortality Based on MELD Score and Cause of Cirrhosis

MELD Score	Group	No. of Patients	Mortality Rate (%)		
			30 d	3 mo	6 mo
≤17	A	39	5.1 (0, 12.1)	7.7 (0, 16.1)	13.9 (2.5, 25.2)
	B	72	5.6 (0.3, 11.0)	14.4 (6.1, 22.7)	21.4 (11.3, 31.4)
≥18	A	14	42.9 (16.9, 68.8)	57.1 (31.2, 83.1)	57.1 (31.2, 83.1)
	B	41	17.9 (5.9, 29.9)	37.9 (22.0, 53.7)	45.9 (28.1, 63.2)

A, alcohol-induced disease; B, non-alcohol-induced disease.

Patients undergoing elective TIPS creation with a MELD score of 18 or more had a significantly lower 3-mo survival rate than those with a MELD score of 17 or less. Numbers in parentheses are 95% confidence intervals.

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Table 13.1-5 Mortality Based on MELD Score and Ascites

MELD Score	Ascites	No. of Patients	Mortality Rate (%)		
			30 d	3 mo	6 mo
≤17	No	48	4.2 (0, 9.8)	4.2 (0, 9.8)	4.2 (0, 9.8)
	Yes	63	6.4 (0.3, 12.5)	18.4 (8.6, 28.3)	31.3 (18.6, 43.9)
≥18	No	18	22.2 (3.0, 41.4)	39.8 (16.8, 62.8)	46.5 (22.6, 70.4)
	Yes	37	25.4 (11.0, 39.8)	44.4 (27.4, 61.4)	48.7 (31.1, 66.3)

Patients undergoing elective TIPS creation with a MELD score of 18 or more had a significantly lower 3-mo survival rate than those with a MELD score of 17 or less. Numbers in parentheses are 95% confidence intervals.

Reprinted with permission from Ferral H, Gamboa P, Postoak DW, et al: Survival after elective transjugular intrahepatic portosystemic shunt creation: prediction with Model for End-Stage Liver Disease Score. *Radiology* 2004;231(1):231-6. Copyright © 2004 Radiological Society of North America.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for TIPS procedures have portal HTN usually 2° end-stage liver disease (ESLD), which will affect the function of a variety of organ systems, as described below.

Respiratory

Abdominal distension → atelectasis → ↑ pulmonary shunting → hypoxemia (hepatopulmonary syndrome in ESLD)

Encephalopathy → hyperventilation → ↓ PaCO₂ (respiratory alkalosis with chronic acidosis as compensation). Pulmonary effusion may be present in 5–10% of patients.

Tests: ✓ CXR (for Sx of atelectasis); ABG and PFT, if indicated

Cardiovascular

Cardiomyopathy (ETOH) and CAD (tobacco) occur at higher incidence in this patient population. Diuretic therapy → hypovolemia + ↓ K⁺ (furosemide) or ↑ K⁺ (spironolactone). Hyperdynamic circulation 2° to ↓ peripheral resistance and ↓ cardiac reserve are common findings and correlate with poor postop outcome.

Tests: ECG; electrolytes; cardiac echo, if indicated from H&P

Neurologic	Symptoms of hepatic encephalopathy range from mild confusion to coma. These patients may be very sensitive to narcotics and sedatives. A characteristic finding in liver failure is asterixis (liver flap). Hyponatremia and hypoglycemia may mimic hepatic encephalopathy. Tests: As indicated from H&P
Hepatic	Drug metabolism may be markedly reduced; anticipate prolonged effect with sedative and narcotic drugs. Drugs with particularly prolonged action include midazolam, meperidine, ranitidine, and lidocaine. Apparent resistance to all muscle relaxants is most likely due to an increased volume of distribution. Succinylcholine effects may be prolonged in patients with severe ESLD (2° ↓ plasma cholinesterase). Tests: Bilirubin; PT; albumin; LFTs
Gastrointestinal	Ascites → ↑ intra-abdominal pressure → ↑ risk of aspiration. Full-stomach precautions and rapid-sequence induction are recommended. Portal HTN → variceal bleeding. If possible, avoid esophageal instrumentation (e.g., TEE, esophageal stethoscope). Gastritis and peptic ulceration may be present.
Renal	Oliguric renal failure may complicate ESLD (hepatorenal syndrome). This may be reversible if liver failure improves. Differential Dx includes prerenal azotemia and acute tubular necrosis. NB: Bilirubin metabolites interfere with creatinine measurement and may mask ↑ creatinine levels. Tests: BUN; Cr; electrolytes
Endocrine	Hypoglycemia may be present in cases of severe cirrhosis. ESLD patients may have ↓ response to catecholamines. Tests: Glucose
Hematologic	Patients may be anemic 2° GI bleeding. The majority of patients will exhibit coagulopathy 2° ↓ hepatic

synthetic function (all factors except VIII and fibrinogen) and ↓ Plt. Patients may require vitamin K (if coagulopathy is present and time permits), FFP (if PT > 2 sec above baseline), or Plt transfusion (if Plt < 100 K) before procedure; these products should be readily available.

Premedication

Tests: CBC; Plt; PT; others as indicated from H&P
Patients with significant ascites will require full-stomach precautions.

NB: Benzodiazepines should be used with caution in liver failure patients.

INTRAOPERATIVE

Anesthetic technique

In some patients (and at some centers), sedation with local anesthesia may be satisfactory; however, balloon dilation of the intrahepatic tract can be exceedingly painful. Remember, these patients may be very sensitive to narcotics → respiratory arrest. GETA is often necessary to provide adequate analgesia and airway protection. The patient will be positioned so that the interventional radiologists can approach both the internal jugular and femoral veins, thus limiting the anesthesiologist's access to the head and airway.

General anesthesia

Induction	Rapid-sequence induction is necessary in patients with encephalopathy, abdominal distention, and recent variceal bleeds (blood in the stomach).	
Maintenance	Standard maintenance. If muscle relaxation is to be maintained, low-dose vecuronium (1–1.5 mcg/kg/min), cisatracurium (3 mcg/kg/min infusion), or rocuronium (5–15 mcg/kg/min) may be used to maintain muscle relaxation.	
Emergence	Extubate when the patient is awake and protective laryngeal reflexes are present. Patient should be transferred to PACU accompanied by anesthesiologist.	
Blood and fluid requirements	Potential for large blood loss	Glucose-containing solutions may be required for patients

	IV: 14–16 ga × 1–2 FFP available Plt available 2–4 U PRBC available NS/BSS (as appropriate) Standard monitors	with hepatic failure ± CHF. ✓ blood glucose levels frequently. Vasopressor response may be impaired.
Monitoring		Ventricular arrhythmias can be provoked by hepatic vein catheterization.
Positioning	✓ and pad pressure points ✓ eyes	Radiology tables usually are not well padded → nerve damage.
Complications	Portal vein rupture Liver capsule perforation Complete heart block CHF	Intra-abdominal hemorrhage may be massive and require emergency surgery. Patients with preexisting LBBB may require a pacemaker pre-TIPS (2° risk of RBBB during procedure). Shunt → ↑↑ venous return → CHF

▼ POSTOPERATIVE

Complications	Portal vein thrombus ↑ encephalopathy Sepsis Bleeding Fluid/electrolyte disturbance	May mimic symptoms of PE or MI 2° ↓ hepatic portal blood flow May require hemodynamic support (e.g., dopamine) Stent insertion → ↑ venous return → ↑ diuresis → electrolyte/fluid imbalance
Pain management	IV opiates	

Suggested Readings

1. Amin AA, Alabsawy EI, Jalan R, et al: Epidemiology, pathophysiology, and management of hepatorenal syndrome. *Semin Nephrol* 2019; 39(1):17–30.
2. Fidelman N, Kwan SW, LaBerge JM, et al: The transjugular intrahepatic portosystemic shunt: an update. *Am J Roentgenol* 2012; 199(4):746–55.
3. Landrigan-Ossar M: Common procedures and strategies for anaesthesia in interventional radiology. *Curr Opin Anaesthesiol* 2015; 28(4):458–63.
4. Riggio O, Nardelli S, Moscucci F, et al: Hepatic encephalopathy after transjugular intrahepatic portosystemic shunt. *Clin Liver Dis* 2012; 16(1):133–46.
5. Shah RP, Sze DY: Complications during transjugular intrahepatic portosystemic shunt creation. *Tech Vasc Interv Radiol* 2016; 19(1):61–73.

ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY

PROCEDURAL CONSIDERATIONS

Description

Endoscopic retrograde cholangiopancreatography (ERCP) was originally used as a diagnostic procedure; however, it is now almost exclusively used for therapeutic purposes. As with many invasive procedures, success (including the absence of complications) is well correlated with operator experience. Basically, ERCP consists of passing an endoscope through the mouth, esophagus, and stomach into the duodenum, where the papilla of Vater is located. A catheter (sometimes preceded by a wire) is inserted through the sphincter of Oddi into the common bile duct or pancreatic duct (see [Fig. 13.1-14](#)). At that point, contrast can be injected, tissue sampled, stones extracted, stents placed, or a sphincterotomy accomplished. The endoscopist may request administration of glucagon (0.25–2 mg iv; side effects, n/v, ↓ BP) to decrease duodenal motility and/or secretin (0.2 mcg/kg iv over 1 min; side effects, flushing, n/v) to help locate the minor papilla.

The overall complication rate is 5–10%, being <5% for simple stone extraction to 20% or more for sphincterotomy. Post-ERCP pancreatitis is the most common complication (up to 40%), being more common in young patients with sphincter of Oddi dysfunction and normal bilirubin levels. Placement of a pancreatic stent may reduce the risk of post-ERCP pancreatitis and is commonly used in high-risk patients. Unfortunately, stent placement itself may cause complications, including ductal and parenchymal injury and

ductal stenosis. Balloon dilation as an alternative sphincterotomy has been associated with a higher risk of pancreatitis.

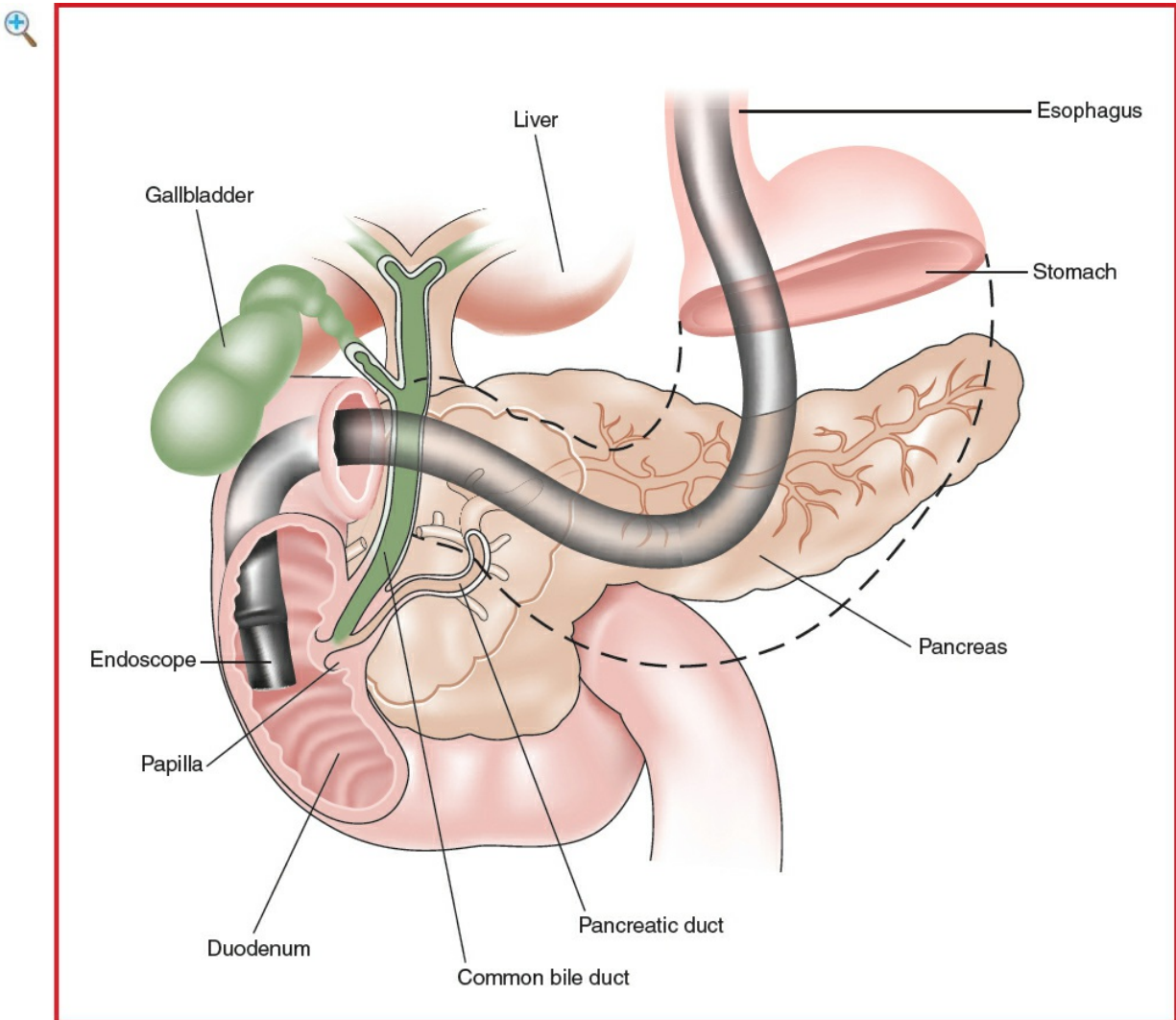


Figure 13.1-14 Anatomy for ERCP.

Usual preop diagnosis

Patients typically present with persistent pain or jaundice as a result of gallstones or other disease processes (e.g., infection, tumor, pancreatitis) that obstruct the bile duct and/or pancreatic duct.

■ SUMMARY OF PROCEDURE

Position	Initially left lateral decubitus, rotating to prone
Incision	None
Special	Endoscope (duodenoscope) and supporting equipment;

instrumentation	mouth guard; fluoroscopy
Unique considerations	X-ray protection will be required; glucagon may be requested (0.25–2 mg iv 10 min before procedure; contraindications, pheochromocytoma, insulinoma)
Antibiotics	Ciprofloxacin 500 mg po 30–60 min before procedure is common.
Procedure time	45–90 min
EBL	Usually minimal
Postop care	PACU → home (occasional overnight stay for stone removal or stent placement)
Mortality	Rare
Morbidity	Pancreatitis: typically 5% (1–40% reported) Bowel or duct perforation: <1% Hemorrhage: 0.1–2% Cholangitis/cholecystitis Aspiration Cardiopulmonary (e.g., myocardial ischemia, respiratory arrest)
Pain score	2–4

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for ERCP most commonly have obstruction of hepatic and/or pancreatic ducts. They may be jaundiced with elevation of both conjugated and unconjugated bilirubin or exhibit Sx of pancreatic disease with associated peritoneal and pleural effusions. As a result of underlying disease processes (typically ASA 3–4), they may be quite frail with limited physiologic reserve. Discuss intraprocedure management with the endoscopist.

Respiratory

Airway evaluation should focus on the need for GETA. Access to the airway may be limited during the procedure. Ascites and pleural fluid accumulation → impaired ventilation, ↑ aspiration risk. Gas insufflated during the procedure may further

	compromise ventilation and ↑ aspiration risk. Rales/percussive dullness suggests pulmonary edema/effusion. Tests: As indicated from H&P
Cardiovascular	ECG changes are not uncommon during this procedure, and a preop ECG may be helpful. Hypotension is not uncommon. Tests: As indicated from H&P
Neurologic	Altered mental status may result from impaired hepatic function.
Hepatic	Hepatic dysfunction → coagulopathy, altered drug metabolism, and other metabolic disturbances. Jaundice is common. Tests: As indicated from H&P
Renal	↑ BUN/Cr suggests volume depletion or acute renal failure. In acute pancreatitis, fluid deficit may be severe; volume resuscitate before proceeding. Tests: As indicated from H&P
Endocrine	↑ glucose (>180 mg% should be treated preop) Tests: As indicated from H&P
Premedication	Morphine may ↑ sphincter of Oddi tone → ↑ ductal pressure + ↑ pain. Midazolam 1–2 mg iv may be appropriate in the anxious patient. Consider aspiration prophylaxis (e.g., Bicitra™ 30 cc po; avoid metoclopramide).

INTRAOPERATIVE

Anesthetic technique

Deep sedation using propofol (avoiding respiratory depression or loss of protective airway reflexes) supplemented with topical anesthesia of the oropharynx is sufficient for many ERCP procedures. GETA is associated with a higher procedure success rate and should be considered for complex ERCP procedures, patients at high risk for aspiration, or uncooperative patients.

General anesthesia

Induction	Rapid-sequence induction is necessary in patients with gastroparesis or abdominal distention. Otherwise, standard induction. Consider etomidate (0.1–0.3 mg/kg) for hemodynamically unstable patients.	
Maintenance	Standard maintenance. An immobile patient will be appreciated by the endoscopist. Manipulation of the endoscope may cause ETT movement → mainstem intubation or extubation.	
Emergence	The duodenum and stomach should be decompressed by the endoscopist before extubation. Standard emergence.	
Blood and fluid requirements	IV: 18 ga × 1 BSS @ 3–5 mL/kg/h	Patients with acute pancreatitis may be hypovolemic.
Monitoring	Standard monitors ± Arterial line	✓ glucose in patients receiving glucagon.
Positioning	Left lateral decubitus → prone ✓ and pad pressure points ✓ eyes	
Complications	Pancreatitis Myocardial ischemia Aspiration	Δ ECG may occur in 25% of patients >65 yr.

POSTOPERATIVE

Complications	Myocardial ischemia Hypoxemia Hemorrhage	
Pain Management	Usually managed with oral analgesics	Avoid morphine (↑ sphincter of Oddi tone).

Suggested Readings

1. ASGE Standards of Practice Committee, Anderson MA, Fisher L, et al. Complications of ERCP. Gastrointest Endosc. 2012; 75(3):467–73.

2. Correia LM, Bonilha DQ, Gomes GF, et al: Sedation during upper GI endoscopy in cirrhotic outpatients: a randomized, controlled trial comparing propofol and fentanyl with midazolam and fentanyl. *Gastrointest Endosc* 2011; 73(1):45–51.
3. Freeman ML: Complications of endoscopic retrograde cholangiopancreatography: avoidance and management. *Gastrointest Endosc Clin N Am* 2012; 22(3):567–86.
4. Garewal D, Vele L, Waikar P: Anaesthetic considerations for endoscopic retrograde cholangio-pancreatography procedures. *Curr Opin Anaesthesiol* 2013; 26(4):475–80.
5. Sharp CD, Tayler E, Ginsberg GG: Anesthesia for routine and advanced upper gastrointestinal endoscopic procedures. *Anesthesiol Clin* 2017; 35(4):669–77.

Suggested Viewing

Links are available online to the following videos:

Wire Guided Cannulation Techniques, S. Edmundowicz, MD: <https://www.youtube.com/watch?v=L3wqtPCeCic>

Dilation Assisted Stone Extraction (DASE), Dr. Abhitabh Patil, MD: <https://www.youtube.com/watch?v=Y1yVsnbosBg>

Animation of Dilation Assisted Stone Extraction (DASE): <http://endosuite.com/videos/dilation-assisted-stone-extraction-dase-procedural-animation>

IMAGING AND IMAGE-GUIDED PROCEDURES

PROCEDURAL CONSIDERATIONS

Description

Projectional imaging includes x-ray fluoroscopy, and cross-sectional imaging includes computed tomography (CT), ultrasound (US), and magnetic resonance imaging (MRI). These techniques have become indispensable in modern diagnosis. In addition, a growing number of invasive procedures are being performed using imaging for guidance, not only for diagnostic purposes but also for therapeutic purposes. Indications include Dx of primary or metastatic tumors; tumor staging; Dx of benign processes, including infections; percutaneous drainage of fluid collections; local-regional treatment of tumors and vascular malformations; endovascular treatment of hemorrhage, vascular malformations, aneurysms, and dissections; percutaneous ablation; catheter-directed thrombolysis and thrombectomy; IVC filter implantation or retrieval; placement of portosystemic shunts; recanalization and angioplasty/stenting of chronically occluded veins; endovascular treatment of varicose veins; dialysis-related interventions; percutaneous placement of feeding tubes; percutaneous treatment of urinary and biliary obstructions; and placement of venous access devices.

Selection of imaging modality depends on ease of identification of the target lesion and resolution of surrounding and intervening structures. Patient compliance is crucial to success because image resolution and spatial accuracy require the patient to be immobile during image acquisition and the procedure itself. In compliant adults, most of these procedures may be done using moderate sedation. For procedures in the pediatric population, as well as for more invasive procedures in adults, GA is frequently necessary.

Diagnostic imaging

Pediatric: In general, moderate sedation is sufficient to ensure pediatric patient compliance for diagnostic CT, US, or MRI, but occasionally, GA is necessary (see [Section 13.2](#)).

Adults: Almost all CT, US, and MRI studies are performed without anesthesia. Approximately 5% of adult patients are too claustrophobic to complete an MRI study. Some of these patients may benefit from sedation, but use of GA is rare. Most patients with implanted PM and ICDs can safely undergo MRI if appropriate precautions are observed (see [Fig. 13.1-15](#)).

Image-guided procedures

Pediatric: Procedures performed on the pediatric patient routinely require GA (see [Section 13.2](#)).

Adults: X-ray fluoroscopy and CT- and US-guided procedures, including biopsies, fluid drainage, and tissue ablation, are routinely performed under moderate sedation. More invasive and painful procedures, such as stent graft repair of aneurysms and dissections, and thermal ablation of unresectable tumors, frequently require GA, spinal, or epidural anesthesia. The required site of access and the positioning of the operators should be considered before the procedure. Position of the ETT and central lines can be immediately confirmed using fluoroscopy. Occasionally, iv injection of iodinated contrast medium is necessary, and adverse reactions—such as urticaria, airway edema, hormone release (e.g., from pheochromocytoma), or anaphylaxis—may occur.

**Section I - This section is to be filled out by the PATIENT:**

The MRI room contains a very strong magnet. Before you go into the room, we must know if you have any metal in your body. Please complete the following:

Pacemaker/Wires*	☐ Yes	☐ No	Recent Stent Placement	☐ Yes	☐ No
* If yes, radiology/cardiology staff must complete section III.					
Aneurysm Clips	☐ Yes	☐ No	Shunt	☐ Yes	☐ No
Surgical Clips	☐ Yes	☐ No	Ear Implant	☐ Yes	☐ No
Eye Implant	☐ Yes	☐ No	IUD	☐ Yes	☐ No
Implantable Defibrillator	☐ Yes	☐ No	Blood Vessel Coil	☐ Yes	☐ No
Bullets, Pellets, etc	☐ Yes	☐ No	Heart Valve	☐ Yes	☐ No
Stimulator/Wires	☐ Yes	☐ No	Artificial Limb	☐ Yes	☐ No
Infusion Pump	☐ Yes	☐ No	Cochlear Device	☐ Yes	☐ No
Penile Prosthesis	☐ Yes	☐ No	Tracheostomy	☐ Yes	☐ No

Do you have kidney disease?	☐ Yes	☐ No
Have you ever been a machinist or metal worker?†	☐ Yes	☐ No
Have you ever had a facial injury from metal?	☐ Yes	☐ No
Have you ever had metal removed from your eyes?	☐ Yes	☐ No
Are you pregnant? Last menstrual period	☐ Yes	☐ No
Do you have any allergies? If yes, please specify	☐ Yes	☐ No
Current medications:		

Weight:

Section II - This section is to be filled out by the RADIOLOGY STAFF for all patients:

†Orbit Films	☐ Yes	☐ No
Anyone attending the patient in the MRI room has been cleared for MRI safety requirements.	☐ Yes	☐ No

Section III - This section is to be filled out by the RADIOLOGY/CARDIOLOGY STAFF for patients with implantable cardiac devices:

Was the pacemaker or ICD generator implanted after 1998 or 2000, respectively?	☐ Yes	☐ No
• If not, cancel MRI		
Were the leads implanted ≥ 6 weeks before today?	☐ Yes	☐ No
• If not, cancel MRI		
Are non-transvenous epicardial or abandoned leads present? (Chest x-ray may be necessary if device implantation records are unavailable)	☐ Yes	☐ No
• If yes, cancel MRI		

☐ Pacemaker	☐ ICD	
Dependent upon pacing?	☐ Yes	☐ No
• If yes, and the device is an ICD, cancel MRI		
• If yes, and the device is a pacemaker, program pacing to VOO/DOO		
• If not, program pacing to VVI/DDI and deactivate tachycardia detection and therapies		

- Deactivate magnet, rate, PVC, noise, ventricular sense, and conducted atrial fibrillation response
- Monitor blood pressure, ECG, oxygen saturation, and symptoms during MRI

Device Manufacturer & model number:	Prior to MRI			After MRI		
	RA	RV	LV	RA	RV	LV
Sensing						
Capture Threshold						
Impedance						
Battery Voltage						

- Keep on ECG monitor after MRI until initial device programming has been restored
- Advise follow-up in device clinic in 3-6 months after MRI

Figure 13.1-15 MRI safety checklist. (Reproduced with permission from Nazarian S, Beinart R, Halperin HR: Magnetic resonance imaging and implantable devices. *Circ Arrhythm Electrophysiol* 2013; 6(2):419–28. Copyright © 2013 American Heart Association, Inc.)

The field of MRI-guided procedures—sometimes referred to as interventional MR (iMRI) or magnetic resonance-guided therapy (MRT)—reflects the emergence of new magnet geometries, allowing physician access to the patient during imaging. These geometries may be C-arm configurations, parallel discs above and below the patient, or dual rings where the patient is placed either through the apertures of the rings or perpendicularly between the rings. Faster image acquisition pulse sequences are allowing near-real-time feedback. In addition to guiding biopsies and drainage, this technology enables more aggressive procedures, such as craniotomies or percutaneous tumor ablations, to be performed with immediate feedback showing the progress of excision or ablation. Many of these procedures require GA and, accordingly, require MRI-compatible monitoring and anesthetic equipment. For safety, the anesthesiologist and proceduralist also are subject to the same restrictions that apply to patients, so having a pacemaker, ICD, ferromagnetic aneurysm clips, or a metallic foreign body in the orbit precludes that person’s suitability to perform these cases. Hybrid systems also are being marketed, combining MRI and x-ray fluoroscopy, CT and x-ray fluoroscopy, or CT and positron emission tomography (PET). These may involve overlapping precautions and risks.

Usual preop diagnosis

Tumor, primary or metastatic; lymphadenopathy; abscess, effusion, empyema, pseudocyst, or other fluid collection; cholecystitis; arterial occlusive disease; aneurysm or pseudoaneurysm; trauma; deep venous thrombosis; chronic venous occlusion; thromboembolic disease; pulmonary embolism; venous insufficiency; implanted IVC filter; biliary obstruction; obstructive uropathy; hydronephrosis; portal hypertension; cirrhosis; renal failure

SUMMARY OF PROCEDURES			
	CT/X-Ray Fluoroscopy	US	MRI/MRT
Position	Supine, prone, lithotomy, or lateral decubitus	⇐	⇐
Unique considerations	Metallic objects should be kept out of the CT imaging field. Exit the procedure room during scanning to avoid	No ionizing radiation. Room lights are	All equipment, including monitors, valves, anesthesia machine, O ₂ tanks, laryngoscopes, etc., must be

	radiation exposure, or use protective lead garment. Contrast reaction possible.	frequently dimmed for better viewing of the screen.	nonferromagnetic. Other metals should be removed from the imaging field to avoid artifact.
Antibiotics	Indicated for open procedures or when draining infected fluid collections, including abscesses, empyemas, and obstructed biliary or urinary systems.	⇐	⇐
Procedure time	0.5–4+ h, depending on complexity of the procedure	⇐	⇐
EBL	Procedure dependent; may be internal hemorrhage, may be preexisting hemorrhage	⇐	⇐
Postop care	PACU → room Level of care dependent on complexity of procedure and patient's overall status	⇐	⇐
Mortality	Rare	⇐	⇐
Morbidity	Contrast reaction Hemorrhage Infection/sepsis Organ injury/perforation Pneumothorax	⇐	⇐

	Pulmonary embolus		
	Arrhythmia		
	Hemodynamic collapse		
Pain score	Procedure dependent	⇐	⇐

■ PATIENT POPULATION CHARACTERISTICS

Age range	All
Male:Female	1:1
Incidence	Common
Associated conditions	Fluid collections: Patients may present with fever, sepsis, pain, or ileus. Mass effect, such as with empyema or pericardial effusion, may also affect respiratory or cardiovascular function. Instrumentation or relief of mass effect may induce a vasovagal response or acute bacteremia and sepsis. Some infections (Echinococcus, Entamoeba, methicillin-resistant Staphylococcus aureus [MRSA], vancomycin-resistant Enterococci [VRE]) may require special precautions, such as respiratory isolation, gown/glove standards, and pre- and postprocedure equipment sterilization. Solid tumors: Mass effect may result in obstruction of airways, blood vessels, GI/biliary tract, or urinary tract or neural impingement. HTN may be seen with significant renal compression and, occasionally, with neuroendocrine tumors. Hematopoietic, hepatic, or renal compromise may result in coagulopathy or Plt dysfunction. Vascular pathologies: Aggressive anticoagulation and/or antiplatelet therapy may be used intraop. Restoration of arterial flow to kidneys may → rapid ↓ of renin production and BP. Interventional procedures for venous thrombosis can → PE. Placement of central vascular access catheters can cause air embolism or cardiac dysrhythmias from right atrial

irritation. Sclerosis of vascular malformations may result in pulmonary embolism, acute pulmonary vasospasm, and HTN.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

The adult patient population requiring anesthesia for imaging is medically quite diverse; however, many have in common the inability or unwillingness to lie still during the procedure. Some of these patients are very ill, requiring the services of an anesthesiologist to maintain cardiopulmonary stability. Adult patients should be NPO (solid food and nonclear liquids) for 6 h preprocedure (elective). In general, US and CT scans present fewer problems for the anesthesiologist than MRI. Regardless of the method of anesthesia chosen, these patients must lie perfectly still, the airway must be protected, and IPPV may be required. Thus, children, the cognitively challenged, claustrophobic, uncooperative, or critically ill patients usually require GA.

Respiratory	As a result of limited access to the patient's airway, the preop examination should focus on the need for elective ET intubation to protect the airway. ET intubation and mechanical ventilation also may be required in trauma patients, the critically ill, or patients with GERD or sleep apnea. An LMA/iGel may be suitable for patients not at risk of aspiration.
Cardiovascular	Presence of a cardiac pacemaker or ICD is not a contraindication to MRI (see Safety Checklist, Fig. 13.1-15). The presence of PA catheter thermistors or retained pacing wires and lead fragments is a relative contraindication.
Neurologic	Patients with ↑ ICP or cranial trauma usually need GA with mechanical ventilation and intubation. The presence of ferromagnetic aneurysm clips and/or coils may be a contraindication to MRI (✓ with surgeon or radiologist). Some aneurysm clips are nonferromagnetic and are, therefore, MRI compatible.

Musculoskeletal	The presence of spinal instrumentation, metal plates, pins, screws, joint replacements, or other prostheses
Premedication	is usually not a contraindication to MRI. Midazolam 1–5 mg iv (titrated to effect) may be appropriate in the very anxious adult patient; alternatively, lorazepam 1–2 mg po/sl 1 h before procedure.

Considerations for MRI/MRT

The MRI/MRT suite poses many challenges to the anesthesiologist. Because of the intense magnetic fields involved in MRI, any equipment containing ferromagnetic components—such as ECG monitors and anesthesia machines—cannot go near the magnet. Thus, MRI-compatible equipment is mandatory in the area of the MRI scanner. The magnetic field will destroy information on credit card/access card magnetic strips and may damage pagers as well as mechanical devices, including wrist watches and infusion pump motors. A microdrip infusion set can be a suitable replacement for an infusion pump.

Noise	May be very distressing for some patients and may be as high as 110 dB in a 1.5-T scanner and >130 dB in a 3-T scanner. Exposure to noise levels of this magnitude should not exceed 2 h/d. Hearing protection is strongly recommended.
Thermal injury	Thermal injury is caused by induced currents in metal implants or in looped conductors in contact with the skin.
Projectile effect	The magnet has a strong attraction for ferromagnetic objects that can then become lethal missiles; therefore, all objects, including pens, scissors, iv poles, O ₂ cylinders, keys, stethoscopes, and eyeglasses or jewelry containing ferrous materials must be removed before entering the scanning room.
Implanted/foreign material	There are several reports describing problems that may occur with cardiac pacemakers (see Safety Checklist; Fig. 13.1-15 ; failure to pace), aneurysm clips (hemorrhage), and intravascular wires (induced currents). Metal workers may be at special risk for

	ocular damage 2° imbedded particles (see Safety Checklist; Fig. 13.1-15).
Contrast agent	Currently, gadolinium chelates are the primary agents in use and have a higher safety margin than iodinated contrast agents. Adverse reactions, however, occur in ~2% of patients and include HA, nausea, dizziness, hemodynamic instability, dysrhythmias, and nephrogenic systemic fibrosis.

INTRAOPERATIVE

Anesthetic technique

Typically, iv/po sedation, often without the services of an anesthesiologist. Patients unwilling or unable to cooperate will require GA.

IV sedation	In patients with a normal airway and no Hx of GERD, sedation can be carried out most easily using dexmedetomidine (1 mcg/kg loading dose over 10 min + 0.2–0.7 mcg/kg/h infusion) or propofol infusion (25–100 mcg/kg/min), ± midazolam (0.025–0.10 mg/kg) titrated to effect. NB: Only MRI-compatible infusion pumps should be used. (A microdrip infusion is a useful alternative.)
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General anesthesia

Induction	Standard induction on an MRI gantry (typically in the magnet anteroom). The anesthetized patient is then transported into the magnet.
Maintenance	Standard maintenance. Most commonly, a propofol or dexmedetomidine infusion provides satisfactory sedation for the procedure (see pump considerations, above). Because continuous muscle relaxation is usually not required, spontaneous ventilation may be safest.
Emergence	Emergence (and extubation) is often accomplished after the patient has been moved back to the anteroom, where additional airway and other support equipment

	should be readily available. The patient should be recovered in the PACU, which may be some distance from the MRI suite. Appropriate monitoring and personnel should accompany the patient during transport.	
Blood and fluid requirements	Usually no blood loss IV: 20 ga × 1 BSS @ TKO	
Monitoring	Standard monitors	Monitoring in the MRI/MRT suite presents special problems, discussed below.
Monitoring, MRI	ECG	ECG may be distorted by magnetic fields. Use MRI-compatible electrodes. Twist leads together to avoid creating loops (↓ artifacts, ↓ burns). V5 and V6 are least likely to develop artifacts.
	Pulse oximetry	MRI-compatible oximeters are available. Locate probe outside bore of the magnet (e.g., toe). Avoid burn injury 2° induced current in looped leads.
	BP (NIBP)	Replace all ferrous connections on cuff and tubing with nylon connectors. Use tubing extensions to keep apparatus away from the field.
	± Arterial line	If an arterial line is medically indicated, keep the transducer close to the patient to avoid recording artifacts. Use MRI-compatible transducers and connectors.

	Precordial/esophageal stethoscope	Recording equipment should have radiofrequency filters. Often unsatisfactory because of magnet noise. An MRI-compatible, infrared, wireless stethoscope is available.
	Temperature	MRI-compatible T monitors are available, although usually not necessary for short procedures in adults.
	Capnography	MRI-compatible capnographs are available. Long sample lines will distort waveforms, and ETCO ₂ concentration may not be accurate; however, this is still useful for measuring RR and relative changes in ETCO ₂ .
	PA catheter	PA catheters with thermistors or retained pacing wires are MRI contraindications.
	Urinary catheter	Catheters with T probes may be MRI compatible up to 3 Tesla, but the thermistor must be disconnected to avoid electrical or burn hazards.
	Verbal/visual	The patient and the monitors may be viewed directly (in procedure room) or through a screened window. Contact should be maintained throughout the procedure.
Positioning	✓ and pad pressure points ✓ eyes	CT and MRI gantries may be poorly padded → potential nerve injury.
Complications	Contrast related	Gadolinium → local and

	Loss of airway	systemic reactions Patient must be promptly extracted from the magnet bore and moved beyond the range of the magnet to permit use of emergency intubation and resuscitation equipment.
	Psychological	Panic attacks and claustrophobia occur in 5–10% of patients. Use of an eye mask may be helpful in selected patients. Heavy sedation or even GA may be necessary.
	Hearing loss	Temporary hearing loss and tinnitus may be expected in 40% of patients. Prevent by using ear plugs or other forms of hearing protection. GA: ↑ risk of hearing damage 2° stapedius muscle relaxation
	Thermal injury	Results from induced current heating of oximeter probe and looping cables

POSTOPERATIVE

Complications	Hearing loss	See discussion above.
	Thermal injury	See discussion above.

Suggested Readings

1. Crute D, Sebeo J, Osborn IP: Neuroimaging for the anesthesiologist. *Anesthesiol Clin* 2012; 30(2):149–73.
2. Duncan JR, Tabriz D: Improving performance during image-guided procedures. *J Patient Saf* 2015; 11(4):230–6.
3. Hofer CK, Mizuguchi AK, Popescu WM: Monitoring the patient at risk c

hemodynamic instability in remote locations. *Int Anesthesiol Clin* 2012; 50(2):141–72.

4. Kim H, Lane J, Schlichter R, et al: Use of anesthesiology services in radiology. *Anesthesiol Clin* 2017; 35(4):601–10.
5. Layne KA, Dargatzis PI, Archer JRH, et al: Gadolinium deposition and the potential for toxicological sequelae—a literature review of issues surrounding gadolinium-based contrast agents. *Br J Clin Pharmacol* 2018; 84(11):2522–34.
6. Munn Z, Jordan Z: Interventions to reduce anxiety, distress and the need for sedation in adult patients undergoing magnetic resonance imaging: a systematic review. *Int J Evid Based Healthc* 2013; 11(4):265–74.
7. Nazarian S, Hansford R, Rahsepar AA, et al: Safety of magnetic resonance imaging in patients with cardiac devices. *N Engl J Med* 2017; 377(26):2555–64.
8. Practice advisory on anesthetic care for magnetic resonance imaging: an updated report by the American Society of Anesthesiologists task force on anesthetic care for magnetic resonance imaging. *Anesthesiology* 2015; 122(3):495–520.
9. Rahmathulla G, Recinos PF, Traul DE, et al: Surgical briefing, checklists, and the creation of an environment of safety in the neurosurgical intraoperative magnetic imaging suite. *Neurosurg Focus* 2012; 33(5):E12.
10. Tagliaferri M, Jankovic J, Pagan F, et al: Safety of MRI in patients with implanted deep brain stimulation devices. *Neuroimage* 2009; 47(Suppl 2):T53–7.
11. Veenith T, Coles JP: Anaesthesia for magnetic resonance imaging and positron emission tomography. *Curr Opin Anaesthesiol* 2011; 24(4):451–8.

TRACHEOBRONCHIAL STENTING

SURGICAL CONSIDERATIONS

Description

Tracheobronchial stenting may be performed for benign or malignant strictures of upper and lower airways that are unsuitable either for surgical reconstruction or as a temporizing measure. Patients may not be considered surgical candidates due to poor medical condition (e.g., comorbidity, recent thoracic surgery, limited life expectancy) or certain characteristics of the stricture (e.g., active disease, airway inflammation, extensive length, or multifocality). The most common cause of tracheobronchial narrowing is bronchogenic carcinoma, with the leading benign cause being stricture secondary to prolonged intubation. Less common causes include radiation-induced stenosis, polychondritis, tracheomalacia, and, in children, extrinsic strictures and 2° vascular malformations.

Since the first lung transplant in 1963, posttransplant bronchial stenosis has joined the list of indications for tracheobronchial stenting. Bronchial stenosis is a relatively common complication of lung transplantation, occurring in single-lung, double-lung, and heart/lung transplant recipients with an overall incidence of 7–40%. It is believed that this complication is 2° the lack of bronchial arterial supply, with resulting airway ischemia. These ischemic stenoses occur at the bronchial suture line and in the more distal airways.

Stent types

There are two primary types of airway stents: silicone based and metallic, with both bare and covered metallic prostheses available. The development of biodegradable airway stents is an active area of research.

Silicone stents

Silicone-based stents (Silastic) are available both as straight, short tubes and as bifurcated Y-shaped devices. Selection of the appropriate size and length is critical. The stent must be long enough to extend across the stricture and of satisfactory diameter to maintain the caliber of the airway. The main **advantage** of silicone stents is that they are easily removed, either when the patient's ventilatory status has recovered sufficiently or when reconstructive surgery is possible. In addition, the stent can be easily repositioned to obtain optimal placement. Bifurcated silicone stents are also available to accommodate the Y-shaped configuration of the carina with extension into the distal trachea and both mainstream bronchi.

Silicone-based stents do have **disadvantages**. Stenotic airways need to be predilated before silicone stent insertion, whereas metallic stents can be placed within a narrow airway lumen and subsequently dilated. Silicone stents frequently become occluded with mucus plugs and granulation tissue or tumor overgrowth; therefore, regular bronchoscopic examination and treatment are necessary to keep the stented airway clear. Silicone stents and covered metallic stents are more likely to migrate than bare stents due to lack of incorporation into the bronchial wall. In general, silicone stents must be placed under GA because of the need for rigid bronchoscopy.

Metallic stents

The main **advantages** of metallic stents are the ease of insertion, a thin wall that rapidly becomes embedded in the airway, and the large gaps in the stent wall that allow normal ciliary function and reduced mucus impaction. The procedure can be performed using flexible bronchoscopy in the interventional room, with sedation. The main **disadvantage** of metallic stents is the difficulty in removing or repositioning these

devices once deployed. Metallic stents become firmly embedded in the wall of the airway and incorporated into the epithelium in <6 wk. Removal can be accomplished by using forceps to grip the wall of the stent and applying a twisting motion to pull the stent away from the wall. Potential complications from this maneuver may be catastrophic, and once these devices are placed, they should be considered permanent. However, if removal becomes necessary, the patient will require TIVA, muscle relaxants, and IPPV. Complications could include tracheal/bronchial disruption, pulmonary artery injury, and pneumothorax and airway obstruction. Another problem associated with metallic stents is the development of granulation tissue either at the ends of the stents or through the interstices. This requires careful follow-up by repeat bronchoscopy and may require subsequent procedures, such as bronchoplasty, restenting, and laser tissue ablation. Because of these potential complications, the FDA issued a warning in 2005 to limit the use of metallic stents in benign airway strictures only after all other treatment options have been explored.

Because of the flexibility and low profile of the deployment systems used for metallic stents, it is feasible to place them under conscious sedation; however, our practice is to utilize GA for tracheal stent placement to reduce patient movement due to the coughing that occurs with tracheal irritation.

Insertion techniques

Silicone stents have low inherent radial force; therefore, strictures need to be dilated before stenting. Rigid bronchoscopy is necessary to allow dilation and subsequent stent placement. Dilation can be performed with the Holinger bronchoscope, which is abutted to the stricture and advanced with a corkscrew motion. Gum-tipped Jackson dilators and various angioplasty balloons also can be used. In patients with tracheal stomas, a T-tube stent can be inserted via either the stoma or the mouth. This extends up to the vocal cords and down as far as the carina. In patients without tracheal stomas, either Y-tubes or straight stents are inserted in a similar fashion. The stent is mounted on the rigid scope, which is advanced across the stricture and then withdrawn, leaving the stent in place. Biopsy forceps are used to advance a limb of the Y-tube into the other bronchus. Placement of stents above the carina in this fashion is relatively straightforward, and the stents can be removed and repositioned until a satisfactory result is achieved. With more distal bronchial stents, the operator's vision is somewhat obscured, and deployment is more difficult.

For insertion of **metallic stents**, cross-sectional and fluoroscopic imaging is used to determine the optimal length and diameter of the stent. During the procedure, selective tracheobronchography may be performed by injecting water-soluble nonionic contrast

through a catheter at the level of stricture. Flexible bronchoscopy helps guide the stent placement with a soft-tipped guide wire being advanced through either the scope or the ETT into the distal airway. In the rare case where the stricture is too narrow and tight to allow passage of a small, flexible bronchoscope, a guide wire is passed, and the stent is positioned before deployment based on fluoroscopy, reconstructed CT images, or bronchography. Using fluoroscopy, the stent is deployed and visually confirmed with bronchoscopy.

Usual preop diagnosis

Bronchial compression 2° carcinoma; posttransplantation anastomotic stricture; radiation-induced stricture; relapsing polychondritis; sarcoidosis

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	None
Special instrumentation	Bronchoscope (flexible or rigid); silicone stent; self-expanding metallic stent (covered or uncovered)
Unique considerations	May have to pull the end of the tracheal tube back to the level of the vocal cords to adequately treat the entire trachea. Alternatively, can stent through LMA
Antibiotics	Not usually administered
Procedure time	1–2 h
EBL	0
Postop care	ICU or step-down unit
Mortality	1–5%
Morbidity	Tracheobronchial irritation (usually temporary). Stent malposition → bronchial occlusion or vocal cord paralysis
Pain score	2–4 (first few hours only)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Respiratory compromise due to airway obstruction may pose a significant management challenge. It is therefore somewhat controversial and institution dependent, whether

these procedures should be performed outside of the OR. A thorough preop workup is necessary before anesthesia. The majority of patients present for palliation of an obstructing pulmonary malignancy or treatment of granulation tissue obstruction of a lung transplant anastomosis. Some may present emergently with impending airway obstruction that precludes a complete preop workup. The choice of anesthesia is primarily a function of the type of stent, surgeon preference, and the comorbid conditions of the patient. In general, silicone stent placement, which requires airway dilation with rigid bronchoscopy, will necessitate GA. Metallic stents can be placed without the need for preceding airway dilation; therefore, either topical anesthesia, with or without conscious sedation, or GA can be used. Immediate improvements in FEV₁, FVC, and PEF can be expected after stenting an obstructed tracheobronchial segment.

Respiratory

Patients with upper or lower airway stricture may present with cough, stridor, dyspnea, and fatigue. Airway obstruction can be classified as dynamic or fixed, depending on whether the obstruction varies with the respiratory cycle. Stridor that worsens during inspiration suggests an extrathoracic dynamic obstruction, whereas expiratory stridor is associated with intrathoracic dynamic obstruction. Other physical findings may include bronchospasm (in COPD patients), clubbing, or cyanosis. Preop fiberoptic bronchoscopic examination and/or high-resolution, thin-section CT scans will help to define the location, size, and extent of the obstruction. In patients suspected of having tracheomalacia, scans are performed at maximal inspiration and maximal expiration to unmask subtle areas of narrowing exacerbated by ↑ intrathoracic pressure on inspiration. Using this information, the appropriate size for an ETT can be estimated and potential problems with tracheal intubation can be anticipated. **Tests:** CXR; PFTs with flow/volume loops (see [Fig. 5-14](#) in Anesthetic Considerations for Mediastinoscopy); CT scans preferably with thin slices and multiplanar reconstructions

Cardiovascular

Directed at any underlying disease process.
Restoration of intravascular volume before induction

	of GA is important in patients with hypovolemia 2° chronic malnutrition from malignancy. Tests: As indicated from H&P
Musculoskeletal	Patients with lung cancer may have myasthenic syndrome (Eaton-Lambert) with ↑ resistance to depolarizing muscle relaxants and ↑ sensitivity to NDMRs. Post–lung transplant patients on certain immunosuppressive therapies (e.g., cyclosporine) may have prolonged muscle blockade from NDMRs.
Hematologic	Blood cross-match not necessary unless high risk of hemorrhage from injury (e.g., rigid bronchoscopy used in patients with friable tumors). Maintaining adequate O₂-carrying capacity is important in patients with poor pulmonary reserve. Tests: CBC; others as indicated from H&P
Laboratory	Other tests as indicated from H&P
Premedication	Patients in respiratory distress are extremely anxious; however, anxiolysis has to be balanced against the risk of impending respiratory arrest in some patients. Careful titration of anxiolytic medications is necessary to avoid oversedation. An antisialagogue (e.g., glycopyrrolate 0.2 mg iv) will help minimize secretions and improve visualization through the bronchoscope. Post–lung transplant patients may be steroid dependent and stress-dose steroid supplement may be required.

INTRAOPERATIVE

Anesthetic technique

Be prepared for an airway emergency. Preparations should be made for emergency rigid bronchoscopy and/or tracheostomy below the lesion. A variety of laryngoscope blades and ETTs of all sizes, including small, uncuffed (5–6 mm) tubes, should be readily available. Endoscopic evaluation of the airway must be performed in a spontaneously breathing patient if the airway lesion has not been defined. Muscle relaxation must be avoided until a detailed examination is completed and the operator is certain an airway can be maintained subsequently. Vocal cord paralysis, which can mimic tracheal

stenosis, can be obscured by muscle relaxation.

Topical anesthesia	Anesthetize palate, pharynx, larynx, vocal cords, and trachea with lidocaine (2–4%), using atomizer (e.g., MADgic™ Atomizer) or having the patient gargle viscous lidocaine (4%). Avoid local anesthetic overdose.
Conscious sedation	Midazolam will increase the seizure threshold in the setting of relative local anesthetic overdose. Other options include dexmedetomidine (0.2–0.7 mcg/kg/h after loading dose) and propofol (10–100 mcg/kg/min), with or without a short-acting opioid (e.g., fentanyl 25–50 mcg; remifentanyl 0.05–0.1 mcg/kg/min). Caution should be exercised when mixing opioids and sedatives to avoid respiratory depression.
Induction	Patients should be well preoxygenated. Patients may be unable to lie flat 2° respiratory distress. Low-density helium-oxygen mixtures (Heliox) reduce airflow resistance past the obstruction, which may be beneficial in ventilating the patient. Ideally, spontaneous ventilation should be maintained until the airway is secured. Inhaled sevoflurane, intravenous ketamine, judicious propofol (without opioid), or awake FOB with topical anesthesia is appropriate. Optimally, the ETT is positioned ~1 cm above the lesion. Flexible bronchoscopy of the distal airways is then performed through the stricture, and the distal extent of the stricture relative to the carina is identified. In patients with a stricture near the vocal cords, the use of an LMA is a viable option, as it avoids passing an airway through the lesion and offers the ability to bronchoscopically examine the entire lesion. Avoid muscle relaxation if possible; if necessary, consider small dose of succinylcholine (or rocuronium → sugammadex). Distal or mild airway stenoses may be appropriately managed with an RSI.
Maintenance	Inhalational anesthesia is unreliable in most cases due to the limited ventilation with a bronchoscope in situ (resistance and leak). TIVA is preferable: use propofol

	(50–100 mcg/kg/min) and remifentanil (0.05–0.2 mcg/kg/min) infusions. Once the airway is secured, muscle relaxation can be given to avoid movement or coughing during the procedure. Succinylcholine drip (0.25–1 g/250 mL NS, titrated to twitch suppression; avoid phase II block by keeping dose <5–6 mg/kg) or intermediate-acting (cisatracurium 0.1 mg iv, vecuronium 0.1 mg/kg iv, rocuronium 0.15–0.3 mg/kg iv) muscle relaxant may be used. Manual IPPV through sidearm of rigid bronchoscope or via the swivel connector if fiber-optic bronchoscope is used. High-flow (up to 20 L/min) O₂ or O₂ flush (barotrauma risk) may be required to compensate for leak if rigid bronchoscope is used. Hyperventilate patient in preparation for periods of apnea. Low-frequency jet ventilation (50 psi at 10/min with I:E 1:2–4) via a Sander injector is an alternative. Barotrauma and dynamic lung hyperinflation are potential problems with this approach.	
Emergence	Patient must be fully awake before extubation with no residual neuromuscular blockade. Emergence can be “stormy.” Patient may cough violently to clear secretions and blood. Wake-up on low-dose (e.g., 0.5 mcg/kg/min) remifentanil infusion tends to be smoother; consider airway suctioning and lidocaine (1 mg/kg iv) to decrease airway reactivity. Postop O₂ supplementation (preferably humidified).	
Blood and fluid requirements	IV: 18 ga × 1 BSS @ 1–2 mL/kg/h	Transfusion unnecessary except to optimize O₂-carrying capacity or to treat hemorrhage.
Monitoring	Standard monitors ± Arterial line	ETCO₂ not accurate during rigid bronchoscopy Art line depending on patient’s comorbidities.
Positioning	Supine ✓ and pad pressure points ✓ eyes	

Complications	Hypoxemia	Suction secretions/blood; airway instrumentation may have to be interrupted and bronchoscope removed to improve oxygenation and ventilation.
	Hypercarbia	Commonly 2° inadequate ventilation. May cause increased sympathetic drive with 2° hypertension/tachycardia/dysrhythmias. Simply ↑ TV and RR.
	Hypertension	Ensure adequate ventilation and sedative/anesthesia.
	Bronchospasm	Rx: bronchodilator (e.g., albuterol puff)
	Stent dislodgement	From coughing or movement. Ensure adequate anesthesia and muscle relaxation.
	Bleeding Tracheobronchial injury Aspiration of debris	More likely with rigid bronchoscope Requires frequent suctioning. Major hemorrhage may require thoracotomy using DLT or BB to isolate and/or tamponade bleeding site. Transfusion. Patient may need to be kept intubated after the procedure.

▼ POSTOPERATIVE

Complications	Airway edema	Rx with corticosteroids (dexamethasone 10 mg iv) and/or racemic epinephrine nebulizer. Impending airway obstruction will require reintubation.
	Airway obstruction	May be catastrophic. Secretion retention or stent dislodgement. Rigorous suction. Reintubation or restenting

	Stent fracture/migration	May be catastrophic. May require emergent reintubation or restenting
	Pneumothorax	Obtain CXR. May require chest tube placement if >20% or enlarging
	Secretion retention	Humidified O ₂
Pain management	None	

Suggested Readings

1. Doyle DJ, Abdelmalak B, Machuzak M, et al: Anesthesia and airway management for removing pulmonary self-expanding metallic stents. *J Clin Anesth* 2009; 21(7):529–32.
2. Flannery A, Daneshvar C, Dutau H, et al: The art of rigid bronchoscopy and airway stenting. *Clin Chest Med* 2018; 39(1):149–67.
3. Folch E, Keyes C: Airway stents. *Ann Cardiothorac Surg* 2018; 7(2):273–83.
4. Jose RJ, Shaefi S, Navani N: Anesthesia for bronchoscopy. *Curr Opin Anaesthesiol* 2014; 27:453–7.
5. Tjahjono R, Chin RY, Flynn P: Tracheobronchial stents in palliative care: a case series and literature review. *BMJ Support Palliat Care* 2018; 8(3):335–9.
6. Wilkey BJ, Alfilie P, Weitzel NS, et al: Anesthesia for tracheobronchial surgery. *Semin Cardiothorac Vasc Anesth* 2012; 16(4):209–19.

THERMAL ABLATION OF TUMORS AND OTHER TARGETS

SURGICAL CONSIDERATIONS

Description

Percutaneous thermal ablation is a minimally invasive procedure often used for treatment of liver, kidney, lung, and bone tumors. It is a treatment option for nonsurgical candidates either in conjunction with or as a standalone therapy to radiation and/or chemotherapy.

The most commonly used thermal ablation platforms are radiofrequency ablation (RFA), microwave ablation (MWA), high-intensity–focused ultrasound (HIFU), and

cryoablation. Irreversible electroporation (IRE) is a nonthermal ablation modality which is reviewed separately. With the exception of HIFU, these thermal ablation systems consist of a generator and applicators that are placed into the target tumor percutaneously through small incisions using imaging guidance (US, CT, or MRI) to deliver cytotoxic thermal energy directly into the target lesion. In contrast, HIFU employs MR guidance to deliver focused US energy from an external source to the target lesion without an incision. MWA, RFA, and HIFU utilize tissue heating to $>60^{\circ}\text{C}$ to achieve lethal coagulation, while cryoablation utilizes tissue freezing to less than -40°C to cause cell death.

Mechanism of tissue destruction

In general, temperatures less than -40°C or more than 60°C cause complete necrosis in most tissues, although cell type and local environment can affect these temperature thresholds. RFA is based on the interaction of high-frequency alternating electric currents (460–500 kHz) within living tissue to generate frictional heat. MWA uses dielectric hysteresis with a rapidly oscillating electromagnetic field, typically in the range of 900–2450 MHz, to force polar molecules, primarily water, to continuously realign in order to generate heat, rendering it more useful in tissues with poor electrical conductance. HIFU uses an acoustic lens to concentrate multiple intersecting beams of US onto a focal target, achieving high acoustic intensities to generate heat.

Heat-based ablation aims to cause acute coagulative necrosis within the target tissues. Initial thermal injury involves the expression of heat shock response with increased thermal resistance in tissues that survive the primary insult. As temperatures increase within the target tissue, irreversible damage occurs with microvascular thrombosis, ischemia, and hypoxia. When temperatures reach more than 60°C , proteins denature and plasma membranes melt, leading to nearly instantaneous cell death.

To achieve rapid cooling, cryoablation devices employ the Joule-Thomson effect, whereby gas, typically argon, cools when rapidly expanded at room temperature after being forced through a narrow opening in the distal end of a cryoprobe. Freezing-based ablations destroy cells at temperatures less than -40°C through ice crystal formation and osmotic shock. Initially, as tissues are cooled, cellular metabolism slows down and ice begins to form outside of the cell, leading to a hyperosmotic extracellular space and cell dehydration. Subsequent thawing can cause fluid influx into the cell with swelling and membrane rupture. When there is rapid cooling, ice crystals can form within the cell, expanding the cell to cause irreversible cell membrane damage.

Techniques of percutaneous thermal ablation

For applicator-based systems, probe number and positioning are often determined based

on preprocedural and periprocedural imaging and manufacturer recommendations. Most commonly, US and/or CT are used to guide probe placement into the target lesion. Some centers use MRI for imaging guidance. The generator or device is activated and output is gradually increased to a predetermined maximum. The treatment time is also programmed into the unit and monitored throughout the procedure. The operator can adjust the ablation time and power during the procedure based on the size of the ablation zone. For cryoablation, two alternating freeze-thaw cycles are typically used to achieve successful tumor necrosis. RFA and MWA ablation zones can be monitored using CT and/or US with varying degrees of success. In contrast, cryoablation allows for direct visualization of a low-density ablation “ice ball” by noncontrast CT.

RFA requires the use of grounding pads typically placed on the patient’s thighs. If a single RFA probe is used, at least two and, preferably, four pads (96 cm² surface area each) or the equivalent (minimum total surface area ~200 cm²) must be used. If a three-probe cluster is used, at least four grounding pads or their equivalent (~400 cm² surface area) must be used. The grounding pads and the treatment probes are connected to the RF generator. MWA does not require grounding pads. Furthermore, newer-generation MWA devices may be more effective for treatment of larger (>3 cm) tumors compared to RFA.

Treatment volume varies according to the size and number of ablation probes, the modality, duration of treatment, and local blood flow. For heat-based ablations, tumor cells adjacent to large blood vessels may not be treated thoroughly due to the heat-sink effect of flowing blood, which carries away the thermal energy once deposited. It is important to note that MWA is less susceptible to heat-sink effects. Cryoablation is thought to be sparing of sensitive adjacent structures, such as nerves, and would be useful in areas such as the chest or abdominal wall as compared to heat-based therapies. Additionally, temperature-sensitive structures (e.g., bowel) near the planned ablation zone can be displaced using hydrodissection, pneumodissection, and/or percutaneous balloon catheter placement.

Liver ablation

The most well-known and best-studied application of RF is in the treatment of primary and metastatic liver tumors. RF has been used widely to treat hepatocellular carcinomas (HCC) as well as liver metastases. Prospective and retrospective studies have demonstrated that complete necrosis can be achieved in 90% of tumors measuring <3 cm in fewer than two sessions. However, the success rate drops markedly in tumors measuring >3.5 cm, with complete necrosis achieved in <50% at two sessions. MWA is increasingly being favored over RFA in this application given shorter ablation times,

higher efficacy in the treatment of larger lesions (>3 cm), and decreased susceptibility to heat-sink effects. Cryoablation is avoided in the liver given its association with increased bleeding risk and a DIC-like reaction termed “cryoshock.”

Renal ablation

Renal tumors smaller than 3 cm in diameter can be ablated successfully with both RFA and cryoablation. Emerging data on MWA for renal tumors are promising. For central renal tumors, cryoablation is favored because of decreased risk for injury to the collecting system and ureter. Overall, the efficacy for percutaneous ablation for small renal tumors is reported to be >90% with a complication rate of ~7%.

Lung ablation

Percutaneous RFA, MWA, and cryoablation in patients with primary lung cancer or lung metastases demonstrate low rates of serious complications. Pneumothorax is seen in ~30% of cases (similar to rates reported for lung biopsies). Efficacy of RFA of lung tumors has been limited due to the inherent poor conduction of electric current and heat in aerated lung. MWA is not susceptible to these limitations and is thus able to create predictable and larger ablation zones in lung. Cryoablation can effectively ablate tumors in the lung and has the advantage of being relatively resistant to the cold-sink effects of ventilation compared with heat-based ablations. Another advantage of cryoablation is seen with tumors in close proximity to the mediastinum or chest wall as it is less likely to cause thermal damage to these structures.

Bone ablation

Percutaneous thermal ablation has been applied to both benign and malignant painful bone tumors. The first use was for thermocoagulation of small, painful osteoid osteomas. The pain caused by osteoid osteoma is related to prostaglandin production by the cells of the small, central vascularized tumor nidus. Pain relief is obtained only after complete removal of the nidus, surgically or percutaneously. RFA has emerged as the method of choice for treating osteoid osteoma. Patients typically have 1–2 d of postprocedure pain that differs from the pain caused by osteoid osteoma. Relief of the osteoid osteoma pain almost always occurs within the first 24–48 h. Cure with one treatment session can be as high as 90%, and retreatment can be performed if the first treatment is not immediately successful or if the patient’s pain recurs. MWA may have a role in the future as antennae with shorter ablation zones become available.

Metastatic disease to the bone can be associated with intractable pain. Preliminary studies of percutaneous thermal ablation of painful metastatic bone lesions for pain relief show promise. The results of these trials suggest that although both heat-based

ablation and cryoablation can be effective, cryoablation is associated with more rapid and durable pain relief.

Prostate ablation

Transrectal ablation of localized prostate cancer using HIFU has shown promise as a non-invasive treatment modality. Although still a subject of clinical trials, it is FDA and Medicare approved. The procedure uses MR and US image guidance for targeting accuracy. Procedural discomfort often requires GA.

Parkinson disease and essential tremor treatment

Transcranial HIFU has been used to mitigate the signs and symptoms of tremor-dominant Parkinson disease and essential tremor. This is accomplished by ablating specific targets in the thalamus, globus pallidus or adjacent areas. Studies suggest that this technique can be as effective as invasive procedures including deep brain stimulation. Most patients will not require the services of an anesthesiologist.

Usual preop diagnosis

Primary or metastatic hepatic, pulmonary and renal malignancies; osteoid osteoma; prostate cancer; Parkinson disease.

■ SUMMARY OF PROCEDURE

Position	Supine, prone, or lateral decubitus
Incision	Over the site to be accessed
Special instrumentation	RFA probes and generator microwave generators and probes; ultrasound probes
Unique considerations	Mild ↑ or ↓ T during procedure; RFA and MWA can be painful, while cryoablation is not.
Antibiotics	Cefazolin 2 g; occasionally ciprofloxacin 500 mg iv
Procedure time	2–4 h
EBL	None
Postop care	Step-down unit
Mortality	<5%
Morbidity	Postprocedure pain at site Hepatic hemorrhage: Rare Pneumothorax (during hepatic dome lesion ablation,

pulmonary ablation)
Hepatic abscess (↑ incidence in patients with previous biliary manipulation)

Pain score 5–8 (first few hours only)

■ PATIENT POPULATION CHARACTERISTICS

Age range	Adults
Male:Female	M > F
Incidence	Uncommon
Etiology	Pain; metastatic cancer; primary malignancies; neurodegenerative disorder
Associated conditions	metastatic cancer; chronic pain

ANESTHETIC CONSIDERATIONS

Patients presenting for percutaneous thermal ablation range from those with end-stage lung cancer to otherwise healthy chronic pain patients. In addition, analgesic/anesthetic requirements are highly variable, depending on the size and location of the target lesions. Close communication between the various care providers is essential to achieving a positive outcome.

Respiratory	Access to the airway may be limited in the typical interventional radiology suite; thus, a patient with a potentially difficult airway may require elective intubation. Appropriate airway adjuncts (LMA/iGels, light wand, fiberoptic cart, etc.) should be readily available. Intraop risks include the possibility of pneumothorax, so needle decompression supplies also should be available. Lung cancer patients presenting for tumor ablation may have compromised pulmonary function and require objective assessment of pathophysiology with PFTs and/or ABG analysis. Tests: PFTs, ABG, as indicated from H&P
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Cardiovascular	HTN and CHF are seen frequently in elderly patients presenting for these procedures. These patients may have a severe disease process, which may be complicated by other significant comorbidities. The potential for cardiac ischemia should be evaluated carefully. Tests: ECG, echo, noninvasive stress testing as indicated from H&P
Neurologic	A thorough neurologic exam is important to document neurologic deficits that are present before the procedure.
Hepatic	Patients may have coagulation defects and mental status changes 2° systemic liver disease. The potential for severe hemorrhage from vascular injury, the risk of aspiration, and the lack of cooperation (encephalopathy) should all be taken into account when planning the anesthesia. Patients with severe liver disease also may exhibit hepatopulmonary or hepatorenal syndrome manifested as hypoxemia or renal failure. Consider the need for preop correction of coagulation and fluid status. Tests: LFTs; ammonia; INR
Endocrine	The possibility of endocrine and metabolic derangements should be assessed. Lung cancer patients may have electrolyte problems 2° SIADH, whereas patients with liver disease are prone to developing hypoglycemia. Tests: As indicated from H&P
Hematologic	Consider the possibility of chronic anemia or hypercoagulability 2° neoplastic disease. Chemotherapy or systemic disease may have depressed bone marrow function and altered the activity of WBCs and Plts. Tests: Hct, Plt count, or CBC, as indicated by H&P
Premedication	Sedation must be adjusted according to patient requirements. Small doses of midazolam, titrated to minimal sedation, are most reasonable (0.5–1 mg

increments). Patients with end-stage disease often require very little premedication.

◆ INTRAOPERATIVE

Anesthetic technique

Sedation or GA, depending on the size and location of lesions, patient comorbidities, patient positioning, and ablation modality. Most ablations are well tolerated with MAC; however, large lesions may require GA. MWA procedures also tend to require GA secondary to pain during ablation.

MAC

MAC cases are performed with fentanyl (25–150 mcg) and midazolam (0.5–2 mg) boluses, combined with a propofol infusion (25–100 mcg/kg/min). The substitution of remifentanyl (0.02–0.1 mcg/kg/min) for fentanyl allows rapid titration of analgesia for the brief periods of intense stimulation.

Induction	Standard induction, preceding placement of an ETT or LMA	
Maintenance	Standard maintenance. Because procedures are of an unpredictable duration, short-acting muscle relaxants (e.g., rocuronium 0.6 mg/kg) are advised. Propofol (25–100 mcg/kg/min) and remifentanyl (0.02–0.1 mcg/kg/min) are an appropriate combination that easily can be titrated to effect. Persistent intraop HTN may be treated with labetalol (5–25 mg) or hydralazine (5–20 mg).	
Emergence	Standard emergence. Short-acting analgesics (e.g., fentanyl 25–150 mcg) should be used in outpatients. Prophylactic treatment with antiemetics (ondansetron 4 mg iv) is beneficial because there is a high incidence of PONV.	
Blood and fluid requirements	IV: 20–14 ga × 1–2 BSS: 2–4 mL/kg/h	Larger access needed for hepatic lesions
Monitoring	Standard monitors ± Arterial line	

	± Urinary catheter	Foley for longer procedures (>2 h)
Positioning	✓ and pad pressure points ✓ eyes	Radiology tables often poorly padded. Underbody Bair Hugger™ may be useful for avoiding hypothermia.
Complications	Hemorrhage	Blood or colloid should be readily available, if the need is anticipated. Large retroperitoneal hemorrhage can develop, yet not be appreciated until blood loss is extensive.
	Pneumothorax Hemothorax	Needle decompression and/or thoracostomy tube placement may be necessary during thoracic or high hepatic procedures.
	Electrical shock Thermal injury	RFA can produce serious injury if not properly grounded.
	Hyperthermia	Patients may develop rapid T increases 2° direct heating of large lesions.

POSTOPERATIVE

Complications	PONV Hemorrhage	Common Occult blood loss may continue for several h after the procedure is terminated.
Pain management	Multimodal pain management	Postprocedure pain is usually well tolerated.
Tests	Postprocedure HCT, as indicated	

Suggested Readings

1. Ahmed M, Brace C, Lee FT Jr, et al: Principles of and advances in percutaneous ablation. *Radiology* 2011; 258(2):351–69.
2. Carrafiello G, Mangini M, Fontana F, et al: Complications of microwave and radiofrequency lung ablation: personal experience and review of the literature. *Radiol Med* 2012; 117(2):201–13.
3. Hinshaw JL, Lubner MG, Ziemlewicz TF, et al: Percutaneous tumor ablation tools microwave, radiofrequency, or cryoablation—what should you use and why? *Radiographics* 2014; 35:1344–62.
4. Kang TW, Rhim H: Recent advances in tumor ablation for hepatocellular carcinoma. *Liver Cancer* 2015; 4(3):176–87.
5. Kwon HJ, Kim PN, Byun JH, et al: Various complications of percutaneous radiofrequency ablation for hepatic tumors: radiologic findings and technical tips. *Acta Radiol* 2014; 55(9):1082–92.
6. Lencioni R, Crocetti L: Image-guided ablation for hepatocellular carcinoma. *Recent Results Cancer Res* 2013; 190:181–94.
7. Moosa S, Martínez-Fernández R, Elias WJ, et al: The role of high-intensity focused ultrasound as a symptomatic treatment for Parkinson's disease. *Mov Disord* 2019 Jul 10. [Epub ahead of print]
8. Wah TM: Image-guided ablation of renal cell carcinoma. *Clin Radiol* 2017; 72(8):636–44.
9. Woodrum DA, Gorny KR, Mynderse LA: MR-Guided Prostate Interventions. *Top Magn Reson Imaging* 2018; 27(3):141–151.

IRREVERSIBLE ELECTROPORATION (IRE ABLATION)

SURGICAL CONSIDERATIONS

Description

Electroporation is a method of exposing cells to ultrashort high-voltage electrical pulses to create small pores in the cell membrane. In reversible electroporation, the induced cell membrane porosity is temporary and cells are able to recover. This transient increase in cell permeability has been used in research for delivery of DNA, RNA, and macromolecules, such as antibodies, drugs, and tracers into cells. With the application of higher electrical voltages, permanent pores are created, resulting in disruption of cell homeostasis and apoptotic cell death. This process is known as **irreversible**

electroporation (IRE) and has emerged as a minimally invasive tissue ablation method. IRE presents several advantages compared to the traditional temperature-based ablation methods (i.e., RF or MWA and cryoablation). Because IRE uses nonthermal energy, it avoids undesirable thermal damage to surrounding tissues. IRE is able to induce cell death in a selective, well-defined manner and has been shown to preserve extracellular structures and connective tissue scaffolds, including large blood vessels and bile ducts. It is also not susceptible to perfusion-mediated heat-sink or cold-sink effects that can lead to suboptimal treatments when using temperature-based ablation methods. As such, IRE may be better suited for ablation of tumors located in close proximity to major portal or hepatic veins and arteries, bile ducts, or bowel.

Mechanism of tissue destruction

IRE utilizes high-voltage pulses ranging from 1000 to 3000 V/cm to create an electric field across the tissue between electrodes placed by the operator. Therefore, the shape and size of the IRE ablation field can be manipulated to include the entire area to be ablated. The intensity of the electric field within a given tissue is dictated by the distance between the electrodes and the voltage difference across the electrodes. The creation of permanent nanopores in the cell membrane irreversibly impairs the cell's ability to maintain homeostasis, resulting in cell death via apoptosis. Apoptosis is a natural, tightly regulated process in which cells are phagocytized by immune cells and then replaced by innate cellular regeneration. This process results in minimal damage to surrounding tissues with minimal fibrosis and scarring. Although the major mechanism of tissue ablation in IRE is electroporation, thermal tissue damage is possible if excessive electric fields are applied. Every electrical field produces thermal energy, which is referred to as the Joule effect. If the parameters of IRE are not chosen correctly, the electrical field generated can cause thermal damage, negating the nonthermal advantages of this therapy.

Techniques of irreversible electroporation

Careful planning is necessary before beginning IRE treatment to determine the proper placement and number of electrodes that will achieve the desired outcome. The commercially available NanoKnife™ IRE system (AngioDynamics; Latham, NY) has treatment planning software program that helps determine the number and placement of electrodes needed for the desired ablation zone. The voltage setting is determined by the distance between each pair of electrodes with the purpose of generating at least 1000 V/cm between them. The number of electrodes used depends on the site and volume of the target area and ease of electrode access. Current flow is determined by tissue resistance and degree of IRE achieved.

With the patient under GA, electrodes are placed in a predetermined pattern in the target area using US and/or CT guidance. To minimize the risk of cardiac dysrhythmias, the patient's electrocardiogram is synchronized to the IRE device such that pulses are delivered during the refractory period of the cardiac cycle. Typically, 70 brief pulses of high-voltage (1500–3000 V) direct current between 20 and 50 A are delivered in groups of 10, at intervals of 70 ms (bipolar) or 100 ms (unipolar). At 2-cm probe spacing and 3000 V delivered, the maximum effective treatment area is 2 cm diameter from each probe (maximum of 4 cm total ablation diameter).

Pancreas IRE: IRE has been shown to be successful in managing surgically unresectable pancreatic tumors encasing the celiac and superior mesenteric vessels.

Liver IRE: Clinical IRE was first attempted in patients diagnosed with HCC with promising clinical results. Liver function tests were normal at 24 h and at 1, 3, and 6 mo postop. There was, however, an expected increase in alanine aminotransferase, a marker for acute hepatocyte damage. Most patients with increased levels will return to baseline levels after 1 mo.

Prostate IRE: IRE has been shown to be a safe and technically feasible primary treatment for localized prostate cancer in carefully selected men.

Usual preop diagnosis

Most centers perform IRE on primary or metastatic malignancies in the liver, kidney, lung, prostate, and pancreas.

■ SUMMARY OF PROCEDURE	
Position	Supine or prone
Incision	Over the site to be accessed determined by preop CT, MRI, or US
Special instrumentation	AngioDynamics IRE electrodes and NanoKnife IRE generator
Unique considerations	Stimulation of skeletal muscles and diaphragm (minimized by the use of muscle relaxants) ↑ cardiac dysrhythmias ↓ T during lengthy procedures in CT room
Antibiotics	Cefazolin 2 g iv; >120 kg = 3 g iv, redose in 4 h
Procedure time	1–2 h

EBL	Usually minimal (although large-volume blood loss may occur during ablation of liver tumors)
Postop care	PACU then step-down posttreatment unit
Mortality	<5%
Morbidity	Post-procedure pain at site Brachial plexus injury (during positioning for liver ablation) Pneumothorax (related to lung ablation, segment VIII, or dome liver ablation) Subcapsular hematoma (related to multiple needle punctures) Intraperitoneal hematoma (related to pancreas ablation)
Pain score	3–8 (first few hours only)

■ PATIENT POPULATION CHARACTERISTICS

Age range	Adults
Male:Female	50:50
Incidence	Uncommon
Etiology	Primary and metastatic tumors
Associated conditions	Liver, lung, kidney, pancreas, prostate cancer

■ ANESTHETIC CONSIDERATIONS

GA is mandatory for IRE procedures. Because the high-voltage electrical pulses can cause intense whole body muscle contractions, complete neuromuscular blockade is required. To minimize the risk of inducing cardiac dysrhythmias, treatment is performed with cardiac synchronization. Defibrillator pads are placed prior to initiation of treatment. The electrocardiogram should be closely monitored.

Respiratory	Lung cancer patients presenting for tumor ablation may have compromised pulmonary function and require objective assessment with PFTs and/or ABG analysis before the procedure. Access to the airway may be limited in the CT/MRI scanning suite. Intraop
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	risks include the possibility of pneumothorax. Tests: PFTs, ABG, as indicated from H&P
Cardiovascular	Patients with underlying cardiac disease are especially susceptible to IRE-induced cardiac dysrhythmias. History of cardiac arrhythmias is a contraindication for IRE. A defibrillator should be available in the procedure room. The potential for myocardial ischemia should be evaluated carefully. Defibrillator pads are preferably placed on the patient prior to initiation of treatment. A cardiac anesthesiologist is not usually required. Tests: ECG, echo, noninvasive stress testing as indicated from H&P
Neurologic	A thorough neurologic exam is important to document preexisting neurologic deficits.
Hepatic	Patients may have coagulation defects and mental status changes related to liver disease. Patients with severe liver disease may exhibit a hepatopulmonary or hepatorenal syndrome manifested as hypoxia or renal failure. Consider the need for preop correction of coagulation and fluid status. Tests: LFTs; ammonia; INR
Endocrine	The possibility of endocrine and metabolic derangements should be assessed. Lung cancer patients may have electrolyte problems 2° SIADH, while patients with liver disease are prone to developing hypoglycemia. Tests: As indicated by H&P
Hematologic	Consider the possibility of chronic anemia or hypercoagulability 2° neoplastic disease. Chemotherapy or systemic disease may have depressed bone marrow function and altered the activity of WBCs and platelets.
Premedication	Consider midazolam 0.5–2 mg iv. Patients with end-stage disease often require very little premedication.

◆ INTRAOPERATIVE

Anesthetic technique

GETA with muscle relaxation is usually required to minimize any involuntary movement caused by IRE ablation. Some small-volume ablations may be tolerated with MAC, and anesthetic requirements should be discussed with the radiologist.

Induction	Standard induction	
Maintenance	Standard maintenance. Procedure duration may be unpredictable. It is important that adequate neuromuscular blockade is maintained due to intense direct muscle stimulation that may be caused by IRE treatment.	
Emergence	Standard emergence. Short-acting analgesics (e.g., fentanyl 25–150 mcg) should be used in outpatients. Prophylactic treatment with antiemetics may be beneficial because there is a high incidence of PONV.	
Blood and fluid requirements	IV: 20–14 ga × 1–2 BSS: 3–5 mL/kg/h	Larger iv access may be needed for procedures with a high risk for, e.g., hepatic lesions.
Monitoring	Standard monitors ± Arterial line ± Foley catheter	↑ BP (20–30 mm Hg) is common following treatment cycles but may be extreme during treatment of upper pole renal tumors (inadvertent adrenal stimulation). Arterial line may be useful for ablation procedures near the heart, for large-volume ablations, and in patients with renal impairment.
Positioning	✓ and pad pressure points ✓ eyes	Extension of the arms above the head may be required for optimal imaging; if prolonged, this may result in

		a brachial plexus injury, particularly when the patient is in a supine position.
Complications	Hemorrhage	Blood should be readily available for high hemorrhage risk procedures. Large retroperitoneal hemorrhage may develop.
	Pneumothorax	Needle decompression and/or thoracostomy tube placement may be necessary during thoracic or high hepatic procedures.
	Muscle stimulation	Intense stimulation of skeletal muscles and the diaphragm may be produced. Movement should be minimized by use of muscle relaxants. Neuromuscular monitoring is useful to maintain a constant level of neuromuscular blockade.
	Dysrhythmias	Application of large currents may cause dysrhythmias. Occasionally, the treatment pulses may capture the ventricles resulting in a short run of ventricular tachycardia, which should resolve at the end of each 10 pulse treatment cycle. Although rarely required, a defibrillator/external pacer should be available.
	Acid-base and electrolyte disturbances	Patients may develop ↑ K during ablation of very large tumors or in the presence of

Hypothermia

preexisting renal impairment. Monitor arterial blood electrolytes during the procedure in those patients. Patients may experience ↓ body temperature during lengthy procedures in the scanning room. Underbody hot-air blankets may be used.

POSTOPERATIVE

Complications	Complications PONV Hemorrhage	Common Occult blood loss may continue for several hours after the procedure has terminated.
Pain management	Standard pain management	Postprocedural pain is usually mild and well tolerated.
Tests	HCT, as indicated	

Suggested Readings

1. Cannon R, Ellis S, Hayes D, et al: Safety and early efficacy of irreversible electroporation for hepatic tumors in proximity to vital structures. *J Surg Oncol* 2013; 107(5):544–9.
2. Distelmaier M, Barabasch A, Heil P, et al: Midterm safety and efficacy of irreversible electroporation of malignant liver tumors located close to major portal or hepatic veins. *Radiology* 2017; 285(3):1023–31.
3. Lee EW, Chen C, Prieto VE, et al: Advanced hepatic ablation technique for creating complete cell death: irreversible electroporation. *Radiology* 2010; 255(2):426–33.
4. Martin RC, Schwartz E, Adams J, et al: Intraoperative anesthesia management in patients undergoing surgical irreversible electroporation of the pancreas, liver, kidney, and retroperitoneal tumors. *Anesth Pain Med* 2015; 5(3):e22786.
5. Wagstaff PG, Buijs M, van den Bos W, et al: Irreversible electroporation: state of the art. *Onco Targets Ther* 2016; 9:2437–46.

Suggested Viewing

Links are available online to the following videos:

TIPSS, Transjugular Intrahepatic Porto-Systemic Shunt: https://www.youtube.com/watch?v=O2u4_hF3234

ONYX embolization of AVM (skip to minute 5 for the animation): <https://www.youtube.com/watch?v=gYTVA3PoeY8>

AVM Embolization Using Onyx-18: <https://www.youtube.com/watch?v=057r1Zo1pj4>

Endovascular Coiling of Unruptured Ophthalmic Artery Aneurysm: <https://www.youtube.com/watch?v=waEtv93XnTU>

ICD Implantable Cardioverter Defibrillator: https://www.youtube.com/watch?v=F_E06rkY4T8

NEJM Procedure: Thrombus Aspiration: https://www.youtube.com/watch?v=_bUMm8E0w6s

AERO Fully Covered Tracheobronchial Stent Video: https://www.youtube.com/watch?v=0IzNZ4_0rNg

CHAPTER 13.2

Non–Operating Room Procedures— Pediatric

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PEDIATRIC RADIATION THERAPY

PROCEDURAL CONSIDERATIONS

Description

Modern pediatric radiation therapy (XRT) requires that the patient be in a stable and reproducible position for daily treatment. Sharply defined beams with secondary collimation are used to irradiate the tumor volume and to spare normal tissue. Patient movement may undermine techniques for sparing normal tissue, and although movement cannot be completely prevented, it must be minimized. In very young children, it is often impossible to prevent movement and achieve adequate cooperation for radiation treatment. In such cases, daily anesthesia is required. Close cooperation of the radiation oncology and anesthesia teams allows for safe and reproducible daily treatment. In general, children older than 3 or 4 yr can be persuaded to lie still for radiation therapy. Children from 2.5 to 4 yr may cooperate during the treatment (which is usually <15 min), but not for the treatment planning and simulation, in which an immobilization-stabilization device is made (often requiring 1–1.5 h). In most infants and young children (<4 yr), anesthesia is essential.

The optimal position for XRT must also be optimal for the anesthesiologist. Ideally, the area to be treated is determined using 3-dimensional conformal techniques to optimize treatment and to minimize normal tissue exposure. This requires a cross-sectional imaging study (e.g., CT scan), with the patient in the identical position as will be used during the radiation treatment. A series of radiographs are taken at the treatment-planning (simulation) appointment, which typically lasts 1–1.5 h and requires GA. It is essential that there be no patient movement between exposures; if the patient moves, the entire procedure must be repeated. Typically, the patient is transported to the CT suite for a 3-dimensional treatment-planning CT scan. Anesthesia preparation may be initiated in the treatment-planning suite or in a nearby preparation room. After scans are obtained, individual beam-shaping devices are made, and a multistep process of contouring begins for treatment planning.

Seven to 10 d following the initial planning session, the patient often has a verification procedure, which is usually of shorter duration—generally requiring only 30 min of anesthesia time. The verification procedure consists of a series of radiographs using the beam-shaping devices, which simulate the treatment to be given. When this procedure is successfully completed, the anesthetized patient is moved to the treatment room. The child is put in the identical position achieved during the planning/verification procedures, and treatment is administered.

The first day or two, and weekly thereafter, a verification x-ray (called a “port film”) is taken to confirm the accuracy of the treatment field. The treatment itself is only a few

minutes in duration for each field; ideally, the entire procedure is completed within 15–30 min. A form of radiotherapy treatment planning and delivery known as intensity-modulated radiation therapy (IMRT) may require slightly longer treatment times due to the larger number and increased complexity of fields treated, while volumetric modulated arc therapy (VMAT) may be of shorter treatment time. A course of treatment may be only a few days or may last for 5–6 wk, generally with treatment given 5 × per week. Occasionally, multiple (2–3) treatments per day are given at 4–8 h (usually 6 h) intervals. At the initial appointment, the patient's optimal position is determined, an immobilization device is constructed, and measurements are taken. The immobilization device is usually a body cradle or cast, and often a head/face mask is made for head and neck or brain treatment. Initially, temporary marks or Band-Aids® are used; however, when the final positioning has been determined, a more permanent mark, such as a tattoo, may be applied. Often, a head holder with a mask is applied to ensure the position for XRT.

In managing certain brain tumors (e.g., medulloblastoma, high-grade intratentorial ependymoma, germ cell tumors, and CNS leukemia), **cranial spinal irradiation (CSI)** may be used. Conventionally, this procedure requires that the patient be placed in the prone position with the head flexed as much as possible to minimize a cervical lordosis. This positioning, however, creates special difficulties for the anesthesiologist. If the child is intubated for the setup, the radiation stabilization device must allow space for the ETT. If the child is not intubated, there must be adequate access to the airway. Current techniques allow patients to be treated with CSI in the supine position facilitating easier airway access for the anesthesiologist, more secure patient immobilization, and faster treatment times.

Fractionation

Pediatric protocols have been testing the efficacy of giving multiple fractions (treatments) of radiation 2–3 × per day, usually at 6-h intervals, to allow higher total radiation doses to be administered with possible less normal tissue morbidity. These schemes have been or are being evaluated for children with central nervous system tumors and total body irradiation (TBI) in preparation for bone marrow transplantation. Until proven to be of increased efficacy, such schemes should remain part of large protocol studies. The timing of radiotherapy may be at 4-, 6-, or 8-h intervals 2–3 × per day, depending on the protocol. These studies provide several challenges for anesthesiologists, radiotherapists, and parents. Radiotherapy under anesthesia, however, has been successfully administered to infants undergoing multiple fractions per day. Attention must be given to potential malnutrition and/or dehydration from prolonged periods of NPO status.

Total body irradiation

Although most TBI techniques are administered with the patient standing, infants and small children must lie prone and supine for the treatment. This positioning requires sedation and/or anesthesia. Gastrointestinal upset, sometimes provoked by the radiation, presents an additional challenge for proper radiotherapy technique, as well as for anesthetic management.

Cranial or body radiation/radiosurgery/IGRT

The technique of using precision, stereotactically localized radiation with a highly collimated radiotherapy photon beam, as generated from a linear accelerator, is currently being employed for select patients with small cranial, base-of-skull, or body tumors. There is increasing enthusiasm for this technique for infants and children with recurrent posterior fossa and cerebral tumors, craniopharyngiomas, optic nerve and chiasmal gliomas, and small AVMs. Radiosurgery has historically required a frame-based technique that requires 6–10 h of continuous anesthesia while a patient undergoes application of a metal frame, CT localization, and multiport radiotherapy treatment. However, newer technology, using image guidance, called image-guided radiation therapy (IGRT) now allows frameless radiation delivery to the cranium or body. This technique requires a 1–1.5-h treatment planning and simulation session followed by a single fraction or limited number of treatment sessions each lasting 1.5–2 h. These approaches require close coordination between the anesthesiologist, neurosurgeon, oncologic surgeon, and radiotherapist.

IMRT/protons/VMAT

Recent developments in external beam radiotherapy techniques have allowed for increasingly conformal treatment using IMRT or proton therapy. However, these complex techniques generally involve image-guided therapy and a greater number of beam directions, thereby resulting in longer overall treatment times. The development of VMAT, in which dose is delivered using a single gantry arc rotation, has allowed for treatment of very conformal volumes with increased speed as well as delivery of fewer monitor units. VMAT is increasingly used in the treatment of pediatric patients, particularly in those who require anesthesia during treatment.

Usual preop diagnosis

Leukemia; brain tumors (benign or malignant); retinoblastoma; solid tumors of childhood

■ SUMMARY OF PROCEDURES

	Standard XRT/IMRT	TBI	Frameless Radiosurgery
Position	Supine or prone	Supine or prone	Supine or prone
Unique considerations	If prone: head flexed for maximal straightening of the C-spine	May be repeated 2–3 ×/day at 4–6 h intervals	Head extended for stabilization of airway
Anesthesia time	Planning: 30–120 min Treatment: <15 min (IMRT: 15–30 min)	45–60 min <20 min	30–90 min 90–120 min
Postop care	PACU → home	PACU → room	PACU → home

■ PATIENT POPULATION CHARACTERISTICS

Age range	Usually ≤4 yr
Male:Female	1:1
Incidence	NA
Associated conditions	Brain tumors: ↑ ICP is of concern in these patients. Postradiation edema following the first few treatments may further ↑ ICP, with potential for brainstem herniation. Some children with brainstem tumors are particularly difficult to anesthetize, perhaps because of disruption of nerve pathways in those areas of the brainstem that are affected by anesthetics. Neuroblastoma: Neuroblastomas are capable of secreting catecholamines and related substances; hence, there is a potential for paroxysmal HTN during anesthesia induction. In these children, the principles of anesthetic management are similar to those for pheochromocytoma. Retinoblastoma: It is imperative that the patient be properly immobilized with no movement, as even a mm of change, as occurs with a sigh, may cause unnecessary radiation to the radiosensitive lens and

anterior chamber. Optimal anesthesia prevents nystagmus and motion of the head. Even minimal lateral or rotary nystagmus may increase the risk of cataract induction.

ANESTHETIC CONSIDERATIONS

Rebecca E. Claire and Brenda Golianu

PREOPERATIVE

Orientation to the XRT suite, positive reinforcement, and play therapy can reduce the number of children requiring anesthesia for XRT; however, the majority of children ≤ 4 yr will require anesthesia. A detailed preanesthesia visit is essential and is also an opportunity to gain the confidence of both child and parents. The majority of these patients will have received chemotherapy and should be evaluated for side effects. The importance of NPO status needs to be stressed repeatedly, discussing the potential danger of emesis during treatment. Written instructions regarding preop protocols are extremely helpful in this context. For children with cancer, prolonged preop fasting for XRT once or twice daily could severely compromise an already marginal nutritional intake. Infants, children, and adolescents should be encouraged to drink clear liquids until 2 h before treatment. Milk and solid foods should be held for an appropriate time interval (6–8 h depending on age; see [Appendix D: Standard Pediatric Anesthetic Management](#)). [Reassessment before each anesthetic is recommended because the patient's medical status may change during the course of radiation therapy. Some children will have Sx of \$\uparrow\$ ICP,](#) which must be taken into account when designing an anesthetic plan to avoid $\uparrow$ PaCO₂ and other factors that may further $\uparrow$ ICP.

The most common diagnoses are primary CNS tumor (28–33%), retinoblastoma (9–26%), acute leukemia (9–26%), neuroblastoma (2–18%), lymphoma (8%), rhabdomyosarcoma (5–7%), Wilms tumor (5%), and Ewing sarcoma (5%). The total number of treatments can range from 1 to 65 (median 20–24). Many patients require a mold of the head and neck. Children can range in age from infants to adolescents, with a median age of 2.4–3.8 yr.

Respiratory

Careful evaluation for respiratory compromise due to chemotherapy. Patients with Sx of URI (rhinorrhea, cough, fever) are commonly seen during XRT treatment. If Sxs are significant, XRT should be

	delayed, and the patient should be evaluated by primary service. Fortunately, most children can be managed without the use of an ETT, which itself can increase the risk of oxygen desaturation, laryngospasm, and bronchospasm. The benefit of XRT vs. delaying treatment must be balanced against the risks of anesthesia. Tests: As indicated from H&P
Cardiac	Careful evaluation for cardiac compromise due to chemotherapy Tests: As indicated from H&P
Neurologic	Patients with intracranial tumors may have ↑ ICP. Postradiation edema following the first few treatments may further ↑ ICP. Sx includes irritability, HA, N/V, and papilledema. Suspicion of ↑ ICP mandates ETT intubation and controlled ventilation to induce hypocarbia.
Laboratory	Tests as indicated from H&P
Premedication	Usually unnecessary in this patient group. Parents usually present for induction, and majority of patients have some form of central venous access or heplock in situ. If inhalational induction is necessary, midazolam 0.5–0.75 mg/kg po may be helpful.

INTRAPROCEDURE

Anesthetic technique

Provision of anesthesia to children at sites remote from the OR is challenging. During XRT, patients must remain immobile so that the tumor can be reliably irradiated while minimizing damage to uninvolved tissue. Anesthetic goals should include patient immobility, rapid onset, brief duration of action, and prompt recovery. The anesthetic should allow maintenance of a patent airway and spontaneous ventilation in a variety of body positions. Propofol bolus and infusion have been shown to be safe and effective for XRT. Higher doses may be needed due to tolerance with repeated treatments. Ketamine is rarely used, but may be effective if there is a history of reaction to propofol. Ketamine is a potent sialagogue and can cause nystagmus, which may interfere with precision radiation of retinoblastomas. It may cause prolonged or unpleasant

emergence and should be avoided in patients with ↑ ICP. Choice of anesthetic technique may be limited by equipment or logistical issues. Anesthesia machines may not be available in all XRT suites.

Induction	IV: Propofol (2–4 mg/kg), titrated slowly to effect. Additional propofol boluses may be needed as patients may develop tolerance. Intubation is usually not needed. Inhalation: Mask induction with sevoflurane is appropriate in children without iv access and may be preferred by some children. The airway can almost always be maintained by careful positioning and extension of the neck. It is essential that this same degree of extension/flexion be maintained for each treatment. A molded immobilization device may be placed over the patient's head and neck, limiting airway access and potentially causing airway obstruction.	
Maintenance	IV: Propofol (100–250 mcg/kg/min) by continuous infusion. Supplemental O₂ should be administered via nasal prongs. Inhalation: Sevoflurane in O₂	
Emergence	Patients awaken rapidly following cessation of propofol infusion or sevoflurane. XRT may cause nausea/vomiting, so antiemetics should be given (ondansetron 0.1–0.15 mg/kg). Avoid dexamethasone (may conflict with chemotherapy protocols and in rare cases may cause tumor lysis syndrome).	
Blood and fluid requirements	IV: Central access BSS: Routine maintenance	Majority of children receiving radiotherapy have some form of central venous access.
Monitoring	Standard monitors	A critical problem is the lack of access to the patient and monitors during XRT. A video camera with a zoom lens can focus on the monitors, and a second camera is trained on the patient. In XRT suites with a viewing window, a

		small marker may be placed on the chest so that the rise and fall of chest motion are seen easily.
Positioning	✓ and pad pressure points ✓ eyes	Careful positioning 2° possible chemotherapy-induced peripheral neuropathy
Complications	Airway obstruction	Respiratory obstruction may occur; it usually responds to nasal or oral airways and careful repositioning.
	Patient movement	Deepen anesthesia

▼ POSTPROCEDURE

Complications	PONV	Many patients have chemotherapy-induced nausea, which may be exacerbated by anesthesia, XRT, and stress.
	Central line sepsis	Patients may be immunocompromised following chemotherapy. Attention to sterility during access of the central venous line is critical because repeated use by multiple health care providers → ↑ risk of catheter contamination. Aseptic preparation of anesthetic iv medications, especially propofol, is important.
	Cerebral edema	In patients with ↑ ICP, XRT can provoke an acute ↑ ICP,

		with consequent ↑ HA, ↑ N/V, ↓ consciousness, and cardiac arrest. These patients should be monitored × 24-h post-
Pain management	Standard approaches	YRT. Radiation treatments are not associated with pain, but narcotics may be useful in relieving pain associated with neoplastic disease.

Suggested Readings

1. Buehrer S, Immoos S, Frei M, et al: Evaluation of propofol for repeated prolonged deep sedation in children undergoing proton radiation therapy. *Br J Anaesth* 2007; 99(4):556–60.
2. Khumi N, Patel P, Koushik A, et al: Anesthesia practice in pediatric radiation oncology: Mayo clinic Arizona's Experience 2014–2016. *Paediatr Drugs* 2018; 20(1):89–95.
3. McFadyen JG, Pelly N, Orr RJ: Sedation and anesthesia for the pediatric patient undergoing radiation therapy. *Curr Opin Anaesthesiol* 2011; 24(4):433–8.
4. Parker WA, Freeman CR: A simple technique for craniospinal irradiation in the supine position. *Radiother Oncol* 2006; 78(2):217–22.
5. Tsai YL, Tsai SC, Yen SH, et al: Efficacy of therapeutic play for pediatric brain tumor patients during external beam radiotherapy. *Childs Nerv Syst* 2013; 29(7):1123–9.
5. Verna V, Beethe AB, LeRiger M, et al: Anesthesia complications of pediatric radiation therapy. *Pract Radiat Oncol* 2016; 6(3):143–54.

PEDIATRIC CARDIAC CATHETERIZATION AND ELECTROPHYSIOLOGY

PROCEDURAL CONSIDERATIONS

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Cardiac catheterization and electrophysiology testing have evolved over the recent decades from purely diagnostic tools to combined diagnostic and therapeutic

procedures. Although the use of GA varies from institution to institution, higher levels of sedation are required at a minimum for critically ill patients, those requiring complex interventional strategies, small children who must remain totally still, and when TEE is used for image-guided therapy. A thorough review of diagnostic and interventional cardiac catheterization and electrophysiology is not possible in this chapter, and the interested reader is referred to the multiple textbooks available on the subject.

The placement of anesthesia equipment for these procedures must allow for (a) proper positioning of the patient, (b) easy access to the head and neck and/or groin for the physician performing the procedure, and (c) rotation and angulation of the imaging equipment. The goal of sedation in all of these procedures is to provide a nontraumatic, safe environment for the patient. Many patients can be cared for adequately and safely using conscious sedation; however, there is a subset of pediatric patients who may require GA. This group may include patients with complex congenital heart disease, ventricular dysfunction, or airway abnormalities.

It is important to understand that certain anesthetic agents may alter cardiac conduction, making arrhythmia inducibility more difficult. Catecholamine-dependent arrhythmias, such as an automatic atrial tachycardia, may be impossible to induce with the patient under GA. Furthermore, the arrhythmia itself may complicate anesthetic care, by causing sudden decreases in BP due to excessively rapid rates. In patients undergoing radiofrequency ablation (RFA) in areas close to other critical structures of the heart, GA may be necessary to keep the patient motionless during application of energy. During anesthesia, the use of muscle relaxants may mask diaphragmatic/phrenic nerve stimulation during mapping and poses the risk of ablation of structures other than the abnormal pathway. In patients returning for a previous failed ablation, controlled ventilation using low tidal volumes to minimize electrophysiologic catheter movement may be helpful.

VASCULAR ACCESS

The modified **Seldinger technique** of cannulating blood vessels percutaneously is used to establish vascular access for cardiac catheterization. (The femoral and IJ veins are most commonly used for venous access.) Transhepatic access to the IVC has been used safely and successfully in patients without femoral venous access. The femoral artery is most commonly used, although the carotid and axillary arteries may be used for specific procedures or when there is bilateral femoral artery occlusion. In newborns, the umbilical artery and vein may be used. Access may be especially difficult in patients who have undergone multiple previous procedures. In the most severe cases,

reconstructive transcatheter techniques, including **balloon angioplasty** and **stent implantation** to rehabilitate the vessels, have been used to allow future catheter-based diagnostic and therapeutic interventions.

Infiltration of the skin and the subcutaneous tissues with a local anesthetic agent to reduce pain is used when the procedure is being performed under conscious sedation. With GA, infiltration of a local anesthetic agent may be helpful as placement of lines can be stimulating. The use of local anesthetic is helpful in minimizing the amount of GA used to produce a similar depth of anesthesia in patients with depressed myocardial function. Local anesthetic may also be deferred to the end of the procedure to alleviate pain and discomfort at vascular access sites during recovery.

HEMODYNAMIC DATA

O₂ sat measurements are made routinely in the various cardiac chambers, venae cavae, and great vessels. These measurements are used to calculate the systemic flow (CO, Q_s), pulmonary flow (Q_p), the ratio of the pulmonary to systemic flow (Q_p:Q_s), and PVR and SVR.

It is ideal to obtain the data with the patient awake and breathing spontaneously in room air. This is rarely possible in pediatric patients. The use of light anesthesia and sedation during the diagnostic part of the study facilitates acquisition of data in as near-normal state as possible. It is important to recognize and limit effects on intracardiac and intrapulmonary pressures and systemic and pulmonary resistances when the procedure is done under GA with IPPV. Though one could argue that the work of breathing and airway apparatus resistance are less with an LMA compared to an ETT, improper seating of the LMA can influence the pressure waveforms. Thus, for patients with obstructive airway disease, a mask supplemented with an oral airway may be more helpful. At a minimum, and when tolerated, baseline hemodynamic measurements should be performed with an FiO₂ as close to 0.21 as possible. In some cases, additional diagnostic information may be collected to study the effects of O₂, NO, vasodilators or inotropes, exercise and balloon occlusion of intracardiac or extracardiac shunts on the CO, pulmonary flow, and PVR and SVR.

From O₂ sat, dissolved O₂ (PO₂), and Hb measurements, the O₂ content (mL/dL) of the mixed venous blood and systemic arterial blood is used to calculate systemic AV O₂ content difference. Pulmonary AV O₂ content difference is similarly estimated by calculating the O₂ content of pulmonary venous and arterial blood. Systemic (Q_s) and

pulmonary flow (Q_p) can then be derived using the **Fick principle**:

$$\text{Flow (Q) (L/min)} = \frac{\text{O}_2 \text{ consumption (mL/min)}}{\text{AV O}_2 \text{ difference (mL of O}_2\text{/L of blood)}}$$

$$\text{Pulmonary Flow (Q}_p\text{)} = \frac{\text{O}_2 \text{ consumption}}{\text{Pulmonary vein SaO}_2 - \text{Pulmonary artery SaO}_2}$$

$$\text{Systemic Flow (Q}_s\text{)} = \frac{\text{O}_2 \text{ consumption}}{\text{Systemic artery SaO}_2 - \text{Mixed venous SaO}_2}$$

Based on Ohm law, which states $V = IR$, where V = voltage (or pressure drop) across a circuit, I = the current (or flow) through the circuit, and R = the resistance in the circuit. SVR and PVR can be calculated as follows:

$$\text{PVR} = (\text{PA}_p - \text{LA}_p) / \text{Q}_p$$

$$\text{SVR} = (\text{Ao}_p - \text{RA}_p) / \text{Q}_s$$

where PA_p = pulmonary artery pressure; LA_p = left atrial pressure; Ao_p = aortic pressure; RA_p = right atrial pressure; PVR = pulmonary vascular resistance; and SVR = systemic vascular resistance.

When the partial pressure of dissolved O_2 is <100 , the PO_2 portion of the equation can be ignored, and the flow can be calculated using the O_2 consumption and O_2 sat measurements alone.

$$\text{Systemic Flow (Q}_s\text{)} = \text{O}_2 \text{ consumption} / (\text{systemic arterial O}_2 \text{ content} - \text{systemic venous O}_2 \text{ content})$$

$$\text{Pulmonary Blood Flow (Q}_p\text{)} = \text{O}_2 \text{ consumption} / (\text{pulmonary venous O}_2 \text{ content} - \text{pulmonary arterial O}_2 \text{ content})$$

$$\text{Systemic Flow (Q}_s\text{)} = \text{O}_2 \text{ consumption} / (13.6 \times \text{Hgb} \times (\text{systemic artery saturation} - \text{mixed venous saturation}))$$

$$\text{Pulmonary Blood Flow (Q}_p\text{)} = \text{O}_2 \text{ consumption} / (13.6 \times \text{Hgb} \times (\text{pulmonary vein saturation} - \text{pulmonary artery saturation}))$$

ANGIOGRAPHY

Biplane cineangiography is performed to delineate intracardiac or vascular anatomy and to evaluate ventricular function. Images are obtained by injection of radiographic contrast agents through angiographic catheters positioned in appropriate locations. The

angiograms may be performed in posteroanterior and lateral projections or by angling the cameras to obtain cranial, caudal, left anterior, or right anterior oblique projections. Based on the site of injection and the information required, the injection may be performed with a power injector, delivering large amounts of contrast quickly, or by hand.

INTERVENTIONAL PROCEDURES

Valvuloplasty

Aortic valvuloplasty

A retrograde approach from the femoral artery generally is used, although an antegrade and transseptal approach from the femoral vein is preferred by some. In either approach, following hemodynamic evaluation and angiographic estimate of the aortic valve annulus, a wire is positioned across the valve, and a balloon catheter is advanced over the wire and positioned across the aortic valve. The balloon is then inflated and deflated quickly. The inflation of the balloon leads to a transient loss of CO, ↓ SBP, and occasionally may be accompanied by ↓ HR. These hemodynamic changes recover quickly on balloon deflation. Although complications of aortic valvuloplasty are rare, the anesthesiologist must be “prepared for the worst,” which includes annular rupture and the creation of significant aortic regurgitation. Supplementation with oxygen and airway intervention, including intubation, should be considered for complicated interventional procedures.

Pulmonic valvuloplasty

After femoral venous or IJ access is obtained, the technique for balloon dilation of the pulmonary valve is nearly identical to that outlined above for the aortic valve. Loss of CO and ↓ HR are seen during the time of balloon inflation with this intervention as well. Although annular rupture is also a potential complication of this procedure, the creation of pulmonary insufficiency is of less concern and better tolerated than aortic insufficiency. Consider supplementing with 100% oxygen during interventional procedures, prior to the period of diminished cardiac output. If complications occur, airway intervention, including intubation, should be considered. Age-appropriate airway equipment and suction should be readily available.

Angioplasty

A number of transcatheter treatment options are available for management of **pulmonary artery stenoses**, including **balloon angioplasty** and **endovascular stent implantation**. Angioplasty has been shown to be highly effective in anatomically appropriate cases with a low complication rate. Hemodynamic and angiographic assessment of the lesion

is obtained, followed by selection of an optimal balloon catheter, based on both the size of the stenosis and surrounding “normal” tissue. Using the same “over-the-wire technique,” a balloon is advanced and centered over the stenosis. Hemodynamic and angiographic data are assessed following each intervention. A high index of suspicion for complications—including dissection or pulmonary artery tear, obstructive intimal flaps, thrombi, and reperfusion pulmonary edema—is justified, as management may require ventilatory manipulations and/or emergent cardiovascular resuscitation. Consider endotracheal intubation prior to intervention since airway management in the presence of pulmonary edema and/or blood in the airway can be difficult to manage.

Balloon angioplasty of coarctation of the aorta may be performed for treatment of native or recurrent coarctation. Angiography of the aorta is performed to delineate the coarctation and estimate the dimension of the coarctated segment and the adjacent aorta. As with other angioplasty techniques, the balloon size is based on the dimensions of the stenotic area and surrounding vessel. Transient loss of lower body perfusion and ↓ HR are to be expected, as with balloon valvuloplasty, on inflation of the balloon. Pressure and angiographic data are obtained to determine adequacy of results and absence of complications. There is a 4–5% incidence of intimal tear and dissection that, in most cases, are nonprogressive. Rarely, aortic disruption may require emergent surgical repair.

Endovascular stent placement

Stent implantation in the pulmonary arteries or for aortic coarctation is used to maintain vessel diameter and decreased gradients in patients unresponsive to balloon dilation. Stents are mounted on balloon catheters, and the balloon/stent combination is advanced over a previously placed wire. A long sheath (originating in the groin or neck) is placed across the area of narrowing to prevent the stent from slipping off the balloon catheter as it makes its way through the heart or vessels. After the stent has been properly positioned, the long sheath is withdrawn to expose the balloon/stent combination. The balloon is inflated to expand the stent and appose it to the vessel wall. Placement of long sheaths, particularly through the right ventricular outflow tract (RVOT), can be difficult and may result in transient bradyarrhythmias and loss of CO.

Closure of congenital defects

Atrial septal defects (ASDs)

As many as 80–85% of secundum ASDs may be amenable to device closure in the cath lab. Most devices currently used include a left atrial disc with an occlusive membrane, a central spool or connecting pin, and a right atrial disc with an occlusive membrane. The membrane occludes flow through the defect, and within months, the device becomes

incorporated into the septum due to endothelialization. Concomitant use of TEE or TTE is indicated and serves three purposes: (a) to ensure an adequate rim for device placement, (b) to confirm appropriate placement of the device so as to avoid valvar structures, and (c) to confirm that once deployed, the device remains seated in the appropriate position. Thus, the use of GETA is preferred. Upon echocardiographic examination and confirmation of an adequate rim for device placement, a prophylactic antibiotic (usually a cephalosporin) is administered by the anesthesiologist. A sizing balloon inflated across the defect permits estimation of the stretched diameter. A long sheath is then placed across the defect over a wire. The device attached to the delivery cable is loaded in the long sheath and advanced to the left atrium. The left atrial disc is opened; the device is withdrawn until the left atrial disc is in contact with the atrial septum; then, the right atrial disc is opened, effectively “sandwiching” the atrial septum between the two discs.

Ventricular septal defect (VSD)

Closure with a device can be performed in the cath lab for isolated or multiple muscular VSDs, as well as perimembranous defects. The technique requires establishment of a continuous AV guide wire loop across the defect. Most often, the wire course is from the femoral vein, through the right atrium, into the RV, across the VSD, out the aortic valve, around the aorta, and out the femoral artery. The device is then deployed via a long sheath placed across the VSD through the RV aspect. Hemodynamic compromise may be seen with tension on the wire if aortic or tricuspid insufficiency is induced. Transient arrhythmias are routine while crossing the VSD and deploying the device. Great care must be taken to avoid entrapment in the mitral, aortic, and tricuspid valves during device deployment. In addition to fluoroscopy, TEE or TTE is used to guide placement of the device. Recent improvements in the devices have significantly reduced the morbidity of this procedure.

Coil occlusion

Aortopulmonary collaterals, AVMs, Blalock-Taussig (B-T) shunts, venous collaterals, coronary artery fistulae, and patent ductus arteriosus (PDA) have all been successfully occluded using the technique of coil embolization. The embolization coils consist of a metal wire, either stainless steel or platinum, and may additionally have Dacron strands, which are available in multiple sizes, lengths and shapes. As with other device procedures, antibiotic prophylaxis is indicated. Although PDA or coronary artery fistula embolization may obviate the need for surgery, most embolizations serve to either reduce the cardiac workload by decreasing the amount of shunting or simplify a planned surgical procedure.

The technique for coil closure of collaterals or other communications is straightforward. A catheter is placed in the vessel to be occluded, and a selective angiogram is done to delineate the anatomy and diameter of the vessel to be closed. Coils that are slightly larger than the diameter of the vessel are used because the vessel will distend when the coil is deployed. Using a long “pusher” wire, the coil is advanced through the catheter and deployed in the vessel. Repeat angiography is performed to confirm complete closure. If residual flow remains, additional coils are placed. Coil dislodgement and embolization to a distal blood vessel are the most common complications. In general, the errant coil can be retrieved in the cath lab without much difficulty and a new coil of a larger size placed to occlude the vessel.

OTHER PROCEDURES

Endomyocardial biopsy is commonly performed for rejection surveillance in patients following cardiac transplantation. It also may be performed in patients presenting with acute onset of cardiomyopathy for histopathologic Dx of myocarditis. The preferred site for obtaining cardiac biopsy is the RV aspect of the intraventricular septum. The specimen is obtained with a biopsy forceps advanced to the RV through a long sheath. It is usual to obtain four to five specimens to improve the diagnostic gain, as the histopathologic changes can be patchy. Complications of endomyocardial biopsy include cardiac perforation and tricuspid valve damage. GA is required in patients with compromised airway and/or cardiopulmonary status from lymphoproliferative disease or obesity 2° to steroid therapy. With right heart catheterization and biopsies, the arms are usually at the patient's side. However, with annual studies, as with all studies requiring left heart catheterization, the arms are pulled upward toward the head. Though this permits observation of the IV site, utmost attention to detail in padding pressure points is needed, specifically to avoid brachial plexus injury.

A variety of other transcatheter therapeutic procedures may be performed in the cardiac cath suite. **Rashkind balloon atrial septostomy, static balloon septoplasty, Brockenbrough transseptal needle puncture, and radiofrequency-assisted perforation** of the pulmonary valve or the atrial septum are all less commonly used than the procedures described above, but routinely are undertaken in high-volume cath labs.

ELECTROPHYSIOLOGY STUDY (EPS)

Patients with atrial or ventricular arrhythmias may require either diagnostic or therapeutic interventions in the cath lab. EP studies are catheterization procedures in which intracardiac electrical signals are recorded via specialized catheters that can

both record electrical activity and stimulate the heart. These studies often are used to make a Dx of the mechanism of arrhythmia, assess the hemodynamic impact of the arrhythmia, assess efficacy of pharmacologic therapy, and map the location of abnormal conduction pathways or automatic foci. Although routine studies usually take 2–3 h, some may be quite lengthy. For all procedures, access is obtained via the RIJ or femoral vein; thus, the head may be turned to the side. Though providing easier access for the cardiologist, it makes airway management more tenuous. Pressure on/near the trachea can promote coughing.

Radiofrequency ablation (RFA) is a procedure in which abnormal electrical conducting pathways or automatic electrical foci (identified by EPS) are destroyed, using the application of RF energy delivered through a deflectable electrode catheter. This procedure was first described in pediatric surgery in 1991, but has rapidly become a preferred therapeutic option for supraventricular tachycardia in this population. On some occasions, RF lesions must be placed close to other critical structures in the heart (e.g., AV node).

With advances in technology and an increased understanding of high-risk pediatric patient populations, **transvenous pacemaker** and **implantable cardioverter defibrillator (ICD)** placements are becoming more common in the pediatric population. Pacemaker placement has become more common as data have accumulated regarding the risk of sudden death in patients with congenital complete heart block, as well as increased survival with postop heart block. New indications for ICD placement in patients with long-QT syndrome, congenital heart disease, and hypertrophic cardiomyopathy have increased the number of ICD implantations in the last 5 yr. These procedures are commonly performed in the cath lab under GA. However, for the placement of loop recorders and subcutaneous ICDs, light sedation can be considered.

Transcatheter valve placement

Pulmonary valve

The most common site for interventional valve replacement is the pulmonary valve. These cases are performed via the femoral or jugular veins. A stiff wire is placed in secure position from the access site to a distal PA branch. A long sheath is then placed across the RV outflow tract and is used for angiography and intervention. There are multiple steps in this procedure involving first pulmonary valvuloplasty of the conduit, bioprosthetic valve, or native pulmonary valve (see pulmonary valvuloplasty above). Contained conduit tear occurs in about 5% and does not have any hemodynamic consequence. The next step involves bare metal stent placement across the site of intended valve deployment in order to open up stenotic and calcified conduits or to

provide a landing zone in a native outflow tract. A covered stent can be used to cover more sizeable tears. The same hemodynamic instability can occur during stent placement as during valvuloplasty with the added risk of stent malposition or embolization. After proper stent placement, the transcatheter valve is then prepared and delivered. The manipulation of the delivery system to properly position the valve can cause short-lived dysrhythmias. Repeat hemodynamic measurements and angiography are performed after valve deployment.

Tricuspid valve

The procedure is similar to PV in that a stiff wire is positioned across the tricuspid valve. Balloon valvuloplasty is performed to open up the stenotic leaflets and to size the bioprosthetic tricuspid valve. The transcatheter valve is then deployed within the tricuspid valve. Again, with balloon angioplasty and valve deployment, there can be a temporary ↓ CO and ↓ O₂ saturations that return once the balloon is deflated.

Aortic valve

This is a much more common procedure in adults; however, AV replacements occasionally do occur in pediatric patients. The approach is usually from the femoral artery although a hybrid procedure can be done with surgeons and an LV apical approach. From the femoral artery, the AV is crossed retrograde and a stiff wire is positioned in the LV. Valvuloplasty may first be performed (see Aortic Valvuloplasty). For valve deployment, rapid ventricular pacing (of the RV) is performed, and TEE is used to confirm proper placement. Potential complications include aortic regurgitation or paravalvular leak, aorta injury secondary to the wire or delivery system, and other vascular injury.

ANESTHETIC CONSIDERATIONS

Claudia Benkwitz and Lisa Wise-Faberowski

PREPROCEDURE

Cardiac catheterization and interventional procedures in children, ranging from those requiring simple diagnostic procedures to those requiring complex interventional procedures, such as balloon angioplasty/valvuloplasty, stent placements, or closure of shunts have facilitated care of younger, smaller, and sicker patients. These patients are often challenging for the anesthesiologist due to their abnormal cardiac anatomy and physiology. Many of them have undergone repeated catheterizations and have had multiple anesthetics. All patients require a thorough preanesthetic evaluation with

emphasis on cardiorespiratory function and associated comorbidities (almost 30% of patients show associated anomalies or syndromes with Down syndrome being the most common). Most children will follow the same NPO protocol as they would for surgery. Particular attention should be paid to patients with single-ventricle physiology (Fontan circulation), who are dependent on their venous return for hemodynamic stability and oxygenation as well as being prone to shunt thrombosis, specifically when dehydrated. In such patients, early establishment of IV access to prevent dehydration is advisable. Clear liquids can be given up to 2 h before surgery. Similarly, Williams syndrome patients with pulmonary, aortic outflow tract, and/or coronary obstruction are at risk of sudden death upon anesthetic induction if they have an inadequate preload.

Previous anesthesia	✓ Surgical, interventional, and radiologic procedures; h/o difficult access; endotracheal tube size and anesthetic techniques; problems with previous anesthetics
Family history	✓ Family history for anesthetic-related problems; h/o malignant hyperthermia
Prematurity	✓ Gestational age. Need for postop admission and apnea monitoring if <55 wk postconceptual age; any associated clinical problems
Neurologic	✓ Head ultrasound in neonates to rule out ischemia and hemorrhage; note preoperative neurologic status due to risk of embolism during procedure; h/o seizures or developmental delay. Heparinization during cardiac catheterization may increase the risk of bleeding.
Craniofacial and airway	✓ H/o difficult airway; any surgery to the airway; h/o tracheal stenosis due to multiple intubations in the past; stridor due to laryngo- or bronchomalacia; h/o loose teeth. DiGeorge (22q11 deletion) and William syndrome patients may present with abnormal airway anatomy. Trisomy 21 patients and post-heart transplant patients on steroids may present with obstructive airway disease. Tests: Recent bronchoscopy, imaging of the airway
Respiratory	✓ Current problem(s): ✓ Respiratory reserve (i.e.,

playing activities, respiratory rate, nasal flaring, intercostal/subcostal retractions, diaphoresis). ✓ H/o cyanosis, chronic cough, steroid use for RAD and other chronic respiratory problems → ensure that they are under optimal control. ✓ H/o recent URI → risk of adverse perioperative events. Note: frequent "URIs" along with FTT may be signs of pulmonary overcirculation in the setting of large L → R shunts as opposed to being of infectious etiology ✓ Patients with pulmonary hypertension may have obstructive sleep apnea. ✓ Cardiac transplant patients can develop lymphoproliferative disease, which results in redundant lymphoid tissue in the pharynx and epiglottis → possible airway obstruction. ✓ Requirement for supplemental O₂, CPAP, BiPAP, nitric oxide. Medications for pulmonary HTN should be continued.

Tests: SpO₂, CXR to r/o infiltrates, CHF and cardiomegaly, PFTs. Compare with baseline. Assess perfusion scan for distribution of PBF if necessary.

Cardiovascular

✓ Current problem(s): ✓ Past surgical history, pacemaker, ICD, etc. If the patient has a pacemaker → check settings, indication(s), underlying rhythm, pacer dependency, and last interrogation. ✓ Existing shunts → may affect anesthetic plan and FiO₂.

✓ Assess cardiopulmonary reserve (e.g., exercise tolerance, diaphoresis, SOB, feeding difficulties, poor weight gain). ✓ H/o syncope, hypoxic spells, and chest pain. Transplant patients may develop CAD. ✓ Arrhythmias, palpitations, murmurs, pulses in all extremities.

Tests: SpO₂, ECG; BP (upper/lower extremity)

✓ Recent echo, cardiac catheterization +/- CT/MRI/MRA reports

Gastrointestinal

✓ Feeding difficulties, reflux due to chronic use of steroids. May have G-tube/Nissen.

✓ Some single-ventricle patients may have hepatic dysfunction and protein-losing enteropathy (↑ risk

	ascites, pleural/pericardial effusion, infection). Alagille patients may also present with hepatic dysfunction and hyperbilirubinemia. SaO ₂ saturation measured using pulse oximetry or NIRS may pose difficulties as high bilirubin levels can affect light absorbance.
Renal	✓ Patients with CHF are frequently on diuretics and/or ACE inhibitors/digoxin → expect hypovolemia and potential electrolyte disturbances (i.e., ↑ ↓ K⁺). ✓ Transplant recipients may develop renal dysfunction due to antirejection medication or HTN. ✓ Patients may have had renal insult during previous surgery. Tests: ✓ BUN and Cr as these patients are also at risk for renal injury due to contrast and also to guide hydration therapy.
Endocrine/steroid	✓ Transplant patients are on steroids. ✓ for other endocrine concerns.
Hematology/immunology	✓ History of previous transfusions and reactions. Immunologic deficiency that may affect the type of blood transfusion (e.g., neonates require CMV-negative blood and products). Tests: Recent hematocrit
Syndromes	✓ For coexisting syndromes and associated anomalies
Ob/Gyn	✓ Last menstrual period
Vascular	✓ H/o thromboembolism. H/o difficult access and vascular occlusion
Current medications	✓ Stop warfarin before cardiac catheterization. Anticoagulate with heparin if needed; d/w cardiology. ✓ Coagulation studies prior to procedure. ✓ Antiarrhythmic drugs are usually stopped prior to EP studies unless the arrhythmia is frequent and poorly tolerated. Hold diuretics and ACE inhibitor starting the night before due to ↑ risk for hypotension

Laboratory	following induction; continue other cardiac meds. ✓ CBC: Use Hgb/HCT baseline to determine need for transfusion; ↑ Hgb/HCT suggests chronic hypoxia; transfusion threshold will vary depending on patient age and comorbidities. Severe anemia should be corrected prior to cath. ✓ Have T&C blood available in the catheterization lab for procedures with risk of increased bleeding. ✓ WBC: if the child has a h/o fever and recent URI. ✓ Electrolytes/Bun and creatinine, especially if the patient is on diuretics and/or digitalis. ✓ Coagulation studies. ✓ Drug levels (e.g., digitalis, antiseizure medications) to ensure therapeutic range. Pregnancy test must be done the morning of surgery if patient has attained menarche.
Premedication	✓ Midazolam 0.5–1.0 mg/kg po or via G-tube. Ketamine 5–8 mg/kg po or via GT may be added in either extremely uncooperative patients or those with significant hemodynamic instability (i.e., severe AS, PAH, ventricular dysfunction), in whom an inhalational induction would not be safe. In patients unwilling to take PO premed, nasal dexmedetomidine 1–3 mcg/kg may be used, however may take 30–45 min to reach full effect. Be cautious in patients with OSA as they are more sensitive to respiratory depression. Cyanotic patients should have SaO₂ saturation monitored before and continuously after premedication. Antibiotics are given to patients as required for SBE prophylaxis per most current (2007) AHA guidelines and for those undergoing interventional procedures. In patients scheduled for EP studies in whom anxiety is a known precipitant of the arrhythmia, preop anxiolysis is a relative contraindication as it may result in a false-negative study.
NPO	ASA fasting guidelines should be followed (see p. D-1, NPO guidelines). Prolonged NPO in single-ventricle physiology and other shunt-dependent lesions should

be avoided, as shunt flow might decrease with dehydration → ↑ viscosity → ↑ shunt thrombosis. Special consideration for patients with hypertrophic cardiomyopathy (HCM) and Williams syndrome patients.

Equipment

Transport patient to and from the catheterization lab with a pulse oximeter and other monitors as appropriate. Equipment to manage difficult airway and hemodynamically unstable patients should be readily available (e.g., difficult airway cart, I-stat, arterial line, blood pump, pumps to infuse vasopressors, pacemaker, defibrillator, NIRS).

INTRAPROCEDURE

Anesthetic technique

Cardiac catheterization can be done under sedation, monitored anesthesia care, and general anesthesia. Intravenous access is usually in an upper extremity as the lower extremities may be needed for unilateral or bilateral femoral access by the cardiologist. Intravenous contrast administered in an IV in the left upper extremity may be requested to confirm vascular patency for transvenous pacemaker wire placement. A spontaneously breathing patient is preferable so as to mimic normal baseline cardiac and respiratory physiologic state and to perform subsequent hemodynamic calculations. General anesthesia with an ETT may be a safer option to provide a secure airway for neonates and younger children as the surgical field often covers the entire patient. However, there is wide variety of practice among different institutions with regard to airway management. Patients with moderate to severe pulmonary hypertension present extra challenges, and care for these patients differs as well among institutions. Because physiologic changes associated with intubation, and more likely extubation, might result in a pulmonary hypertensive crisis in these patients, sedation is favored by some anesthesiologists. However, others prefer muscle relaxation and ETT placement, as hypoxia-hypercarbia related to an unsecured airway can also lead to life-threatening increases in PAP. Older children or patients undergoing procedures of shorter duration with no other significant comorbidities may tolerate incremental IV sedation with local anesthesia.

A large variety of drug regimens can be used to provide anesthesia/sedation for children undergoing cardiac catheterization procedures. These include propofol, ketamine, remifentanyl, dexmedetomidine, midazolam, fentanyl, and morphine infusions either

alone or in combination. While all have been used successfully, bradycardia/sinus arrest, heart block, hypotension, transient increase in systemic and pulmonary pressures (SVR > PVR) after rapid bolus injection, as well as insufficient anesthesia/analgesia and prolonged recovery have been reported.

Several approaches have proven safe and effective. Currently, there is no consensus on what constitutes optimal anesthetic management. Propofol and ketamine, propofol and dexmedetomidine, dexmedetomidine and ketamine, as well as volatile anesthetics, in combination with spontaneous ventilation or GETA, have all been safely employed. Often, there is institutional preference for one or the other, based on previous clinical experience.

A few sample regimes are reviewed here:

- Propofol 10 mg/kg/ketamine 1 mg or “ketofol” at a dose of 50–200 mcg/kg/min
- Propofol (50–200 mcg/kg/min) and dexmedetomidine (0.5 mg to 2 mcg/kg/h). This regimen eliminates concerns about ketamine-related secretions with the attendant need for glycopyrrolate, but may delay discharge from PACU due to prolonged sedation.
- Dexmedetomidine (2 mcg/kg/h) and ketamine (20 mcg/kg/min) supplemented by additional ketamine +/- midazolam as needed

Independent of what technique is ultimately used, normoxia and normocarbida are key for these procedures, since hypoxemia, hyperoxemia, acidosis/alkalosis, and PPV all exert significant disturbances in pulmonary vascular resistance. PPV decreases systemic venous return, particularly in the setting of depleted intravascular volume. This is of particular relevance in those patients with restrictive RV or single-ventricle physiology, where pulmonary blood flow is mostly passive and depends on adequate preload. When possible, patients with Glenn and Fontan physiology should be allowed to breathe spontaneously. This improves systemic venous return, pulmonary blood flow and hence systemic blood flow, and cardiac output. Both circulations are dependent on passive pulmonary blood flow as opposed to a shunt physiology (BT or central), which provides active pulmonary blood flow. Pulmonary blood flow and adequate cardiac output depend on PVR.

ECMO backup should be considered in patients with poor ejection fraction and critically ill neonates, as well as for some patients with severe PAH or Williams Syndrome, who have a significantly higher mortality regardless of the anesthetic technique. Rarely, for certain interventional procedures, surgical backup and OR availability may have to be prioritized. Significant preparation and coordination are necessary to transport a newborn from the delivery room to the catheterization lab for

emergent procedures such as an atrial septostomy.

Induction	Standard inhalation or iv induction (see p. D-1). As in all patients with CV morbidities having anesthesia, the choice of anesthetic drugs must include careful consideration of their effects on myocardial contractility, preload and afterload, pulmonary and systemic vascular resistance, and respiration. FiO_2 should also be carefully controlled in ductal or shunt-dependent lesions where a balance of $Q_p:Q_s$ is desired.
Maintenance	General anesthesia with volatile agents (e.g., sevoflurane) or TIVA techniques as previously described may be used. O_2 supplementation via mask/nasal cannula might be appropriate in many cases; however, an ETT/LMA to protect airway should be strongly considered in high-risk patients and/or interventions. For EP procedures, the Supreme LMA with the orogastric port allows for ease in placement of a transesophageal pacing probe if needed. EP procedures are long; thus, an LMA provides for not only a protected airway during IJ access, but allows for controlled ventilation when minimal catheter movement is desired. Most catheterizations require the patient to be maintained on room air for accurate SaO_2 sat and pressure measurements. With higher FiO_2 (>0.3), firstly, it will be necessary to consider the dissolved oxygen fraction when calculating Q_p with the Fick equation. Secondly, high inspired FiO_2 will likely result in a reduction of PVR, thereby increasing the Q_p/Q_s ratio. More importantly, any FiO_2 changes should be communicated with the interventionist so that $Q_p:Q_s$ calculations can be corrected. Similarly, routine use of antisialagogues should be avoided so that the heart rate remains at baseline. The same holds true for other vasoactive drugs, such as ephedrine and phenylephrine. If deemed necessary, and after other measures have failed, (e.g., adjustment of anesthetic depth, volume resuscitation, etc.), their intended use should be

communicated prior to administration. EP studies are best managed with iv propofol 75–200 mcg/kg/min, because this agent has the least ability to induce dysrhythmias or to interfere with the induction of tachyarrhythmias for diagnostic purposes. Sevoflurane or isoflurane (<1 MAC) as well as a combination of propofol and ketamine (7.5–20 mcg/kg/min) is also acceptable. In contrast, remifentanyl and dexmedetomidine have both been shown to significantly depress sinoatrial and atrioventricular nodal function and are therefore not desirable in patients undergoing EPS for ablation of SVT or in patients with prolonged QTc. In patients with a history of prolonged QT interval, TIVA (propofol/remifentanyl) has been recommended since both drugs have been shown to have none to minimal effects on the QTc interval. Caution should be exercised with reversal agents and various antiemetics, including 5-HT₃ receptor antagonists, since they can also prolong the QT interval. Since the electrophysiologist is often aware of the likely underlying mechanism of the arrhythmia, this allows for adaptation of the anesthesia technique that is the least likely to interfere with the patient's electrophysiology. Heparin 50–100 U/kg is given for patients with arterial sheaths. The activated clotting time (ACT) should be in the range of >200–300 sec during the procedure and <175 sec when the catheters are removed. Once the sheath is removed, a considerable amount of time is needed to attain hemostasis. NO +/- inhaled prostacyclins may be required for the Rx of pulmonary HTN. In the latter case, check with RT to ensure the appropriate setup for administration via either ventilator or nasal cannula is available. A special injector is needed when administering NO via nasal cannula.

Emergence

If the patient's clinical status allows for deep extubation, it can be done to prevent coughing. Interventional procedures and EP/ablation likely require hospital admission for observation. For neonates with SV physiology or other patients with complex underlying

disease, comorbidities or significant hemodynamic instability, a prearranged ICU bed might be advisable.

Blood and fluid requirements

IV: 20–22 ga × 1
BSS
PRBCs
Fluid warmer

A second volume iv may be desirable for interventional cases. The cardiologist will also have a venous access line that can be used if necessary. Interventional procedures may cause tearing of vessels and/or myocardium and can cause abrupt hemorrhage; therefore, blood must be available in the room. Neonates can lose significant blood during access and sampling and may also require transfusion. Follow serial Hcts. In patients who are TPN dependent or receiving G-tube feeds, hypoglycemia may occur. Glucose monitoring is helpful and the administration of glucose containing fluids may be indicated.

Monitoring

Standard monitors, including
± arterial line
± Foley catheter
± NIRS

Respiratory monitoring can be performed via oxygen masks or nasal cannulae adapted to measure CO₂ flow in a spontaneously breathing patient. Discuss with the cardiologist if A-line access is required. Frequently, the femoral artery is cannulated for the catheterization and may be available for use. However, access may be limited at critical times (e.g., stenting or coiling) and not available during a critical induction; therefore, a peripheral

arterial line may be desirable. ST-segment analysis (requiring proper lead placement) is useful in patients with a tenuous hemodynamic state, ductal-dependent lesions, or Williams syndrome. CAD is the leading cause of death in posttransplant patients. For cases of long duration, consider Foley catheter placement. Monitor temperature. In ductal-dependent lesions, monitoring of both pre- and postductal saturations may be helpful. Cerebral oximeter monitoring is recommended in single-ventricle and in other cyanotic lesions as pulse oximetry is a poor predictor of changes in cerebral saturation. Arterial line also plays an important role in patients with poor cardiac function and/or placement of melody valve.

Positioning

✓ pad pressure points

Supine, arms may be flexed above the head to allow fluoroscopy of the chest. Adolescent patients are at risk of brachial plexus injury. Keep extension <90°.

✓ cover eyes

All ECG leads and monitoring wires must be cleared from axilla and chest to allow fluoroscopy. Corneal abrasions are not uncommon as often the patient's face is covered by surgical drapes and cardiologists may rest their arms, hands, and instruments on

		the patient's face.
Complications	Airway obstruction	Rx: Airway support, oral/nasal airway, LMA. Intubate if necessary.
	Aspiration	Rx: Suction, and secure airway if needed. Bronchodilators, H ₁ - and H ₂ -blockers if needed. CXR. O ₂ monitoring. Admit if postop oxygen requirement persists.
	Arrhythmias	Determine etiology, whether hemodynamically stable or not. Cardiovert/defibrillate/emergency drugs as necessary. Check electrolytes and ABG. Consult EP.
	Hemorrhage (access site, rupture of vessels, tamponade)	Rx: Fluid resuscitation with crystalloid, 5% albumin and/or blood; vasopressors as needed. Secure airway, 100% FiO ₂ . Check Hct.
	Valvular incompetence (severe regurgitation/annular rupture)	Rx: Secure airway, volume resuscitation, vasopressor support, consult surgeon and proceed with emergent surgical repair if needed.
	Hypoxemia	Try to determine the cause. Check Hct in single-ventricle patients. Rx: Airway support; increase FiO ₂ . Intubate and ventilate as needed.
	Pulmonary hypertensive crisis	Avoid hypoxia, hypercarbia, acidosis, and pain/agitation. Rx: FiO ₂ 100%, deepen anesthetic if tolerated. Consider NO and systemic vasopressors.
	Contrast reaction/anaphylaxis	Rx: Support airway, 100% FiO ₂ , intubate if needed: phenylephrine, volume

		resuscitation, steroid, antihistamine, H ₂ -blocker.
	Hypothermia	Especially neonates. Rx: Forced-air warming device, heat lamp. Increase environmental temperature.
	Air embolism	Rx: 100% FiO ₂ ; identify and occlude source; Trendelenburg; fluid resuscitation; vasopressor support. Aspirate air from central access if possible.
	Positioning injury	Especially brachial plexus injury. Careful documentation, consult neurologist. Prevention = diligent positioning is key.
	Vascular thrombosis/stroke	Support airway, 100% FiO ₂ , intubate if needed; obtain emergent brain CT; consult neuro/stroke team.

POSTPROCEDURE

Complications	Hematoma at access site Ischemic limb Emergence delirium Pericardial tamponade Corneal abrasion Recurrent arrhythmia
Pain management	Postop analgesia is usually managed with acetaminophen, or fentanyl; rarely are long-acting opioids (morphine, hydromorphone) needed; care should be taken to avoid hypoxia or hypercarbia. The access sites are commonly infiltrated with local anesthetic by cardiologist. Use of ketorolac should be discussed with the cardiologist.
Tests	CXR, if indicated

TTE, if clinical picture is suggestive of tamponade

Suggested Readings

1. Bobhate P, Guo L, Jain S, et al: Cardiac catheterization in children with pulmonary hypertensive vascular disease. *Pediatr Cardiol* 2015; 36:873–9.
2. Devroe S, Meeusen R, Gewillig M, et al: Xenon as an adjuvant to sevoflurane anesthesia in children younger than 4 years of age undergoing interventional or diagnostic cardiac catheterization. *Paediatr Anaesth* 2017; 27:1210–9.
3. Friesen RH, Twite MD, Nichols CS, et al: Hemodynamic response to ketamine in children with pulmonary hypertension. *Paediatr Anaesth* 2016; 26:102–8.
4. Grunwell JR, Travers C, Stormorken AG, et al: Pediatric procedural sedation using the combination of ketamine and propofol outside of the emergency department: a report from the Pediatric Sedation Research Consortium. *Pediatr Crit Care Med* 2017; 18:e356–63.
5. Joshi VS, Kollu SS, Sharma RM: Comparison of dexmedetomidine and ketamine versus propofol and ketamine for procedural sedation for procedural sedation in children undergoing minor cardiac procedures in cardiac catheterization laboratory. *Ann Card anaesth* 2017; 20:422–6.
6. Lam JE, Lin EP, Alexy R, et al: Anesthesia and the pediatric cardiac catheterization suite: a review. *Paediatr Anaesth* 2015; 25:127–34.
7. Lin CH, Desai S, Nicolas R, et al: Sedation and anesthesia in pediatric and congenital cardiac catheterization: a prospective multicenter experience. *Pediatr Cardiol* 2015; 36:1363–75.
8. Odegard KC, Vincent R, Baijal RG, et al: SCA/CCAS/SPA Expert Consensus Statement for Anesthesia and Sedation Practice Recommendations for patients undergoing diagnostic and therapeutic procedures in the Pediatric and Congenital Cardiac Catheterization Laboratory. *Anesth Anal* 2016; 123:1201–9.
9. Riveros R, Makarova N, Riveros-Perez E, et al. Utility and clinical profile of Dexmedetomidine in Pediatric Cardiac Catheterization Procedures: a matched controlled analysis. *Semin Cardiothorac Vasc Anesth* 2017;21;330–40.
10. Taylor KL, Laussen PC. Anaesthesia outside of the operating room: the paediatric cardiac catheterization laboratory. *Curr Opin Anaesthesiol* 2015;28:453–7.
11. Tirotta CF, Nguyen T, Fishberger S, et al. Dexmedetomidine use in patients undergoing electrophysiologic study for supraventricular tachyarrhythmias. *Paediatr Anaesth* 2017; 27:45–51.

PEDIATRIC ONCOLOGIC PROCEDURES

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Pediatric oncology patients represent a unique population of patients who come for frequent procedures requiring anesthesia over the course of several years. These patients require anesthesia for a variety of procedures: diagnostic imaging, surgical resection, radiation therapy, central line placement, and multiple minor procedures for staging and intrathecal chemotherapy. The pediatric anesthesiologist is in the position of caring for these patients for long periods from initial diagnosis, through years of induction and maintenance therapy, and finishing with end of therapy procedures. More than in other patient populations, the anesthesiologist plays a critical role in a child's experience over several years in the hospital environment.

Pediatric oncology patients now often have a 5-yr survival rate of over 80%. It is important to keep in mind the possible effects of anesthesia and sedation on the developing brain. Current recommendations are primarily based on extrapolating animal studies to humans. The main concerns are in patients under 3 yr of age who are undergoing procedures over 3 hours or are having multiple anesthetics. Oncology patients face a unique combination of stressors from illness, hospitalization, and multiple painful procedures that may affect psychoemotional development. Anesthesiologists and oncologists need to be mindful of these issues and be prepared to discuss the risks versus benefits of anesthesia. Conscious sedation may be an alternative to GA; however, most children under 12 yr of age will be better served by a brief GA for each procedure.

When patients first present for evaluation of possible malignancy, either solid organ tumors or hematologic cancers, they may undergo multiple imaging studies such as computed tomography scans of the chest, abdomen, and pelvis, bone scintigraphy, and magnetic resonance imaging scans. These studies are not painful, but require patients to be immobile for anywhere from a few minutes to over an hour. Sedation or general anesthesia is usually required, especially in patients under 5 yr of age.

For solid organ tumors of the brain, chest, abdomen, pelvis, and bone, pediatric patients may undergo daily doses of radiation therapy over the course of several weeks before or after resection. Although radiation therapy may only be a few minutes long, the specific anatomic dosing requires the patient to be immobile, again requiring multiple brief anesthetics. Solid organ tumors involve major surgical resection, or diagnostic tissue biopsy, either before or after concomitant radiation therapy or chemotherapy.

In the initial evaluation of hematologic diseases, patients must undergo evaluation of bone marrow with bone marrow aspirates and/or biopsies. Involvement of the cerebrospinal fluid is evaluated with a diagnostic lumbar puncture (LP). These patients then go on to require repeat bone marrow procedure and serial LPs for staging and for

intrathecal chemotherapy dosing when indicated. Approximately 50% of leukemia patients will develop CNS disease without intrathecal prophylaxis.

In addition to the above diagnostic and treatment procedures, the majority of pediatric oncology patients undergo placement of indwelling central venous catheters, including percutaneous intravenous central catheters (PICC), Hickman or Broviac catheters, or portacaths. With the exception of PICC placements, most of these catheters are inserted under anesthesia and in a sterile operating room environment. When full therapy is completed, central venous line removal again requires sedation or general anesthesia.

PREPROCEDURE

Pediatric oncology patients are followed very closely during their treatment, undergoing at least monthly physical exams to evaluate for side effects of therapy to determine whether proceeding with the prescribed therapy is indicated. When these patients require invasive procedures, it is essential to work with the primary oncology team to evaluate whether the requested procedure is elective or urgent. All patients undergo routine periodic blood counts to determine the safety of continuing chemotherapy dosing. Occasionally, a procedure, such as LP for intrathecal chemotherapy, may be postponed if blood counts have dropped too low. Checking with the primary team is important to determine if the procedure is to proceed as planned. It is important to remember that the toxicity of chemotherapeutic drugs may affect every organ in the body beyond the target cancer tissue.

A thorough preanesthetic history and physical exam should be performed for all major and minor cases and the standard NPO protocol applied.

Respiratory and airway

Major surgical procedures for patients with symptoms of upper respiratory infection (e.g., cough, fever) are postponed unless there are circumstances (CNS disease, bone marrow relapse) that require the treatment to proceed. In those circumstances, LPs and BMAs can be performed either with sedation or with the use of local anesthesia. A number of chemotherapeutic agents have been associated with pulmonary toxicity: bleomycin (2–5%), BNCU (20–50%), busulfan (2.5–11.5%), cyclophosphamide (rare), and methotrexate (rare). The use of these agents requires follow-up evaluation to make certain

the patient is able to proceed with his/her procedure. Patients who present with an anterior mediastinal mass may present with signs and symptoms of superior vena cava compression or with superior mediastinal syndrome with tracheal compression. It is critical to recognize and evaluate patients with a mediastinal mass prior to inducing general anesthesia. The adolescent patient presenting with a new diagnosis of Hodgkin or non-Hodgkin lymphoma or acute lymphoblastic leukemia should be carefully assessed for the possibility of a mediastinal mass. Younger patients may also present with symptoms of anterior mediastinal compression. Careful preop evaluation should be made for respiratory symptoms or compromise, as well as obtaining a chest x-ray, and possibly even chest CT scan. Careful consideration should be given to performing diagnostic procedures under minimal sedation, since general anesthesia in these patients can result in significant cardiorespiratory complications and fatality. Preop treatment with irradiation or corticosteroids may reduce tumor size; however, this may also change tissue pathology. Key features in GA management, if absolutely required, are the maintenance of spontaneous ventilation, no paralysis, having a rigid and flexible bronchoscope immediately available, and preparing for postop ICU monitoring.

Routine airway assessment is essential.

Chemotherapy may result in severe, painful oral mucositis or stomatitis. Patients are also more susceptible to infectious viral oral lesions, such as herpetic lesions, and may require isolation. Frequent URI may occur due to immune compromise. Fever, loose cough, wheezing, or coarse BS requires evaluation and likely postponement of any procedures.

Tests: All pediatric oncology patients undergo

weekly complete blood counts to evaluate for toxicity from therapy. In addition, patients who have received any of the agents associated with pulmonary toxicity undergo pulmonary function tests at regular intervals. Although some chemotherapy agents are associated with pulmonary toxicity, the cumulative doses of those agents are such that most patients are asymptomatic. CXR may be indicated to evaluate upper respiratory symptoms.

Cardiovascular

Anticancer treatment can produce a wide range of cardiac toxicity. Anthracycline-induced cardiomyopathy, while rare, can lead to significant cardiovascular compromise. This complication is associated with the use of cumulative doses >300 mg/m.

Cardiomyopathy is insidious in origin and most commonly presents many years following treatment with doxorubicin, daunorubicin, or amacrine. The use of high doses of radiotherapy is also associated with cardiac complications, including constrictive pericarditis, atherosclerosis, and early myocardial infarction. Signs of CV toxicity may include SOB, ECG changes, rhythm disturbances, pericarditis, or CHF. Preop hemodynamic status needs to be evaluated. Patients on steroid therapy may develop steroid-induced hypertension. It is important to obtain a careful history of any antihypertensive medication, especially a record of medications taken on the day of the procedure or surgery. Conversely, patients who are dehydrated or anemic may become significantly hypotensive after induction of general anesthesia. These patients may require preop hydration, and vasoactive medication should be available for intraop use. Patients actively receiving chemotherapy may also have 3rd spacing and may need aggressive hydration and/or pressor support.

Tests: Patients receiving any agents associated with cardiac toxicity are monitored at yearly interval with

	echocardiograms and electrocardiograms. These tests do not need to be performed routinely prior to a procedure requiring GA unless history and physical exam suggest a change in the patient's clinical condition.
Hematologic	Bone marrow suppression occurs with all cytotoxic drugs. The anesthesiologist and oncologist need to work together to evaluate anemia and thrombocytopenia. Due to myelosuppression, anemia is common in oncology patients. In assessing the degree of anemia in a given patient, it is important to assess his or her overall clinical appearance and hemodynamic compensation. A very low hemoglobin level <6–7 increases the patient's risk of significant BP instability during GA. For the procedure planned, can the patient tolerate the stress from anesthesia, and is there further blood loss expected during the procedures? Patient assessment with expectations for the surgical/diagnostic procedure will determine whether red blood cell or platelet transfusion is required preop. If no transfusion is planned and the patient is anemic, prepare fluid replacement, colloid (e.g., albumin), and/or vasoactive support. Tests: Complete blood counts are performed once to twice a week in all pediatric oncology patients. These tests help determine whether continuing with the prescribed therapy is indicated. Baseline platelet and hemoglobin counts need to be measured before proceeding with lumbar punctures and bone marrow aspirates.
Neurologic	Peripheral neuropathies due to vincristine, vinblastine, cisplatin, and procarbazine may be present. Patient should be carefully positioned and padded. These patients may require additional attention to preop pain management and anxiolysis, with supplemental intraop and postop pain control.
GI/Hepatic	Many chemotherapy agents cause significant nausea

	and vomiting. Patients may present with these symptoms preop or may be dehydrated due to vomiting and poor fluid intake. Hepatotoxicity may also occur from chemotherapy or radiation therapy. Asparaginase can induce severe pancreatitis. ✓LFT, amylase, lipase.
Endocrine	Most cancer patients are treated with corticosteroids at some point in their therapy. Steroids can lead to weight gain, suppression of the stress response, and rarer complications such as psychosis. In addition, steroid-induced hyperglycemia may require pre- and postop glucose monitoring and insulin therapy.
Premedication	For major procedures, individualized premedication is recommended. Since oncology patients often have an indwelling intravenous catheter, or are familiar with having venipunctures and iv placed, iv induction is frequently used. Intravenous midazolam may be easily administered when required for anxiolysis. Some patients may require premedication with oral midazolam or lorazepam, if they do not have iv access. However, many radiology and minor oncology procedures may be performed in a less restrictive environment, such as a procedure room, where the parents may stay with their child through induction. For the very anxious patient, or a child who has undergone significant emotional trauma in the hospital environment, preop counseling with a child life specialist or psychologist can be very important and helpful in alleviating fears and helping the child adjust to the stress of frequent hospital/procedure visits.

INTRAPROCEDURE

Anesthetic technique

For major surgical procedures, GETA management is indicated. Below are recommendations for pediatric oncology patients undergoing minor procedures.

Procedures such as LPs and bone marrow aspirates are usually short, 10–20 min. Adolescent patients may be evaluated by their provider or the anesthesiologist for possible conscious sedation. Younger patients usually require brief general anesthesia—inhalation or intravenous.

Induction	For lumbar punctures in the patient without iv access, a brief inhalation anesthetic may be adequate. However, since most patients will have iv access, intravenous anesthesia with propofol provides adequate anesthesia, causes less nausea, and allows a rapid recovery. Standard protocol should include meticulous sterile technique in using the central venous line and confirmation of blood aspiration and catheter flow, both prior to beginning induction and after repositioning the patient in a lateral decubitus position. On positioning a patient for lumbar puncture in the left lateral position, it is possible to dislodge the portacath needle, with resulting extravasation of fluid or propofol into the subcutaneous tissue.
Maintenance	Intermittent boluses of propofol 0.5–1 mg/kg may be used or an infusion of 200 mcg/kg/min. N₂O or low-dose sevoflurane may be added. For BMA, add fentanyl 1 mcg/kg or hydromorphone 1–2 mcg/kg for postop pain management. The provider performing the procedures infiltrates the area with lidocaine. For Broviac catheter removals, heparin must be removed from the catheter and discarded before use for induction. A sufficient bolus of propofol (e.g., 0.5–1 mg/kg) must be given just prior to pulling the catheter. Additional iv usually is not warranted because the procedure is normally ≤10 min, but may be desirable if postop pain medication is anticipated. Occasionally, the catheter may break during attempted removal, and the surgeon will have to make a skin incision to allow for removal of the internal fragment. In such cases, a peripheral iv may be inserted quickly to facilitate administration of additional propofol, or mask anesthesia may be given. If the SpO₂ decreases,

	blow-by O₂ with gentle head and neck positioning are generally sufficient. For more significant obstruction, deepening of anesthetic with additional propofol and positive pressure ventilation may be needed.	
Emergence	Allow child to awaken in procedure room. Ondansetron 0.15 mg/kg for patients receiving intrathecal chemotherapy during their lumbar puncture. Avoid dexamethasone, as steroids may be part of the patient's chemotherapy protocol or may precipitate tumor lysis syndrome.	
Blood and fluid requirements	Patients may require blood or factor supplementation prior to or during procedure based on preop laboratory values.	It is important to be aware of patients where fluid restriction may be required, such as in a recent bone marrow transplant patient, or patient with graft vs. host disease, who may have complications of fluid overload. Sodium bicarbonate infusion is common during chemotherapy and should be continued during anesthesia when possible.
Monitoring	Standard monitors	
Positioning	Lateral decubitus (lumbar puncture) Supine, seated, or prone (bone marrow aspiration, biopsy) Supine (Broviac)	
Complications	Airway obstruction Retained central venous catheter	Head and neck repositioning, jaw thrust May require open surgical procedure
Postprocedure pain management	Lumbar punctures are not generally painful, whereas bone marrow aspirations and biopsies can be. Preop care includes the application of EMLA cream to the LP or bone marrow site, as well as having the oncologist infiltrate	

with lidocaine after induction. For pain management, low-dose fentanyl 1 mcg/kg or hydromorphone 2 mcg/kg may be used and repeated in the PACU if needed. Check with the oncologist before administering acetaminophen. Acetaminophen is not routinely given for pain prophylaxis due to the possibility of masking a fever. Acetaminophen is not given unless needed and should never be given as a rectal suppository to the immunocompromised patient. Some patients may have chronic or acute pain syndromes due to tumor bone pain or as a side effect from chemotherapy. These patients need to be evaluated on an individual basis for their pain management. After lumbar punctures, the patient should remain in a horizontal position for at least 30 min to prevent post-dural puncture headache.

Suggested Readings

1. Androupolos DB, Greene MF: Anesthesia and developing brains—implications of the FDA warning. *N Engl J Med* 2017; 376:905–7.
2. Anghelescu DL, Burgoyne LL, Liu T, et al: Clinical and diagnostic imaging findings predict anesthetic complications in children presenting with malignant mediastinal masses. *Pediatr Anesth* 2007; 17:1090–8.
3. Christensen J, Fatchett D: Promoting parental use of distraction and relaxation in pediatric oncology patients during invasive procedures. *J Pediatr Oncol Nurs* 2002; 19(4):127–32.
4. Cravero JP, Beach ML, Blike GT, et al: The incidence and nature of adverse events during pediatric sedation/anesthesia with propofol for procedures outside the operating room: a report from the Pediatric Sedation Research Consortium. *Anesth Analg* 2009; 108(3):795–804.
5. Einaudi S, Bertorello N, Masera N, et al: Adrenal axis function after high-dose steroid therapy for childhood acute lymphoblastic leukemia. *Pediatr Blood Cancer* 2008; 50:537–41.
6. Evans P, Chisholm D: Anaesthesia and paediatric oncology. *Curr Anaesth Crit Care* 2008; 19:50–8.
7. Ghadami YA, Ayatollahi V, Hashemi A, et al: Effect of two different concentrations of propofol and ketamine combinations (Ketofol) in pediatric patients under lumbar puncture or bone marrow aspiration. *Iran J Ped Hematol Oncol* 2013; 3(1):187–92.
8. Ghasemi A, Gharavi Fard M, Sabzevari A: General anesthesia for lumbar puncture and bone marrow aspiration/biopsy in children with cancer. *Iran J Ped Hematol*

Oncol 2013; 3(2):54–8.

9. Latham GJ: Anesthesia for the child with cancer. *Anesthesiology Clin* 2014; 32:185–213.
10. Pinyavat T, Warner DO, Flick RP, et al: Summary of the update session on clinical neurotoxicity studies. *J Neurosurg Anesthesiol* 2016; 28:356–60.
11. Styczynski J: Conscious sedation or general anaesthesia for lumbar puncture in children in Poland. *Lancet* 2017; 390(10099):1025.
12. Whitlow PG, Saboda K, Roe DJ, et al: Topical analgesia treats pain and decrease propofol use during lumbar punctures in a randomized pediatric leukemia trial. *Pediatr Blood Cancer* 2015; 62(1):85–90.

UPPER/LOWER GI ENDOSCOPY

Ann Ming Yeh, William Berquist, Rebecca E. Claure, and Brenda Golianu

Upper and lower endoscopy allows for accurate diagnosis and treatment of gastrointestinal diseases. Diagnostic indications for upper endoscopy include dysphagia; odynophagia; persistent vomiting; abdominal pain with weight loss/anorexia; acid peptic disease, dyspepsia, and heartburn despite medical therapy; diarrhea, hematemesis, or melena; obtaining samples or biopsies; and diagnosis of infection or disease (graft vs. host disease, inflammatory disorders, malignancy) especially in an immunocompromised patient. Indications for colonoscopy include evaluation of gross or occult bleeding, symptom evaluation of new-onset constipation or unexplained diarrhea or abdominal pain, screening or surveillance for inflammatory bowel disease, and screening for colorectal neoplasia. Therapeutic indications for gastrointestinal endoscopy include esophageal dilation, foreign body removal, sclerotherapy or banding of esophageal varices, hemostasis with directed thermal or injection therapy, percutaneous endoscopic gastrostomy tube placement, decompression of sigmoid volvulus or megacolon, polypectomy, PEG placement, and removal or change of gastrostomy or GJ tubes. Children presenting for endoscopy may have congenital and acquired abnormalities of the GI tract. Examples include repaired tracheoesophageal fistula (TEF) with esophageal dysmotility or strictures, ingestion of a caustic material, and presence of a foreign body.

Many endoscopy procedures are done with sedation only. Requests for anesthesia are dependent on the gastroenterologist's preference, as well as the severity of the patient's underlying illnesses. All patients must receive a complete preanesthesia H&P and follow the same NPO protocol used for surgery (see [Appendix D: Standard Pediatric](#)

Anesthetic Management). Patients undergoing colonoscopy require bowel preparation to have the colon free of solid stool for adequate visualization. Common bowel preparation regimens include a combination of magnesium citrate, bisacodyl, and polyethylene glycol.

Absolute contraindications of upper and lower endoscopy include suspected perforation and peritonitis in a toxic patient. Relative contraindications include coagulopathy, neutropenia, fulminant colitis, bowel obstruction and toxic dilation with increased risk of perforation, torrential colonic bleeding, poor bowel preparation, and cardiopulmonary instability.

Overall complication rates of upper GI endoscopy are generally rare, with complications estimated at 0.3% risk of bleeding following an EGD with biopsy and risk of perforation in <0.3% of cases. Hypoxia secondary to sedation, hematoma, missed lesions, and infections are also potential infrequent complications. Complications of colonoscopy occur in <2% of patients, with bleeding and perforation reported in <0.3% and 0.3%, respectively.

ANESTHETIC CONSIDERATIONS

Gastrointestinal	Carefully evaluate for esophageal dysfunction and Sx of GERD. The stomach and bowel will be insufflated with air during the procedure, possibly increasing the likelihood of reflux.
Laboratory	As indicated from H&P. Hct, plt, INR, T+S, if GI bleeding
Premedication	If inhalation induction is planned, administration of midazolam 0.5–0.75 mg/kg po may be helpful for anxiolysis. Consider midazolam for younger children requiring an iv induction (1–2 mg).

INTRAPROCEDURE

Anesthetic technique

Upper GI endoscopy for esophageal dilation, foreign body removal, severe GERD gastrointestinal bleeding, and banding/sclerotherapy of esophageal varices requires GETA. For children ≥ 2 , with no contraindications, supplemental O₂ via nasal cannula, spontaneous respiration, and propofol infusion can be considered. Children ≤ 2 have a

higher incidence of complications when not intubated due to the relatively large endoscope. Complications include partial airway obstruction, tracheal compression, laryngospasm, bronchospasm, and desaturation. Placement of the endoscope in the mouth precludes the ability to use a mask or LMA. For lower GI endoscopy (e.g. colonoscopy), supplemental O₂ via nasal cannula or mask, spontaneous respiration, and propofol infusion may be preferable.

Induction	If rapid-sequence indication is indicated, iv catheter should be placed prior to induction. Proceed with preoxygenation, cricoid pressure, propofol (2–3 mg/kg), and succinylcholine (1–2 mg/kg) or rocuronium (1 mg/kg). ETT should be well secured to the side of the mouth. If risk of aspiration is relatively low, a standard pediatric iv or mask induction is appropriate (see Appendix D: Standard Pediatric Anesthetic Management).	
Maintenance	GETA with inhalational agents or iv infusion of propofol (100–250 mcg/kg/min), ± remifentanyl (0.05–0.10 mcg/kg/min). Alternatively, GA/sedation may be continued with supplemental O ₂ via nasal cannula, spontaneous respiration, and infusion of propofol (100–250 mcg/kg/min), ± remifentanyl (0.05–0.10 mcg/kg/min). If intubated, careful observation and/or holding the ETT prevents inadvertent extubation during removal of the endoscope. If spontaneously breathing, careful attention is needed to observe for breath holding, laryngospasm, and/or desaturation.	
Emergence	If intubated, extubate awake. Emergence in procedure room. Patients usually transported to and recovered in PACU.	
Blood and fluid requirements	No blood loss. IV: 22 or 24 ga × 1 BSS at maintenance	
Monitoring	Standard monitors (see Appendix D: Standard Pediatric Anesthetic Management)	
Positioning	✓ and pad pressure	Lateral decubitus for upper

	points	and lower endoscopy
Complications	eyes Airway obstruction Hypoxemia 2° gastric insufflation Inadvertent extubation Pulmonary aspiration GI perforation	Due to endoscopist working in the mouth and pharynx

POSTPROCEDURE

Pain management	Acetaminophen (35–40 mg/kg pr or 10–15 mg/kg po 10 mg/kg iv), hydromorphone (0.01–0.015 mg/kg iv), or morphine (0.05–0.1 mg/kg iv)
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Suggested Readings

1. Dilos BM: Anesthesia for pediatric airway endoscopy and upper gastrointestinal endoscopy. *Int Anaesthesiol Clin* 2009; 47(3):55–62.
2. Disma N, Astuto M, Rizzo G, et al: Propofol sedation with fentanyl or midazolam during oesophago-gastroduodenoscopy in children. *Eur J Anaesthesiol* 2005; 22(11):848–52.
3. Friedt M, Welsch S: An update on pediatric endoscopy. *Eur J Med Res* 2013; 18:24.
4. Oliva S, Cucchiara S, Cohen SA: Recent advances in pediatric gastrointestinal endoscopy: an overview. *Expert Rev Gastroenterol Hepatol* 2017; 11(7):643–50.
5. Sury M, Smith JH: Deep sedation and minimal anesthesia. *Paediatr Anaesth* 2008; 18(1):18–24.

ANESTHESIA FOR THE PATIENT WITH EPIDERMOLYSIS BULLOSA (EB)

Erin Bushell and Louise Furukawa

Introduction: Care of the patient with epidermolysis bullosa (EB) is always challenging, requiring familiarity with the disease and its sequelae and how it affects the ability to perform the procedure and anesthetic. This section will focus on recessive

dystrophic EB (RDEB), as it is the most common form in patients presenting for esophageal dilation. Esophageal stricture is present in 50% of patients by age 10 with a cumulative risk of stricture of 95% by age 45. Generally speaking, procedures performed out of the OR take the patient and provider away from resources and help, adding additional risk and complexity. Out-of-OR procedures have been identified as higher risk for adverse events, particularly in shared airway cases such as endoscopy. Furthermore, a primary tenet of Crisis Resource Management is “know your environment,” which can be difficult if the procedure takes place in an infrequently utilized location. EB patients often require multiple procedures at specialty institutions. The cost of travel and caregiver loss of wages can be prohibitive; therefore, coordinating multiple procedures under the same anesthetic may be desirable. The care of the EB patient in a multidisciplinary clinic is optimal.

PREPROCEDURE CONSIDERATIONS

Education of team members	• EB clearly known beforehand and indicated on schedule • Staff (including anesthesia techs) familiar with care of EB patient • Special equipment or supplies available
Past anesthetic history	• Vascular access history • Past anesthetic records with previous intubation techniques (disease is progressive so old records may not predict current airway status) • History of PONV, postop pain/sore throat
Craniofacial/airway	• Review of past records • In-person visit or video interview important. Document detailed oral exam. Important to include pictures in preop note • H/o stridor concerning for extensive pharyngeal scarring • H/o corneal abrasions/scarring. Note eye care regimen. Careful dental exam—poor dentition common. Esophageal stricture may compromise the ability to swallow oral secretions.

Respiratory	Severe kyphoscoliosis or extensive cicatricial scarring may impair pulmonary mechanics and cause restrictive disease.
Cardiac	- Cardiomyopathy may be present with mean age of onset 9–12 yr.⁵ - Signs and symptoms similar to patients without EB. Active patients will have DOE. Most will display orthopnea and edema.
Gastrointestinal	Stricture, poor nutrition, hypoalbuminemia, and iron deficiency anemia are common.
Renal	Some patients have renal failure—increases risk for cardiomyopathy (cause unknown)^{5,6}
Hematology	- Multifactorial anemia common - Many receive Fe infusion vs. transfusion⁵
Oncologic	- Squamous cell carcinoma (SCCA) can arise as early as the second decade of life. - Metastatic SCCA is the principal cause of death in patients with RDEB.
Chronic pain	Chronic pain is experienced by every patient due to extensive wounds and arduous dressing changes. Some use narcotics frequently.
Psychological	- Chronic pain, anxiety, PTSD, and regression are common. - Most patients have a very close attachment to caregivers. - Caregivers may have difficulty relinquishing control and should be allowed to participate in the preoperative process.
Current medications	

	• Patients should continue essential preoperative medications.
Lab tests	• CBC; hematologist may ask for additional tests such as iron studies. • If vascular access is difficult, labs are often drawn after anesthetic induction. • Chemistries if renal failure is present
Premedication	• Some patients allow IV access preop. • Many (even older) patients may have significant psychological trauma and request mask induction. • ELA-max (lidocaine) cream is useful. • Ultrasound-guided access is helpful and US gel is safe. Oral or GT versed OK to use. • IM premedication may also be indicated.
NPO	• Standard NPO guidelines apply. • Many patients have gastrostomy tubes and should be hydrated with clear liquids until NPO cutoff time.
Check-in process	• Staff involved with checking in the patient should be aware of disease and have modified equipment for monitoring vital signs. • Sheepskin should be placed on the gurney and follow patient throughout procedure. • Caregiver should be able to stay with patient and assist preoperative preparation.

INTRAOPERATIVE ANESTHETIC CONSIDERATIONS

Induction: IV IV induction— TIVA	Adults and pediatric patients with IV access: A standard IV induction followed by a TIVA is appropriate. Preoxygenation can be accomplished by careful application of the face mask (as below) or using a heated, humidified high flow nasal cannula such as the Optiflow™. Most awake patients tolerate flows of 30
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No IV— inhalational

lpm, which then can be increased to 70 lpm after induction to maintain oxygenation during intubation. IV access is often difficult. Asking the patient and caregivers about which veins have been successfully used in the past and use of ultrasound is advised. Elastic tourniquets should generally be avoided. However, if a hand tourniquet is not sufficient, an elastic tourniquet may be carefully placed over Webril™. The area may be gently cleaned with baby shampoo on a moist gauze or an alcohol swab, taking care to avoid any shear stress. The IV must be secured with care. Mepilex^R transfer foam can be used to pad under the IV hub, Mepitel^R may be placed over the catheter, and a self-adherent wrap such as Coban™ is used to secure the IV by wrapping it circumferentially around extremity. It is suggested that an extension such as a T-piece be used so the IV may be disconnected during patient positioning and transfers without risk of displacement.

Once patient is induced and an IV has been placed and secured, a TIVA allows for maintenance of anesthesia while securing the airway.

Pediatric patients without IVs: An inhalational induction with sevoflurane and N₂O (see [p. D-1](#)) is appropriate for these patients. Great care must be taken to avoid trauma from the face mask and jaw thrust. Petroleum jelly or other petroleum-based lubricants applied to a well-inflated mask and petroleum jelly gauze beneath the mask can reduce the risk of mask-related facial trauma. Mepilex^R transfer foam can be used to pad the area over the angle of the mandible to reduce trauma related to jaw thrust. Again, avoidance of shear stresses is of primary importance. (Note: These patients may have psychosocial issues due to their illness and psychological trauma related to past procedural experiences, and this may necessitate inhalational induction in patients older than one traditionally would consider this approach.)

Intubation: Patients with EB often have ankyloglossia,

limited mouth opening, and intraoral and pharyngeal mucosal scarring and adhesions. They tend to have patent natural airways, so ventilation is not difficult, but intubation can be both difficult and potentially traumatic to friable tissues. Therefore, fiberoptic intubation with a smaller tube than would normally be used is recommended. Microlaryngeal tubes (MLT)—pediatric diameter tubes that are adult size in length, and have a standard adult cuff—can be used for both oral and nasal intubations. Once placed, the ETT is tied with umbilical tape placed over petroleum jelly gauze to avoid trauma to the face and lips. There is no way to truly secure the ETT in these patients, and it is, therefore, recommended that the anesthesiologist manually hold the tube during times when the risk of accidental extubation is high such as when the dilating catheters and endoscopes are being placed or removed. Nasal intubation may decrease the risk of accidental extubation and thus should be considered for complex cases. It is strongly suggested that during the time out, the team review the importance of having the proceduralist notify the anesthesiologist prior to inserting or removing anything from the mouth so the anesthesiologist may take appropriate steps to prevent unintentional extubation.

Maintenance

TIVA with short-acting agents such as propofol (100–250 mcg/kg/min) and remifentanyl (0.1–3 mcg/kg/min). Muscle relaxation is not necessary.

Emergence

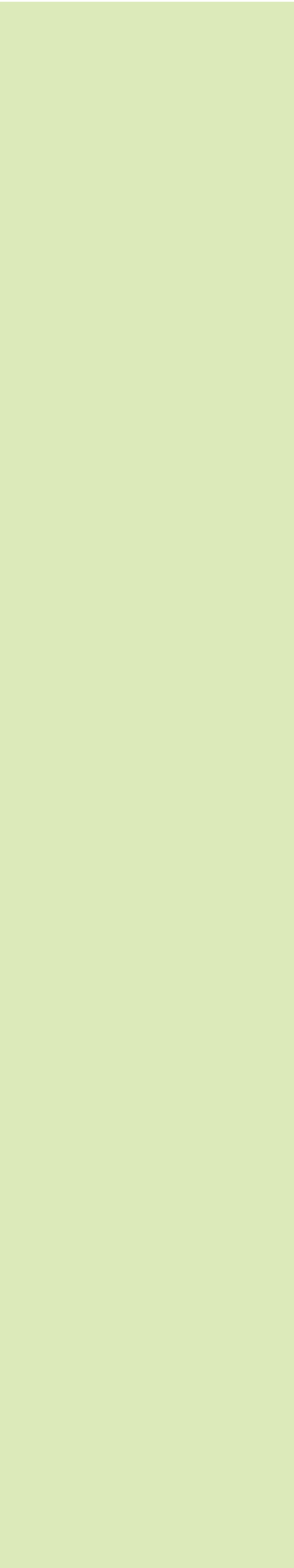
Patients awaken rapidly after discontinuation of TIVA. A narcotic should be employed to address postop pain and an antiemetic such as ondansetron (0.1–0.15 mg/kg for children, 4 mg for adults) should be given. Dexamethasone (0.1 mg/kg for children, 4 mg for teens, or 4–8 mg for adults) will reduce nausea and the edema that occurs following these procedures. Deep extubation should be considered as it reduces coughing on tube in these patients already prone to sore throat and trauma.

Blood and fluid

No blood loss.

Routine maintenance fluids

requirements	IV: 22 or 24 ga × 1 BSS at maintenance	are required unless a major complication such as esophageal perforation occurs.
Monitoring	Standard ASA monitors (see p. D-1). Applied without adhesives to avoid shear injury	Standard monitors must be applied in a nonstandard fashion to avoid injury to the patient. ⁵ EKG leads may be either applied to gel pads such as Defib-Pads (3M™) covered with moistened 2 × 2 gauze to prevent the gel from drying out and sticking to the patient's skin and secured with Coban™ if on an extremity or merely placed on patient's chest. Adhesive pulse oximetry probes should be rendered nonadhesive by placing transparent film dressing such as a Tegaderm™ adhesive side to adhesive side and trimming away the excess. The probe may then be secured to a digit (or stump if pseudosyndactyly is present) with Coban™. The blood pressure cuff may be placed over the patient's existing dressings or over Webriil™ to avoid skin trauma due to cuff inflation.
Positioning	✓ and pad pressure points ✓ eyes Sheepskin under patient	A sheepskin pad underneath the patient can provide protection as well as a means to move the patient in a way



that avoids shear stress. Instead of utilizing a roller to move the patient between bed and gurney, which can create a significant amount of shear stress, the patient is lifted directly up using the sheepskin, moved through the air, and placed carefully down upon the bed. The majority of EB patients, including adults, are of low weight and small physical size and can be easily moved in this manner by the operative team.

A moisturizing cream or ointment such as Albolene[®] or Eucerin[®] applied to the cheeks and lips can reduce the risk of a facial trauma due to airway management. Careful attention must be paid to the eyes as the majority of these patients have corneal blistering and other ocular complications associated with EB and are at extremely high risk for perioperative eye injury. An ophthalmic ointment or methylcellulose eye drops should be applied to both eyes, which should then be closed to the extent possible and covered with Mepitel[™] and moistened 2 × 2 gauze.⁶ Prior to draping the patient, it

is advisable to carefully examine the patient by head-to-toe checking to ensure that all pressure points are padded and that no iv lines or monitoring cables are against the patient's skin. NB: Drapes should not touch the skin directly, as they may stick to the skin and create new lesions when removed.

Complications

Esophageal perforation and bleeding are the most common procedure-related complications, and the patients should be intubated to provide airway protection should this occur. If perforation does occur, it is considered a surgical emergency and the patient should proceed to the OR.

The majority of complications are related to the patient's disease and injury from positioning. Careful attention to protect the patient from these injuries is paramount.

- Esophageal perforation/rupture/mediastinitis.
- Contrast esophagram may be performed during the procedure to rule out esophageal injury.
- Formation of new lip, intraoral, pharyngeal, or esophageal blisters from manipulation during anesthesia or pharyngeal/esophageal trauma during the dilation procedure.

Pain

- Sore throat due to intubation may be caused by traumatic intubation but also by excessive endotracheal tube cuff pressures. It is therefore important to use MLT tubes instead of non-MLT small tubes with excessively inflated cuffs.
- Sore throat and odynophagia caused by the dilation procedure are also common in the postoperative period.
- May use narcotics as well as nonnarcotic adjuvants.
- Oral budesonide mixed with honey or sucralose may help ease postoperative discomfort.

PONV	• Postoperative nausea and vomiting are common. • Dexamethasone, ondansetron, and adequate hydration are helpful.
Restenosis	• A recent study suggests that regular treatment with budesonide slurry may slow the development of restenosis.

▼ POSTPROCEDURE

Pain management	Acetaminophen (35–40 mg/kg pr or 10–15 mg/kg po 10 mg/kg iv), hydromorphone (0.01–0.015 mg/kg iv), or morphine (0.05–0.1 mg/kg iv)
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Suggested Readings

1. Dures E, Morris M, Gleeson K, et al: The psychosocial impact of epidermolysis bullosa. *Qual Health Res* 2011; 21(6):771–82.
2. Fine J, Mellerio JE: Extracutaneous manifestations and complications of inherited epidermolysis bullosa. Part I. Epithelial associated tissues. *J Am Acad Dermatol* 2009; 61(3):367–84.
3. Fine J, Mellerio JE: Extracutaneous manifestations and complications of inherited epidermolysis bullosa. Part II. Other organs. *J Am Acad Dermatol* 2009; 61(3):387–402.
4. Gustafsson IM, Lodenius A, Tunelli J, et al: Apnoeic oxygenation in adults under general anaesthesia using transnasal humidified rapid-insufflation ventilatory exchange (THRIVE)—a physiological study. *Br J Anaesth* 2017; 118(4):610–7.
5. Humphreys S, Lee-archer P, Reyne G, et al: Transnasal humidified rapid-insufflation ventilatory exchange (THRIVE) in children: a randomized controlled trial. *Br J Anaesth* 2017; 118(2):232–8.
6. Metzner J, Domino KB: Risks of anesthesia or sedation outside the operating room: the role of the anesthesia care provider. *Curr Opin Anaesthesiol* 2010; 23:523–31.
7. Metzner J, Posner KL, Domino KB: The risk and safety of anesthesia at remote locations: the US closed claims analysis. *Curr Opin Anaesthesiol* 2009; 22:502–8.
8. Nandi R, Howard R: Anesthesia and epidermolysis bullosa. *Dermatol Clin* 2010; 28:319–24.
9. Patel A, Nouraei SA: Transnasal humidified rapid-insufflation ventilatory exchange

(THRIVE): a physiological method of increasing apnoea time in patients with difficult airways. *Anaesthesia* 2015; 70:323–9.

10. Zanini A, Guez S, Salera S, et al: Oral viscous budesonide as a first-line approach to esophageal stenosis in epidermolysis bullosa: an open-label trial in six children. *Paediatr Drugs* 2014; 16(5):391–5.

CROSS-SECTIONAL IMAGING (CT, MRI)

Rebecca E. Claire and Brenda Golianu

PREPROCEDURE

A complete preanesthetic H&P should be performed in all cases and the usual NPC protocol followed (see [Appendix D: Standard Pediatric Anesthetic Management](#)). For MRI, thorough questioning regarding potentially ferromagnetic implants (neurostimulators, cochlear implants, pacemakers, surgical clips, infusion pumps, etc.) is critical. Direct questions regarding suitability of implants for scanning to radiologist.

Respiratory	Procedures for patients with Sx of URI (nasal congestion, cough, fever) are usually postponed for 3–4 wk unless the study is urgent. If Sx are minor, the study may proceed.
Cardiovascular	Pacemakers are usually contraindicated. Prosthetic heart valves may be contraindicated (identify the type of valve present and consult with radiologist). Fresh surgical clips (e.g., recent PDA ligation) may also represent a contraindication for MRI. For cardiovascular patients, hold ACE inhibitors the morning of the procedure.
Neurologic	Head CT or MRI may be performed in patients with seizure disorders or brain tumors. Review seizure meds and ✓ serum levels if appropriate. Recent drug levels may not be necessary in children with well-controlled or stable seizure disorders. If ↑ ICP or ↓ intracranial compliance are present (e.g., 2° tumors or hydrocephalus), the anesthesia plan should include tracheal intubation with controlled

	ventilation. Metal surgical clips or coils are usually contraindications for MRI. Children with head trauma may have concurrent C-spine injuries.
Laboratory	As indicated from H&P. Usually none required
Premedication	If inhalation induction is planned, administration of midazolam 0.5–0.75 mg/kg po may be helpful for anxiolysis. If iv induction is planned, apply topical anesthesia (see Appendix D: Standard Pediatric Anesthetic Management).

INTRAPROCEDURE

Unique considerations for MRI

See Adult Out-of-OR Procedures, [p. 1686](#).

Anesthetic technique

CT scans: With current scanners, many studies can be completed in <10 min. Each set of data takes 2–3 min to collect, allowing many children to complete scans without any sedation. In some institutions, children undergo noninvasive CT scans with sedation by radiology personnel. For more complex cases, sedation/GA is provided by an anesthesiologist. Most commonly, noninvasive CT scans are performed with inhalation anesthesia via mask or with a continuous infusion of propofol (100–250 mcg/kg/min). For patients undergoing invasive CT-guided procedures (needle biopsy, placement of drainage tubes, such as thoracostomy tube, etc.), GETA is usually performed. Techniques include inhalational anesthesia and/or continuous infusion of propofol (100–200 mcg/kg/min) and remifentanyl (0.05–0.10 mcg/kg/min).

MRI: MRI requires GA more often than noninvasive CT because prolonged immobility is required for up to 1–2 h. Options include inhalational anesthesia with an MRI-compatible anesthesia machine or continuous propofol infusion with an MRI-compatible infusion pump. Hypothermia frequently occurs in small children. The cooling fan for the magnet can sometimes be turned off to decrease heat loss (discuss with MRI technician). One blanket over patient is usually permissible. The iv, airway circuit, and monitors require extensions. Ferromagnetic objects in the MRI environment are a hazard. Objects can be propelled toward the magnet with sufficient speed and force to result in serious or fatal injury to the patient and/or health care provider.

Induction	Standard pediatric mask or iv induction (see Appendix D:
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	Standard Pediatric Anesthetic Management) can be accomplished in the CT scanner room or in a special induction area outside the MRI scanner. After induction and stabilization of the airway, the patient and all personnel entering the MRI scanner should undergo a second check for removal of metal objects and equipment. Once check completed, transport patient into scanner and resume monitoring.	
Maintenance	GA continued with supplemental O₂ via nasal cannula, spontaneous respiration, and infusion of propofol (100–250 mcg/kg/min). In children with an LMA or ETT, maintenance with inhalational agent or propofol infusion. Typically, these patients do not require muscle relaxants, and spontaneous breathing is appropriate unless controlled ventilation is required to treat ↑ ICP or for protocols involving breath holding for prolonged periods of time (e.g., several minutes for abdominal scans). Most infusion pumps need to remain a specific distance from the MRI scanner. The appropriate distance is dependent on the degree of shielding of each particular magnet. Adequate ear protection should be routinely used during MRI.	
Emergence	Transport MRI patient back to induction area for emergence so that airway equipment is readily available. Patients are transported to and recovered in PACU.	
Blood and fluid requirements	No blood loss. IV, if required: 22 or 24 ga × 1 BSS @ maintenance	
Monitoring	Standard MRI-compatible monitors	For infants, use a child-size NIBP cuff on the leg.
Positioning	Ear protection necessary ✓ and pad pressure points ✓ eyes	
Complications	Airway obstruction Hypothermia	Repositioning, nasal or oral airway Minimize duration of scan in

Burn injury

IV contrast reaction
Hearing loss (MRI)

younger patients. Have available warming devices—blankets, Bair Hugger, warming pads, etc. From inappropriate placement of pulse oximeter probe or EKG. Avoid coiling of wires and use only MRI-compatible monitors.

▼ POSTPROCEDURE

Pain management

Noninvasive procedures are painless.

Acetaminophen (35–40 mg/kg pr, 10–15 mg/kg po, or 10 mg/kg iv), hydromorphone (0.01–0.015 mg/kg iv), or morphine (0.05–0.1 mg/kg iv) as appropriate for invasive procedures

Suggested Readings

1. Ahmad R, Hu HH, Krishnamurthy R, et al: Reducing sedation for pediatric body MR using accelerated and abbreviated imaging protocols. *Pediatr Radiol* 2018; 48(1):37–49.
2. De Sanctis Briggs V: Magnetic resonance imaging under sedation in newborns and infants: a study of 640 cases using sevoflurane. *Paediatr Anaesth* 2005; 15(1):9–15.
3. Salerno S, Granata C, Trapenese M, et al: Is MRI imaging in pediatric age totally safe? A critical appraisal. *Radiol Med* 2018; 123(9):695–702. doi:10.1007/s11547-018-0896-1.
4. Serafini G, Zadra N: Anaesthesia for MRI in the pediatric patient. *Curr Opin Anaesthesiol* 2008; 21(4):499–503.
5. Usher AG, Kearney RA, Tsui B: Propofol total intravenous anesthesia for MRI in children. *Paediatr Anaesth* 2005; 15(1):23–8.

SURGICAL CONSIDERATIONS FOR ECMO

PROCEDURAL CONSIDERATIONS

Gary Hartman and Claudia Mueller

Extracorporeal membrane oxygenation (ECMO) for prolonged periods (3–21 d) allows cardiopulmonary support for newborns and children with reversible respiratory failure. The most common indications are meconium aspiration and pulmonary HTN associated with congenital diaphragmatic hernia. The procedure is performed in the NICU with OF techniques. Patients are given anticoagulants prior to cannulation. Subsequently, repair of the diaphragmatic hernia also may be performed in the NICU on ECMO support before decannulation. (Fig. 13.2-1 shows ECMO schematic.)

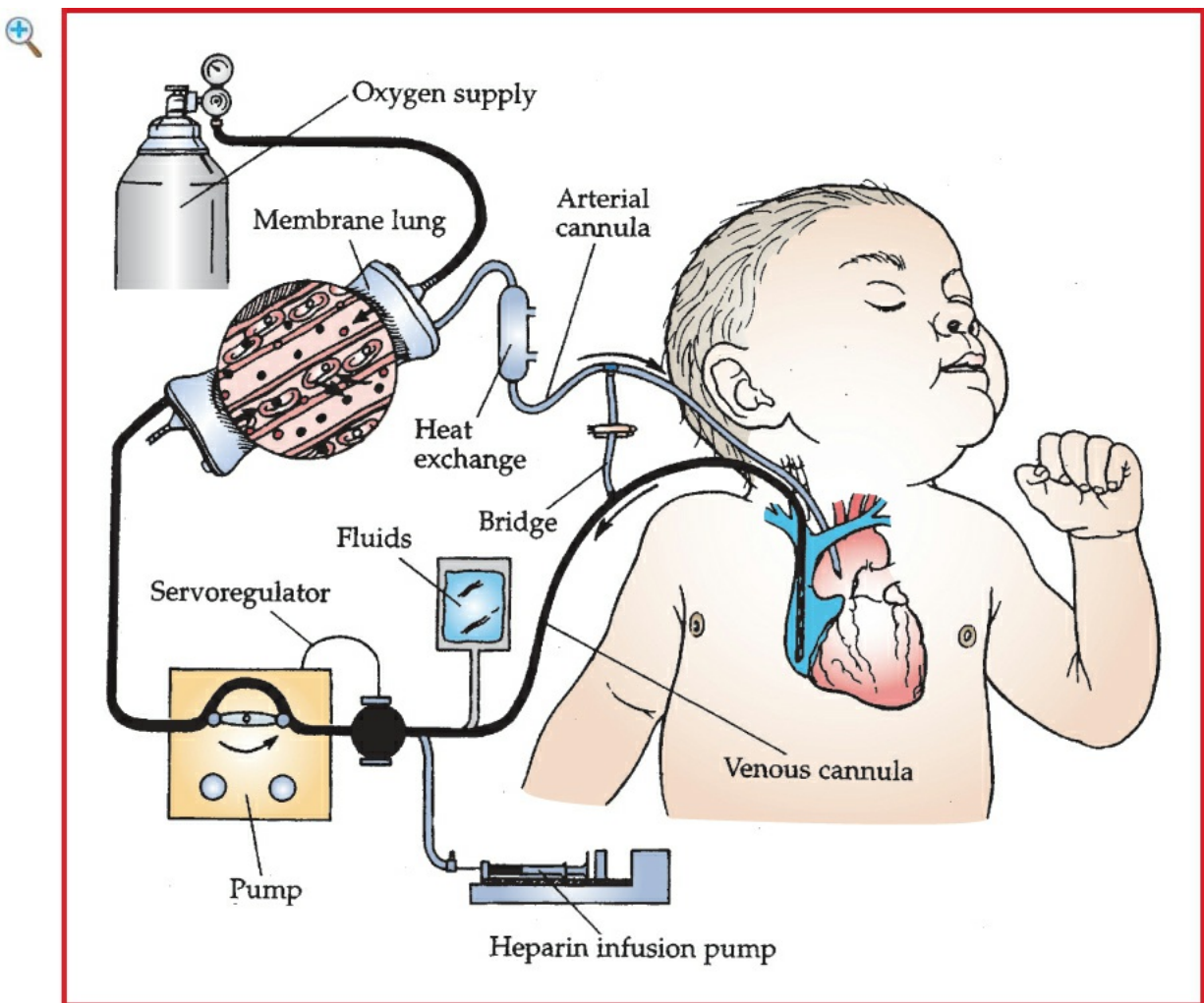


Figure 13.2-1 ECMO circuit illustrating venoarterial (VA) cannulation. Venous blood is withdrawn by gravity through a servoregulator to prevent pump from actively siphoning venous return. A pump delivers blood back to the arterial cannula after it passes through the membrane oxygenator and heat exchanger. Venous return is from the right atrium, whereas arterial infusion is into the aortic arch in double-cannula (VA) or right atrium (venovenous) techniques. (Reproduced with permission from Baker RJ, Fischer JE: *Mastery of Surgery*, Vol. I, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

The potential detrimental effects of the diaphragmatic repair on respiratory function can be managed with increased circuit flow. ECMO also has been helpful in some newborns with cardiopulmonary failure following correction of congenital cardiac defects. Vascular access is accomplished with one (venovenous) or two (venoarterial) cannulae. The IJ vein is cannulated in both methods with the tip of the cannula in the right atrium. In venoarterial ECMO, the common carotid artery is used with the tip of the cannula at the aortic arch. The wound is closed around the cannulae, which are secured to the infant's scalp.

Usual preop diagnosis

Meconium aspiration; diaphragmatic hernia

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Right neck incision; subcostal for hernia repair
Special instrumentation	ECMO circuit (Fig. 13.2-1)
Unique considerations	Anticoagulation
Antibiotics	Broad spectrum based on underlying condition
Surgical time	1–2 h
Closing considerations	Assess for changes in ventilation (e.g., PIP, pre- and postductal ABG); assess hemostasis.
EBL	>5–10 mL/kg
Postop care	Paralysis maintained; fentanyl infusion at 2–4 mcg/kg/h
Mortality	Based on underlying conditions
Morbidity	Respiratory failure Sepsis
Pain score	6–7

■ PATIENT POPULATION CHARACTERISTICS	
Age range	Primarily neonatal with increasing use in older children

Male:Female Incidence	1–2:1 1:4000 live births
Etiology	Unknown
Associated conditions	For diaphragmatic hernia: congenital heart disease (15%); renal anomalies; intestinal atresia; CNS abnormalities

ANESTHETIC MANAGEMENT FOR SURGICAL PROCEDURES UNDER ECMO

Claudia Benkwitz and Lisa Wise-Faberowski

ECMO has become a standard therapy for refractory neonatal and pediatric cardiorespiratory failure. Over the past 2 decades, the indications for ECMO have changed. The use of ECMO for neonatal respiratory failure peaked in the early 1990s but its use has now decreased with the development of alternative strategies such as surfactant, high-frequency oscillation, and inhaled nitric oxide. In contrast, cardiac indications for ECMO have continued to increase especially for cardiopulmonary arrest (ECPR) refractory to standard CPR management. Improved patient outcomes with ECMO have been achieved by combining new technologies with refined management strategies and improved training of personnel.

Anesthesiologists may find themselves encountering an increasing number of neonatal and pediatric patients on ECMO support for various surgical procedures and imaging studies. Successful management of these patients includes meticulous planning, comprehensive teamwork, communication, and attention to detail. Intrahospital transport of these patients often requires multiple personnel from various medical specialties, and therefore, it is important to elect a team leader prior to transport and to allocate specific tasks (e.g., protection of ECMO cannulae) in order to avoid any confusion or conflict.

The delivery of anesthetic agents can be via a TIVA technique or a combination of intravenous and inhalation agents. However, the oxygenator membrane of the bypass circuit can sequester varying amounts of lipid-soluble intravenous agents, such as fentanyl, altering the plasma levels and resulting in unpredictable hemodynamic changes. Volatile anesthetics are not routinely available on ECMO circuits due to difficulties with scavenging. Anesthetic agents cause preload and afterload changes, and therefore additional volume (blood products and colloids) and vasoactive agents should

be readily available in order to maintain adequate filling and blood pressure.

It is important to keep in mind that altering the surgical table height and thus the patient's height will have an impact on the venous return to the ECMO circuit (passive gravity-assisted drainage); hence, the perfusionist must be informed before undertaking this maneuver.

Diaphragmatic hernia repair is the most common noncardiac procedure undertaken on ECMO support. ECMO therapy protects the pulmonary alveoli from ventilator associated barotrauma and oxygen toxicity while maintaining tissue oxygenation. Prior to surgery, a trial period off bypass is used to assess the adequacy of conventional ventilation independent of ECMO. If the neonate is unable to be weaned, then the surgery occurs on full ECMO support. ACT is maintained at ~200 sec by the addition of heparin to the ECMO circuit. The potential detrimental effects of diaphragmatic repair on respiratory function can be managed with increased circuit flow.

Monitoring includes heart rate, arterial blood pressure, temperature, and pump flow rate with frequent ABGs and VBGs.

The majority of practitioners employ VA cannulations for ECMO support (see [Fig. 13.2-1](#)) although some centers prefer VV ECMO support if hypoxemia is the primary indication.

Suggested Readings

1. Harvey C: Cannulation for neonatal and pediatric extracorporeal membrane oxygenation for cardiac support. *Front Pediatr* 2018; 6:17.
2. Lasa JJ, Jain P, Raymond TT, et al: Extracorporeal cardiopulmonary resuscitation in the pediatric cardiac population: in search of a standard of care. *Pediatr Crit Care Med* 2018; 19:125–30.
3. Lorusso R, Taccone FS, Belliato M, et al: Brain monitoring in adult and pediatric ECMO patients: the importance of early and late assessments. *Minerva Anestesiol* 2017; 83:1061–74.